1 BF operating

1.1 Pulverized coal injection system - Coal injection pipeline - Operation monitoring

1.1.1 Pressure of 1# pulverized coal injection pipe: The static pressure detected in the No.1 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.2 Pressure of 2# pulverized coal injection pipe: The static pressure detected in the No.2 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.3 Pressure of 3# pulverized coal injection pipe: The static pressure detected in the No.3 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.4 Pressure of 4# pulverized coal injection pipe: The static pressure detected in the No.4 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.5 Pressure of 5# pulverized coal injection pipe: The static pressure detected in the No.5 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.6 Pressure of 6# pulverized coal injection pipe: The static pressure detected in the No.6 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.7 Pressure of 7# pulverized coal injection pipe: The static pressure detected in the No.7 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.8 Pressure of 8# pulverized coal injection pipe: The static pressure detected in the No.8 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.9 Pressure of 9# pulverized coal injection pipe: The static pressure detected in the No.9 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.10 Pressure of 10# pulverized coal injection pipe: The static pressure detected in the No.10 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.11 Pressure of 11# pulverized coal injection pipe: The static pressure detected in the No.11 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.12 Pressure of 12# pulverized coal injection pipe: The static pressure detected in the No.12 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.13 Pressure of 13# pulverized coal injection pipe: The static pressure detected in the No.13 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.14 Pressure of 14# pulverized coal injection pipe: The static pressure detected in the No.14 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.15 Pressure of 15# pulverized coal injection pipe: The static pressure detected in the No.15 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.16 Pressure of 16# pulverized coal injection pipe: The static pressure detected in the No.16 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.17 Pressure of 17# pulverized coal injection pipe: The static pressure detected in the No.17 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.18 Pressure of 18# pulverized coal injection pipe: The static pressure detected in the No.18 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.19 Pressure of 19# pulverized coal injection pipe: The static pressure detected in the No.19 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.20 Pressure of 20# pulverized coal injection pipe: The static pressure detected in the No.20 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.21 Pressure of 21# pulverized coal injection pipe: The static pressure detected in the No.21 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.22 Pressure of 22# pulverized coal injection pipe: The static pressure detected in the No.22 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.23 Pressure of 23# pulverized coal injection pipe: The static pressure detected in the No.23 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.24 Pressure of 24# pulverized coal injection pipe: The static pressure detected in the No.24 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.25 Pressure of 25# pulverized coal injection pipe: The static pressure detected in the No.25 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.26 Pressure of 26# pulverized coal injection pipe: The static pressure detected in the No.26 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.27 Pressure of 27# pulverized coal injection pipe: The static pressure detected in the No.27 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.28 Pressure of 28# pulverized coal injection pipe: The static pressure detected in the No.28 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.29 Pressure of 29# pulverized coal injection pipe: The static pressure detected in the No.29 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.30 Pressure of 30# pulverized coal injection pipe: The static pressure detected in the No.30 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.31 Pressure of 31# pulverized coal injection pipe: The static pressure detected in the No.31 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.32 Pressure of 32# pulverized coal injection pipe: The static pressure detected in the No.32 pulverized coal injection branch pipe of a blast furnace, characterizing the conveying pressure of pulverized coal in this pipe.

1.1.33 Pressure of outlet about injection tank: The pressure value measured at the discharge outlet of the pulverized coal injection tank, reflecting the output pressure of pulverized coal from the storage tank.

1.1.34 Pressure of main injection pipe: The total pressure of the main pipeline for pulverized coal injection in a blast furnace, representing the overall conveying pressure of the injection system.

1.1.35 Pressure of main nitrogen pipe: The operating pressure of the main nitrogen pipeline used for purging, conveying and pressurizing in the pulverized coal injection system.

1.1.36 Pressure of distributor - 1: The pressure detected at the No.1 pulverized coal distributor, which distributes pulverized coal to each branch injection pipe.

1.1.37 Pressure of distributor - 2: The pressure detected at the No.2 pulverized coal distributor, which distributes pulverized coal to each branch injection pipe.

1.1.38 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Pulverized coal injection system - Coal injection pipeline - Operation monitoring”.

1.2 Pulverized coal injection system - Coal injection pipeline - Pulverized coal flow control

1.2.1 Weight of pulverized coal of 1# injection tank: The real-time mass of pulverized coal stored in the No.1 pulverized coal injection tank.

1.2.2 Weight of pulverized coal of 2# injection tank: The real-time mass of pulverized coal stored in the No.2 pulverized coal injection tank.

1.2.3 Weight of pulverized coal of 3# injection tank: The real-time mass of pulverized coal stored in the No.3 pulverized coal injection tank.

1.2.4 Hourly volume of pulverized coal injection: The total mass of pulverized coal injected into the blast furnace per hour.

1.2.5 Flow rate of secondary supplemental air about pulverized coal injection: The flow rate of auxiliary air used to assist pulverized coal conveying in the injection system.
1.2.6 Weight setpoint value of hourly volume of pulverized coal injection: The preset target value for the hourly pulverized coal injection volume.
1.2.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Pulverized coal injection system - Coal injection pipeline - Pulverized coal flow control”.

1.3 Gas system - Gas analyzer - Composition detection

1.3.1 CO content: The volume percentage of carbon monoxide in blast furnace top gas.
1.3.2 CH4 content: The volume percentage of methane in blast furnace top gas.
1.3.3 H2 content: The volume percentage of hydrogen in blast furnace top gas.
1.3.4 CO2 content: The volume percentage of carbon dioxide in blast furnace top gas.
1.3.5 N2 content: The volume percentage of nitrogen in blast furnace top gas.
1.3.6 Gas utilization rate: The efficiency of carbon and hydrogen in blast furnace gas participating in the reduction reaction.
1.3.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Gas system - Gas analyzer - Composition detection”.

1.4 Furnace top system - Integrated terminal - Operation monitoring

1.4.1 Stockline level: The vertical height of the burden surface inside the blast furnace shaft.
1.4.2 Temperature of upper seal seat ring: The operating temperature of the upper sealing seat ring of the blast furnace charging device.
1.4.3 Temperature of lower seal seat ring: The operating temperature of the lower sealing seat ring of the blast furnace charging device.
1.4.4 Temperature of valve box: The ambient temperature of the valve box for blast furnace charging and sealing valves.
1.4.5 Liquid level of oil tank in hydraulic station: The height of hydraulic oil stored in the oil tank of the blast furnace hydraulic power unit.
1.4.6 Temperature of oil tank in hydraulic station: The temperature of hydraulic oil inside the oil tank of the blast furnace hydraulic power unit.
1.4.7 Temperature of secondary equalizing pressure about nitrogen: The temperature of nitrogen used for secondary pressure equalization in the blast furnace charging system.
1.4.8 Temperature of furnace top steam: The temperature of steam medium at the blast furnace top for sealing or purging.
1.4.9 Weighing net weight: The net mass of blast furnace burden after deducting tare weight in weighing.
1.4.10 Pressure of secondary equalizing pressure about nitrogen: The pressure of nitrogen used for secondary pressure equalization in the blast furnace charging system.
1.4.11 Pressure of hoisting bucket: The internal pressure of the blast furnace charging hoisting bucket during operation.
1.4.12 Pressure of 1# valve box: The pressure inside the No.1 valve box of the blast furnace charging system.
1.4.13 Pressure of 2# valve box: The pressure inside the No.2 valve box of the blast furnace charging system.
1.4.14 Differential pressure of valve box: The pressure difference between the inside and outside of the blast furnace charging valve box.
1.4.15 Pressure of main helium pipe: The operating pressure of the main helium pipeline for blast furnace

detection or sealing.

1.4.16 Pressure of furnace top steam: The pressure of steam medium at the blast furnace top for sealing or purging.

1.4.17 Flow rate secondary equalizing pressure about nitrogen: The flow rate of nitrogen used for secondary pressure equalization in the blast furnace charging system.

1.4.18 Pressure of 1# hoisting bucket: The internal pressure of the No.1 blast furnace charging hoisting bucket.

1.4.19 Pressure of 2# hoisting bucket: The internal pressure of the No.2 blast furnace charging hoisting bucket.

1.4.20 Temperature A of gas-tight box: The temperature at point A of the blast furnace gas-tight box for charging equipment.

1.4.21 Temperature B of gas-tight box: The temperature at point B of the blast furnace gas-tight box for charging equipment.

1.4.22 Temperature C of gas-tight box: The temperature at point C of the blast furnace gas-tight box for charging equipment.

1.4.23 Pressure of furnace top: The static pressure of gas at the blast furnace top.

1.4.24 Flow rate of nitrogen to gas-tight box: The flow rate of nitrogen supplied to the blast furnace gas-tight box.

1.4.25 Flow rate of nitrogen to lower seal box: The flow rate of nitrogen supplied to the lower sealing box of blast furnace charging equipment.

1.4.26 Pressure of nitrogen to gas-tight box: The pressure of nitrogen supplied to the blast furnace gas-tight box.

1.4.27 Pressure of nitrogen pipe about lower valve box: The pressure of nitrogen in the pipeline supplying the lower valve box of blast furnace charging equipment.

1.4.28 Temperature of nitrogen to lower seal box: The temperature of nitrogen supplied to the lower sealing box of blast furnace charging equipment.

1.4.29 Daily total batch count of coke: The total number of coke batches charged into the blast furnace per day.

1.4.30 Daily total batch count of ore: The total number of ore batches charged into the blast furnace per day.

1.4.31 Upper limit value of stockline height: The maximum allowable height of the blast furnace burden surface.

1.4.32 Lower limit value of stockline height: The minimum allowable height of the blast furnace burden surface.

1.4.33 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Integrated terminal - Operation monitoring”.

1.5 Furnace top system - Steam pipeline - Operation monitoring

1.5.1 Temperature of furnace top steam: The temperature of steam medium at the blast furnace top for sealing or purging.

1.5.2 Pressure of furnace top steam: The pressure of steam medium at the blast furnace top.

1.5.3 Opening of flow control valve for secondary equalizing pressure: The opening degree of the control valve for nitrogen secondary equalizing pressure flow.

1.5.4 Opening of pressure control valve for steam: The opening degree of the control valve for blast furnace top steam pressure.

1.5.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Steam pipeline - Operation monitoring”.

1.6 Furnace top system - Hydraulic station - Operation monitoring

1.6.1 Enable mode for main pump local control: The activation state of the local control mode for the hydraulic station main pump.

1.6.2 Enable mode for valve remote control: The activation state of the remote control mode for blast furnace hydraulic valves.

1.6.3 Ultimate lower limit value of oil level: The critical minimum oil level of the hydraulic station oil tank to prevent pump damage.

1.6.4 Lower limit value of oil level: The normal minimum alarm oil level of the hydraulic station oil tank.

1.6.5 Upper limit value of oil level: The normal maximum alarm oil level of the hydraulic station oil tank.

1.6.6 1# return oil filter clogged: Alarm state indicating the No.1 hydraulic return oil filter is blocked.

1.6.7 2# return oil filter clogged: Alarm state indicating the No.2 hydraulic return oil filter is blocked.

1.6.8 1# circulation oil filter clogged: Alarm state indicating the No.1 hydraulic circulation oil filter is blocked.

1.6.9 2# circulation oil filter clogged: Alarm state indicating the No.2 hydraulic circulation oil filter is blocked.

1.6.10 Ultimate upper limit value of oil temperature: The critical maximum oil temperature of the hydraulic station to prevent system failure.

1.6.11 Upper limit value of oil temperature: The normal maximum alarm oil temperature of the hydraulic station.

1.6.12 Lower limit value of oil temperature: The normal minimum alarm oil temperature of the hydraulic station.

1.6.13 Ultimate lower limit value of oil temperature: The critical minimum oil temperature of the hydraulic station to ensure fluidity.

1.6.14 Fault of hydraulic station: The overall fault state of the blast furnace hydraulic power unit.

1.6.15 Pump stop due to low level: Automatic pump stop caused by hydraulic oil level being too low.

1.6.16 Pump stop due to high temperature: Automatic pump stop caused by hydraulic oil temperature being too high.

1.6.17 Pump stop due to low temperature: Automatic pump stop caused by hydraulic oil temperature being too low.

1.6.18 Pump stop due to high pressure: Automatic pump stop caused by hydraulic system pressure being too high.

1.6.19 Pump stop due to low pressure: Automatic pump stop caused by hydraulic system pressure being too low.

1.6.20 Low pressure of furnace top accumulator: Alarm state of insufficient pressure in the blast furnace top hydraulic accumulator.

1.6.21 Low pressure of furnace top nitrogen: Alarm state of insufficient nitrogen pressure at the blast furnace top.

1.6.22 Low pressure of blow-off valve accumulator: Alarm state of insufficient pressure in the blow-off valve hydraulic accumulator.

1.6.23 Low nitrogen pressure A of blow-off valve: Alarm state of insufficient nitrogen pressure at point A of the blow-off valve.

1.6.24 Low nitrogen pressure B of blow-off valve: Alarm state of insufficient nitrogen pressure at point B of the blow-off valve.

1.6.25 Fault of 1# motor: Fault state of the No.1 hydraulic station driving motor.

1.6.26 Fault of 2# motor: Fault state of the No.2 hydraulic station driving motor.

1.6.27 Ultimate upper limit value of system pressure: The critical maximum pressure of the hydraulic system to prevent pipeline rupture.

1.6.28 Pressure of accumulator: The operating pressure of the hydraulic accumulator for blast furnace valves.

1.6.29 Pressure of nitrogen: The general operating pressure of nitrogen for hydraulic system sealing and auxiliary.

1.6.30 Pressure of blow-off valve accumulator: The operating pressure of the hydraulic accumulator dedicated to the blow-off valve.

1.6.31 Nitrogen pressure of 1# blow-off valve: The nitrogen pressure of the No.1 blast furnace blow-off valve.

1.6.32 Nitrogen pressure of 2# blow-off valve: The nitrogen pressure of the No.2 blast furnace blow-off valve.

1.6.33 System pressure of hydraulic station: The main operating pressure of the blast furnace hydraulic power unit.

1.6.34 Current of 1# motor: The operating current of the No.1 hydraulic station driving motor.

1.6.35 Current of 2# motor: The operating current of the No.2 hydraulic station driving motor.

1.6.36 Current of motor of circulation pump: The operating current of the hydraulic circulation pump motor.

1.6.37 Liquid level of hydraulic station oil tank: The height of hydraulic oil in the hydraulic station oil tank.

1.6.38 Temperature of hydraulic station oil tank: The temperature of hydraulic oil in the hydraulic station oil tank.

1.6.39 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Hydraulic station - Operation monitoring”.

1.7 Furnace top system - Pressure valve group - Pressure control

1.7.1 Fault of 1# oil pump: Fault state of the No.1 hydraulic oil pump.

1.7.2 Fault of 2# oil pump: Fault state of the No.2 hydraulic oil pump.

1.7.3 Fault of circulation pump: Fault state of the hydraulic circulation pump.

1.7.4 Upper Limit value of pressure of hydraulic station: The normal maximum alarm pressure of the hydraulic station system.

1.7.5 Lower Limit value of pressure of hydraulic station: The normal minimum alarm pressure of the hydraulic station system.

1.7.6 Upper Limit value of temperature of hydraulic station: The normal maximum alarm temperature of the hydraulic station.

1.7.7 Lower Limit value of temperature of hydraulic station: The normal minimum alarm temperature of the hydraulic station.

1.7.8 Lower Limit value of liquid level of hydraulic station: The normal minimum alarm oil level of the hydraulic station oil tank.

1.7.9 Ultimate lower limit value of liquid level of hydraulic station: The critical minimum oil level of the hydraulic station oil tank.

1.7.10 Alarm for differential pressure of filter: Alarm state caused by excessive pressure difference of hydraulic oil filter.

1.7.11 Alarm for power failure: Alarm state of power supply interruption for the hydraulic station.
1.7.12 Running of 1# oil pump: The operating state of the No.1 hydraulic oil pump.
1.7.13 Running of 2# oil pump: The operating state of the No.2 hydraulic oil pump.
1.7.14 Running of circulation pump: The operating state of the hydraulic circulation pump.
1.7.15 Running of heater: The operating state of the hydraulic oil tank heater.
1.7.16 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Furnace top system - Pressure valve group - Pressure control".

1.8 Furnace top system - Pressure regulator - Operation monitoring

1.8.1 System pressure: The general operating pressure of the blast furnace auxiliary hydraulic system.
1.8.2 System liquid level: The general liquid level of the blast furnace auxiliary hydraulic system oil tank.
1.8.3 System temperature: The general temperature of the blast furnace auxiliary hydraulic system oil.
1.8.4 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Furnace top system - Pressure regulator - Operation monitoring".

1.9 Furnace top system - Pressure regulator - System configuration

1.9.1 Lower limit setpoint value of pressure: The preset minimum pressure threshold for hydraulic system control.
1.9.2 Upper limit setpoint value of pressure: The preset maximum pressure threshold for hydraulic system control.
1.9.3 Lower limit setpoint value of temperature: The preset minimum temperature threshold for hydraulic system control.
1.9.4 Upper limit setpoint value of temperature: The preset maximum temperature threshold for hydraulic system control.
1.9.5 Upper limit setpoint value of liquid level: The preset maximum liquid level threshold for hydraulic system control.
1.9.6 Lower limit setpoint value of liquid level: The preset minimum liquid level threshold for hydraulic system control.
1.9.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Furnace top system - Pressure regulator - System configuration".

1.10 Furnace top system - Ascension pipe - Operation monitoring

1.10.1 Temperature of riser conduit (Southeast): The temperature of the blast furnace southeast riser conduit for gas discharge.
1.10.2 Temperature of riser conduit (Southwest): The temperature of the blast furnace southwest riser conduit for gas discharge.
1.10.3 Temperature of riser conduit (Northwest): The temperature of the blast furnace northwest riser conduit for gas discharge.
1.10.4 Temperature of riser conduit (Northeast): The temperature of the blast furnace northeast riser conduit for gas discharge.
1.10.5 Flow rate of furnace top water injection: The flow rate of cooling water injected into the blast furnace

top.

1.10.6 Mean temperature of furnace top: The average temperature of gas and equipment at the blast furnace top.

1.10.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Ascension pipe - Operation monitoring”.

1.11 Furnace top system - Airtight box - Operation monitoring

1.11.1 Fault of 1# pump: Fault state of the No.1 blast furnace cooling water pump.

1.11.2 Fault of 2# pump: Fault state of the No.2 blast furnace cooling water pump.

1.11.3 Fault of make-up water pump: Fault state of the blast furnace soft water make-up pump.

1.11.4 Emergency stop of ESD: Emergency shutdown state of the blast furnace emergency shutdown system.

1.11.5 Switch closing of normal power: The closing state of the main power supply switch for blast furnace cooling system.

1.11.6 Switch closing of standby power: The closing state of the standby power supply switch for blast furnace cooling system.

1.11.7 Auto mode of 1# make-up water shut-off valve: The automatic control mode state of the No.1 make-up water shut-off valve.

1.11.8 Auto mode of 2# make-up water shut-off valve: The automatic control mode state of the No.2 make-up water shut-off valve.

1.11.9 Auto mode of 3# drain ditch shut-off valve: The automatic control mode state of the No.3 drain ditch shut-off valve.

1.11.10 Manual mode of 1# make-up water shut-off valve: The manual control mode state of the No.1 make-up water shut-off valve.

1.11.11 Manual mode of 2# make-up water shut-off valve: The manual control mode state of the No.2 make-up water shut-off valve.

1.11.12 Manual mode of 3# drain ditch shut-off valve: The manual control mode state of the No.3 drain ditch shut-off valve.

1.11.13 Alarm for 1# low liquid level: Alarm state of low liquid level in the No.1 cooling water tank.

1.11.14 Alarm for 2# low liquid level: Alarm state of low liquid level in the No.2 cooling water tank.

1.11.15 Switch of flow: The on-off state of the cooling water flow control switch.

1.11.16 Temperature of nitrogen to gas-tight box: The temperature of nitrogen supplied to the gas-tight box.

1.11.17 Pressure of nitrogen to gas-tight box: The pressure of nitrogen supplied to the gas-tight box.

1.11.18 Flow rate of nitrogen to gas-tight box: The flow rate of nitrogen supplied to the gas-tight box.

1.11.19 Temperature of water about gearbox outlet: The outlet water temperature of the gearbox cooling system for blast furnace equipment.

1.11.20 Temperature of heater inlet: The inlet water temperature of the blast furnace cooling water heater.

1.11.21 Temperature of valve box: The ambient temperature of the valve box for blast furnace charging and sealing valves.

1.11.22 Temperature of 1# gas-tight box: The temperature of the No.1 blast furnace gas-tight box.

1.11.23 Temperature of 2# gas-tight box: The temperature of the No.2 blast furnace gas-tight box.

1.11.24 Temperature of 3# gas-tight box: The temperature of the No.3 blast furnace gas-tight box.

1.11.25 Pressure of gas-tight box: The internal pressure of the blast furnace gas-tight box.

1.11.26 Flow rate of return water about soft water-cooling plate: The return water flow rate of the soft water cooling plate for blast furnace.

1.11.27 Flow rate of soft water to gas-tight box: The supply flow rate of soft water to the blast furnace gas-tight box.

1.11.28 Temperature of water about gearbox inlet: The inlet water temperature of the gearbox cooling system for blast furnace equipment.

1.11.29 pH value of circulation water: The pH value of the blast furnace closed circulation cooling water.

1.11.30 Flow rate of soft water make-up: The flow rate of soft water supplemented to the blast furnace cooling system.

1.11.31 Flow rate of return industrial cooling water: The return flow rate of industrial cooling water for blast furnace.

1.11.32 Daily total flow rate of soft water cooling: The total flow rate of soft water cooling per day.

1.11.33 Monthly total flow rate of soft water cooling: The total flow rate of soft water cooling per month.

1.11.34 Daily total auto water injection count by make-up shut-off valve: The total number of automatic water injections by the make-up valve per day.

1.11.35 Daily total manual water injection count by make-up shut-off valve: The total number of manual water injections by the make-up valve per day.

1.11.36 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Airtight box - Operation monitoring”.

1.12 Furnace top system - Throat cross temperature measuring device - Operation monitoring

1.12.1 Center temperature-A0: The central temperature A0 of the blast furnace shaft cross temperature measurement.

1.12.2 1# Southeast temperature: The No.1 temperature point in the southeast direction of blast furnace shaft.

1.12.3 2# Southeast temperature: The No.2 temperature point in the southeast direction of blast furnace shaft.

1.12.4 3# Southeast temperature: The No.3 temperature point in the southeast direction of blast furnace shaft.

1.12.5 4# Southeast temperature: The No.4 temperature point in the southeast direction of blast furnace shaft.

1.12.6 5# Southeast temperature: The No.5 temperature point in the southeast direction of blast furnace shaft.

1.12.7 6# Southeast temperature: The No.6 temperature point in the southeast direction of blast furnace shaft.

1.12.8 7# Southeast temperature: The No.7 temperature point in the southeast direction of blast furnace shaft.

1.12.9 1# Southwest temperature: The No.1 temperature point in the southwest direction of blast furnace shaft.

1.12.10 2# Southwest temperature: The No.2 temperature point in the southwest direction of blast furnace shaft.

1.12.11 3# Southwest temperature: The No.3 temperature point in the southwest direction of blast furnace shaft.

1.12.12 4# Southwest temperature: The No.4 temperature point in the southwest direction of blast furnace shaft.

1.12.13 5# Southwest temperature: The No.5 temperature point in the southwest direction of blast furnace shaft.

1.12.14 6# Southwest temperature: The No.6 temperature point in the southwest direction of blast furnace shaft.

1.12.15 7# Southwest temperature: The No.7 temperature point in the southwest direction of blast furnace shaft.

1.12.16 1# Northwest temperature: The No.1 temperature point in the northwest direction of blast furnace shaft.

1.12.17 2# Northwest temperature: The No.2 temperature point in the northwest direction of blast furnace shaft.

1.12.18 3# Northwest temperature: The No.3 temperature point in the northwest direction of blast furnace shaft.

1.12.19 4# Northwest temperature: The No.4 temperature point in the northwest direction of blast furnace shaft.

1.12.20 5# Northwest temperature: The No.5 temperature point in the northwest direction of blast furnace shaft.

1.12.21 6# Northwest temperature: The No.6 temperature point in the northwest direction of blast furnace shaft.

1.12.22 7# Northwest temperature: The No.7 temperature point in the northwest direction of blast furnace shaft.

1.12.23 1# Northeast temperature: The No.1 temperature point in the northeast direction of blast furnace shaft.

1.12.24 2# Northeast temperature: The No.2 temperature point in the northeast direction of blast furnace shaft.

1.12.25 3# Northeast temperature: The No.3 temperature point in the northeast direction of blast furnace shaft.

1.12.26 4# Northeast temperature: The No.4 temperature point in the northeast direction of blast furnace shaft.

1.12.27 5# Northeast temperature: The No.5 temperature point in the northeast direction of blast furnace shaft.

1.12.28 6# Northeast temperature: The No.6 temperature point in the northeast direction of blast furnace shaft.

1.12.29 7# Northeast temperature: The No.7 temperature point in the northeast direction of blast furnace shaft.

1.12.30 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Throat cross temperature measuring device - Operation monitoring”.

1.13 Furnace top system - Furnace top pressure monitor - Operation monitoring

1.13.1 Southeast pressure: The pressure in the southeast direction of blast furnace shaft or gas area.

1.13.2 Southwest pressure: The pressure in the southwest direction of blast furnace shaft or gas area.

1.13.3 Northwest pressure: The pressure in the northwest direction of blast furnace shaft or gas area.

1.13.4 Northeast pressure: The pressure in the northeast direction of blast furnace shaft or gas area.

1.13.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Furnace top pressure monitor - Operation monitoring”.

1.14 Furnace top system - Furnace top pressure valve group - Pressure control

1.14.1 Opening of remote-control valve: The opening degree of the blast furnace remote control regulating valve.

1.14.2 Opening of range valve: The opening degree of the blast furnace range regulating valve.

1.14.3 Opening of automatic valve: The opening degree of the blast furnace automatic control valve.

1.14.4 Opening of quick-opening valve: The opening degree of the blast furnace quick-opening valve.

1.14.5 Opening of stator vane: The opening degree of the stator vane for blast furnace gas or air flow control.

1.14.6 Mode of remote-control valve: The control mode of the blast furnace remote control valve.

1.14.7 Mode of range valve: The control mode of the blast furnace range valve.

1.14.8 Mode of automatic valve: The control mode of the blast furnace automatic valve.

1.14.9 Mode of quick-opening valve: The control mode of the blast furnace quick-opening valve.

1.14.10 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Furnace top pressure valve group - Pressure control”.

1.15 Furnace top system - Backblow valve group - Mode control

1.15.1 Mode of timed back-blowing: The control mode of timed back-blowing for blast furnace dust removal equipment.

1.15.2 Mode of differential pressure back-blowing: The control mode of differential pressure back-blowing for blast furnace dust removal equipment.

1.15.3 Mode of manual back-blowing: The control mode of manual back-blowing for blast furnace dust removal equipment.

1.15.4 Mode emergency manual: The emergency manual operation mode of blast furnace dust removal system.

1.15.5 Mode of system idle: The idle operation mode of blast furnace dust removal system.

1.15.6 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Backblow valve group - Mode control”.

1.16 Furnace top system - Main relief valve group - Operation monitoring

1.16.1 Start of 1# blow-off pipe: The starting/operating state of the No.1 blast furnace blow-off pipe.

1.16.2 Start of 2# blow-off pipe: The starting/operating state of the No.2 blast furnace blow-off pipe.

1.16.3 Start of 3# blow-off pipe: The starting/operating state of the No.3 blast furnace blow-off pipe.

1.16.4 Dedusting valve of furnace top: The operating state of the blast furnace top dust removal valve.

1.16.5 Start of 250 blow-off pipe: The starting/operating state of the 250-type blast furnace blow-off pipe.

1.16.6 Start of 400 blow-off pipe: The starting/operating state of the 400-type blast furnace blow-off pipe.

1.16.7 Shut-off valve: The on-off state of the blast furnace general shut-off valve.

1.16.8 Running of vibrator: The operating state of the blast furnace burden or dust removal vibrator.

1.16.9 Normal current status of pneumatic valve: The normal operating current state of blast furnace pneumatic valves.

1.16.10 Normal status of 1# star valve: The normal operating state of the No.1 blast furnace star unloader valve.

1.16.11 Normal status of 2# star valve: The normal operating state of the No.2 blast furnace star unloader valve.

1.16.12 Normal status of 1# pneumatic dome valve: The normal operating state of the No.1 pneumatic dome valve.

1.16.13 Normal status of 2# pneumatic dome valve: The normal operating state of the No.2 pneumatic dome valve.

1.16.14 Normal status of 1# pneumatic damper valve: The normal operating state of the No.1 pneumatic damper valve.

1.16.15 Normal status of 2# pneumatic damper valve: The normal operating state of the No.2 pneumatic damper valve.
1.16.16 Pressure of furnace top: The static pressure at the blast furnace top.
1.16.17 Low temperature of lower cone ash level at north side: Alarm state of low temperature at the north lower cone ash level of dust removal equipment.
1.16.18 Low temperature of lower cone ash level at south side: Alarm state of low temperature at the south lower cone ash level of dust removal equipment.
1.16.19 High temperature of lower cone ash level at north side: Alarm state of high temperature at the north lower cone ash level of dust removal equipment.
1.16.20 High temperature of lower cone ash level at south side: Alarm state of high temperature at the south lower cone ash level of dust removal equipment.
1.16.21 Upper limit value of pressure when opening blow-off value: The maximum allowable pressure for opening the blast furnace blow-off valve.
1.16.22 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Furnace top system - Main relief valve group - Operation monitoring".

1.17 Furnace top system - Water injection pipeline - Operation monitoring

1.17.1 Flow rate of water injection: The real-time flow rate of blast furnace cooling water injection.
1.17.2 Flow of accumulated total single water injection: The total accumulated flow of a single water injection process.
1.17.3 Time of accumulated total single water injection: The total duration of a single water injection process.
1.17.4 Accumulated value of daily flow: The total accumulated flow of water injection per day.
1.17.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Furnace top system - Water injection pipeline - Operation monitoring".

1.18 Furnace top system - Water injection valve group - Operation monitoring

1.18.1 Summer: The summer operation mode parameter of blast furnace cooling system.
1.18.2 Winter: The winter operation mode parameter of blast furnace cooling system.
1.18.3 Start of 1# nitrogen valve: The starting/operating state of the No.1 nitrogen control valve.
1.18.4 Start of 2# nitrogen valve: The starting/operating state of the No.2 nitrogen control valve.
1.18.5 Start of 1# drain valve: The starting/operating state of the No.1 drain valve.
1.18.6 Start of 1# inner ring valve: The starting/operating state of the No.1 inner ring water injection valve.
1.18.7 Start of 2# inner ring valve: The starting/operating state of the No.2 inner ring water injection valve.
1.18.8 Start of main water injection valve: The starting/operating state of the main water injection valve.
1.18.9 Flow rate of main water injection pipe: The flow rate in the main water injection pipeline.
1.18.10 Temperature of main water injection pipe: The temperature in the main water injection pipeline.
1.18.11 Pressure of main water injection pipe: The pressure in the main water injection pipeline.
1.18.12 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Furnace top system - Water injection valve group - Operation monitoring".

1.19 Furnace top system - Water injection valve group - System configuration

1.19.1 Start of inner ring: The starting/operating state of the inner ring water injection system.

1.19.2 Start of outer ring: The starting/operating state of the outer ring water injection system.

1.19.3 Leak detection value of main water injection pipe: The leakage detection parameter of the main water injection pipeline.

1.19.4 Automatic draining time of drain water valve: The preset duration of automatic draining by the drain valve.

1.19.5 Single ring setpoint value of water injection for furnace top temperature: The preset water injection flow for single ring to control furnace top temperature.

1.19.6 Dual ring setpoint value of water injection for furnace top temperature: The preset water injection flow for dual ring to control furnace top temperature.

1.19.7 Setpoint value of Stopping Water Injection: The preset threshold for stopping water injection at furnace top.

1.19.8 Alarm setpoint value of water injection time: The preset duration threshold for water injection time alarm.

1.19.9 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Water injection valve group - System configuration”.

1.20 Blast furnace system - Pressure monitor - Furnace body pressure difference monitoring

1.20.1 Differential pressure of blast furnace upper section: The pressure difference between upper measuring points of the blast furnace shaft.

1.20.2 Differential pressure of blast furnace upper-middle section: The pressure difference between upper and middle measuring points of the blast furnace shaft.

1.20.3 Differential pressure of blast furnace middle section: The pressure difference between middle measuring points of the blast furnace shaft.

1.20.4 Differential pressure of blast furnace lower section: The pressure difference between lower measuring points of the blast furnace shaft.

1.20.5 Whole furnace differential pressure of blast furnace: The total pressure difference between furnace top and furnace hearth of blast furnace.

1.20.6 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Pressure monitor - Furnace body pressure difference monitoring”.

1.21 Blast furnace system - Pressure monitor - Furnace body static pressure monitoring

1.21.1 Pressure of furnace top: The static pressure at the blast furnace top.

1.21.2 1# Static pressure of section 11: The No.1 static pressure value at section 11 of blast furnace shaft.

1.21.3 2# Static pressure of section 11: The No.2 static pressure value at section 11 of blast furnace shaft.

1.21.4 3# Static pressure of section 11: The No.3 static pressure value at section 11 of blast furnace shaft.

1.21.5 4# Static pressure of section 11: The No.4 static pressure value at section 11 of blast furnace shaft.

1.21.6 Static pressure-Average of section 11: The average static pressure value at section 11 of blast furnace shaft.

1.21.7 1# Static pressure of section 13: The No.1 static pressure value at section 13 of blast furnace shaft.

1.21.8 2# Static pressure of section 13: The No.2 static pressure value at section 13 of blast furnace shaft.

1.21.9 3# Static pressure of section 13: The No.3 static pressure value at section 13 of blast furnace shaft.

1.21.10 4# Static pressure of section 13: The No.4 static pressure value at section 13 of blast furnace shaft.

1.21.11 Static pressure-Average of section 13: The average static pressure value at section 13 of blast furnace shaft.

1.21.12 1# Static pressure of section 15: The No.1 static pressure value at section 15 of blast furnace shaft.

1.21.13 2# Static pressure of section 15: The No.2 static pressure value at section 15 of blast furnace shaft.

1.21.14 3# Static pressure of section 15: The No.3 static pressure value at section 15 of blast furnace shaft.

1.21.15 4# Static pressure of section 15: The No.4 static pressure value at section 15 of blast furnace shaft.

1.21.16 Static pressure-Average of section 15: The average static pressure value at section 15 of blast furnace shaft.

1.21.17 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Pressure monitor - Furnace body static pressure monitoring”.

1.22 Blast furnace system - Production daily report - Theoretical material consumption calculation

1.22.1 Theoretical consumption of sinter: The calculated theoretical usage of sinter per ton of hot metal.

1.22.2 Theoretical consumption of basic pellet: The calculated theoretical usage of basic pellet per ton of hot metal.

1.22.3 Theoretical consumption of acidic pellet: The calculated theoretical usage of acidic pellet per ton of hot metal.

1.22.4 Theoretical consumption of lump ore: The calculated theoretical usage of lump ore per ton of hot metal.

1.22.5 Theoretical consumption of hot-pressed lump ore: The calculated theoretical usage of hot-pressed lump ore per ton of hot metal.

1.22.6 Ore consumption per ton iron: The total ore usage per ton of hot metal produced.

1.22.7 Theoretical consumption of coke: The calculated theoretical usage of metallurgical coke per ton of hot metal.

1.22.8 Theoretical consumption of nut coke: The calculated theoretical usage of nut coke per ton of hot metal.

1.22.9 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Production daily report - Theoretical material consumption calculation”.

1.23 Blast furnace system - Production daily report - Operational material consumption monitoring

1.23.1 Actual consumption of sinter: The real-time usage of sinter per ton of hot metal.

1.23.2 Actual consumption of basic pellet: The real-time usage of basic pellet per ton of hot metal.

1.23.3 Actual consumption of acidic pellet: The real-time usage of acidic pellet per ton of hot metal.

1.23.4 Actual consumption of lump ore: The real-time usage of lump ore per ton of hot metal.

1.23.5 Actual consumption of hot-pressed lump ore: The real-time usage of hot-pressed lump ore per ton of hot metal.

1.23.6 Actual ore consumption per ton iron: The real-time total ore usage per ton of hot metal.

1.23.7 Actual consumption of coke: The real-time usage of metallurgical coke per ton of hot metal.

1.23.8 Actual consumption of nut coke: The real-time usage of nut coke per ton of hot metal.

1.23.9 Internal control rate: The internal control index of blast furnace smelting operation.

1.23.10 Smelting intensity: The smelting rate per cubic meter of blast furnace effective volume per day.

1.23.11 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Production daily report - Operational material consumption monitoring”.

1.24 Blast furnace system - Production daily report - Burden structure operational monitoring

1.24.1 Count of batch: The total number of burden batches charged into the blast furnace.
1.24.2 Batch weight of coke: The mass of coke in a single charging batch.
1.24.3 Batch weight of ore: The mass of ore in a single charging batch.
1.24.4 Proportion of sinter: The mass percentage of sinter in the ore burden.
1.24.5 Proportion of acidic pellet: The mass percentage of acidic pellet in the ore burden.
1.24.6 Proportion of PB lump ore: The mass percentage of PB lump ore in the ore burden.
1.24.7 Proportion of hot-pressed lump ore: The mass percentage of hot-pressed lump ore in the ore burden.
1.24.8 Proportion of stockpile: The proportioning ratio of different stockpiles in blast furnace burden.
1.24.9 Smelting cycle: The time required for one complete blast furnace smelting process.
1.24.10 Theoretical content of CaO: The calculated theoretical calcium oxide content in blast furnace slag.
1.24.11 Theoretical content of Si: The calculated theoretical silicon content in blast furnace hot metal.
1.24.12 Theoretical value of R2/Basicity: The calculated theoretical binary basicity of blast furnace slag.
1.24.13 Theoretical value of S load: The calculated theoretical sulfur load in blast furnace burden.
1.24.14 Theoretical value of coke ratio: The calculated theoretical coke consumption per ton of hot metal.
1.24.15 Theoretical value of lump coke ratio: The calculated theoretical lump coke ratio in blast furnace fuel.
1.24.16 Theoretical content of nut coke: The calculated theoretical nut coke content in blast furnace fuel.
1.24.17 Theoretical value of small coke ratio: The calculated theoretical small coke ratio in blast furnace fuel.
1.24.18 Theoretical value of coke load: The calculated theoretical ore-to-coke ratio of blast furnace burden.
1.24.19 Theoretical value of yield of hot metal per batch: The calculated theoretical hot metal output per burden batch.
1.24.20 Theoretical value of slag-to-iron ratio: The calculated theoretical slag mass per ton of hot metal.
1.24.21 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Production daily report - Burden structure operational monitoring”.

1.25 Blast furnace system - Production daily report - Key parameter operational monitoring

1.25.1 Hourly count of batch: The number of burden batches charged into the blast furnace per hour.
1.25.2 Daily average value of cold air flow: The daily average flow rate of cold air supplied to blast furnace.
1.25.3 Daily average value of hot blast pressure: The daily average pressure of hot blast supplied to blast furnace.
1.25.4 Daily average value of whole furnace differential pressure: The daily average total pressure difference of blast furnace.
1.25.5 Daily average value of permeability index: The daily average permeability index of blast furnace burden column.
1.25.6 Daily average value of oxygen-enriched volume: The daily average flow rate of oxygen-enriched for blast furnace.
1.25.7 Daily average value of hot blast temperature: The daily average temperature of hot blast supplied to blast furnace.

1.25.8 Daily average value of bag temperature: The daily average temperature of blast furnace gas bag filter.
1.25.9 Daily average value of temperature of furnace top: The daily average temperature at blast furnace top.
1.25.10 Daily average value of whole furnace temperature difference: The daily average temperature difference of blast furnace shaft.
1.25.11 Daily average value of Hourly volume of pulverized coal injection: The daily average hourly pulverized coal injection rate.
1.25.12 Daily average value of coal ratio: The daily average pulverized coal consumption per ton of hot metal.
1.25.13 Daily average value of fuel ratio: The daily average total fuel consumption per ton of hot metal.
1.25.14 Daily average value of H2 content: The daily average hydrogen content in blast furnace top gas.
1.25.15 Daily average value of gas utilization rate: The daily average gas utilization rate of blast furnace.
1.25.16 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Production daily report - Key parameter operational monitoring”.

1.26 Blast furnace system - Production daily report - Tuyere equipment operational monitoring

1.26.1 1# Tuyere diameter: The inner diameter of the No.1 blast furnace tuyere.
1.26.2 2# Tuyere diameter: The inner diameter of the No.2 blast furnace tuyere.
1.26.3 3# Tuyere diameter: The inner diameter of the No.3 blast furnace tuyere.
1.26.4 4# Tuyere diameter: The inner diameter of the No.4 blast furnace tuyere.
1.26.5 5# Tuyere diameter: The inner diameter of the No.5 blast furnace tuyere.
1.26.6 6# Tuyere diameter: The inner diameter of the No.6 blast furnace tuyere.
1.26.7 7# Tuyere diameter: The inner diameter of the No.7 blast furnace tuyere.
1.26.8 8# Tuyere diameter: The inner diameter of the No.8 blast furnace tuyere.
1.26.9 9# Tuyere diameter: The inner diameter of the No.9 blast furnace tuyere.
1.26.10 10# Tuyere diameter: The inner diameter of the No.10 blast furnace tuyere.
1.26.11 11# Tuyere diameter: The inner diameter of the No.11 blast furnace tuyere.
1.26.12 12# Tuyere diameter: The inner diameter of the No.12 blast furnace tuyere.
1.26.13 13# Tuyere diameter: The inner diameter of the No.13 blast furnace tuyere.
1.26.14 14# Tuyere diameter: The inner diameter of the No.14 blast furnace tuyere.
1.26.15 15# Tuyere diameter: The inner diameter of the No.15 blast furnace tuyere.
1.26.16 16# Tuyere diameter: The inner diameter of the No.16 blast furnace tuyere.
1.26.17 17# Tuyere diameter: The inner diameter of the No.17 blast furnace tuyere.
1.26.18 18# Tuyere diameter: The inner diameter of the No.18 blast furnace tuyere.
1.26.19 19# Tuyere diameter: The inner diameter of the No.19 blast furnace tuyere.
1.26.20 20# Tuyere diameter: The inner diameter of the No.20 blast furnace tuyere.
1.26.21 21# Tuyere diameter: The inner diameter of the No.21 blast furnace tuyere.
1.26.22 22# Tuyere diameter: The inner diameter of the No.22 blast furnace tuyere.
1.26.23 23# Tuyere diameter: The inner diameter of the No.23 blast furnace tuyere.
1.26.24 24# Tuyere diameter: The inner diameter of the No.24 blast furnace tuyere.
1.26.25 25# Tuyere diameter: The inner diameter of the No.25 blast furnace tuyere.
1.26.26 26# Tuyere diameter: The inner diameter of the No.26 blast furnace tuyere.
1.26.27 27# Tuyere diameter: The inner diameter of the No.27 blast furnace tuyere.
1.26.28 28# Tuyere diameter: The inner diameter of the No.28 blast furnace tuyere.

1.26.29 29# Tuyere diameter: The inner diameter of the No.29 blast furnace tuyere.
1.26.30 30# Tuyere diameter: The inner diameter of the No.30 blast furnace tuyere.
1.26.31 31# Tuyere diameter: The inner diameter of the No.31 blast furnace tuyere.
1.26.32 32# Tuyere diameter: The inner diameter of the No.32 blast furnace tuyere.
1.26.33 1# Tuyere length: The length of the No.1 blast furnace tuyere.
1.26.34 2# Tuyere length: The length of the No.2 blast furnace tuyere.
1.26.35 3# Tuyere length: The length of the No.3 blast furnace tuyere.
1.26.36 4# Tuyere length: The length of the No.4 blast furnace tuyere.
1.26.37 5# Tuyere length: The length of the No.5 blast furnace tuyere.
1.26.38 6# Tuyere length: The length of the No.6 blast furnace tuyere.
1.26.39 7# Tuyere length: The length of the No.7 blast furnace tuyere.
1.26.40 8# Tuyere length: The length of the No.8 blast furnace tuyere.
1.26.41 9# Tuyere length: The length of the No.9 blast furnace tuyere.
1.26.42 10# Tuyere length: The length of the No.10 blast furnace tuyere.
1.26.43 11# Tuyere length: The length of the No.11 blast furnace tuyere.
1.26.44 12# Tuyere length: The length of the No.12 blast furnace tuyere.
1.26.45 13# Tuyere length: The length of the No.13 blast furnace tuyere.
1.26.46 14# Tuyere length: The length of the No.14 blast furnace tuyere.
1.26.47 15# Tuyere length: The length of the No.15 blast furnace tuyere.
1.26.48 16# Tuyere length: The length of the No.16 blast furnace tuyere.
1.26.49 17# Tuyere length: The length of the No.17 blast furnace tuyere.
1.26.50 18# Tuyere length: The length of the No.18 blast furnace tuyere.
1.26.51 19# Tuyere length: The length of the No.19 blast furnace tuyere.
1.26.52 20# Tuyere length: The length of the No.20 blast furnace tuyere.
1.26.53 21# Tuyere length: The length of the No.21 blast furnace tuyere.
1.26.54 22# Tuyere length: The length of the No.22 blast furnace tuyere.
1.26.55 23# Tuyere length: The length of the No.23 blast furnace tuyere.
1.26.56 24# Tuyere length: The length of the No.24 blast furnace tuyere.
1.26.57 25# Tuyere length: The length of the No.25 blast furnace tuyere.
1.26.58 26# Tuyere length: The length of the No.26 blast furnace tuyere.
1.26.59 27# Tuyere length: The length of the No.27 blast furnace tuyere.
1.26.60 28# Tuyere length: The length of the No.28 blast furnace tuyere.
1.26.61 29# Tuyere length: The length of the No.29 blast furnace tuyere.
1.26.62 30# Tuyere length: The length of the No.30 blast furnace tuyere.
1.26.63 31# Tuyere length: The length of the No.31 blast furnace tuyere.
1.26.64 32# Tuyere length: The length of the No.32 blast furnace tuyere.
1.26.65 Service days of 1# tuyere: The continuous service days of the No.1 blast furnace tuyere.
1.26.66 Service days of 2# tuyere: The continuous service days of the No.2 blast furnace tuyere.
1.26.67 Service days of 3# tuyere: The continuous service days of the No.3 blast furnace tuyere.
1.26.68 Service days of 4# tuyere: The continuous service days of the No.4 blast furnace tuyere.
1.26.69 Service days of 5# tuyere: The continuous service days of the No.5 blast furnace tuyere.
1.26.70 Service days of 6# tuyere: The continuous service days of the No.6 blast furnace tuyere.
1.26.71 Service days of 7# tuyere: The continuous service days of the No.7 blast furnace tuyere.
1.26.72 Service days of 8# tuyere: The continuous service days of the No.8 blast furnace tuyere.

1.26.73 Service days of 9# tuyere: The continuous service days of the No.9 blast furnace tuyere.
1.26.74 Service days of 10# tuyere: The continuous service days of the No.10 blast furnace tuyere.
1.26.75 Service days of 11# tuyere: The continuous service days of the No.11 blast furnace tuyere.
1.26.76 Service days of 12# tuyere: The continuous service days of the No.12 blast furnace tuyere.
1.26.77 Service days of 13# tuyere: The continuous service days of the No.13 blast furnace tuyere.
1.26.78 Service days of 14# tuyere: The continuous service days of the No.14 blast furnace tuyere.
1.26.79 Service days of 15# tuyere: The continuous service days of the No.15 blast furnace tuyere.
1.26.80 Service days of 16# tuyere: The continuous service days of the No.16 blast furnace tuyere.
1.26.81 Service days of 17# tuyere: The continuous service days of the No.17 blast furnace tuyere.
1.26.82 Service days of 18# tuyere: The continuous service days of the No.18 blast furnace tuyere.
1.26.83 Service days of 19# tuyere: The continuous service days of the No.19 blast furnace tuyere.
1.26.84 Service days of 20# tuyere: The continuous service days of the No.20 blast furnace tuyere.
1.26.85 Service days of 21# tuyere: The continuous service days of the No.21 blast furnace tuyere.
1.26.86 Service days of 22# tuyere: The continuous service days of the No.22 blast furnace tuyere.
1.26.87 Service days of 23# tuyere: The continuous service days of the No.23 blast furnace tuyere.
1.26.88 Service days of 24# tuyere: The continuous service days of the No.24 blast furnace tuyere.
1.26.89 Service days of 25# tuyere: The continuous service days of the No.25 blast furnace tuyere.
1.26.90 Service days of 26# tuyere: The continuous service days of the No.26 blast furnace tuyere.
1.26.91 Service days of 27# tuyere: The continuous service days of the No.27 blast furnace tuyere.
1.26.92 Service days of 28# tuyere: The continuous service days of the No.28 blast furnace tuyere.
1.26.93 Service days of 29# tuyere: The continuous service days of the No.29 blast furnace tuyere.
1.26.94 Service days of 30# tuyere: The continuous service days of the No.30 blast furnace tuyere.
1.26.95 Service days of 31# tuyere: The continuous service days of the No.31 blast furnace tuyere.
1.26.96 Service days of 32# tuyere: The continuous service days of the No.32 blast furnace tuyere.
1.26.97 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Production daily report - Tuyere equipment operational monitoring”.

1.27 Blast furnace system - Production daily report - Output index operational monitoring

1.27.1 Theoretical yield of hot metal: The calculated theoretical output of hot metal per unit time.
1.27.2 Daily inventory: The daily inventory of blast furnace hot metal products.
1.27.3 Daily deficiency amount of yield of hot metal: The daily difference between actual and theoretical hot metal output.
1.27.4 Daily productivity coefficient of blast furnace: The daily hot metal output per cubic meter of blast furnace effective volume.
1.27.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Production daily report - Output index operational monitoring”.

1.28 Blast furnace system - Production daily report - Charging system configuration

1.28.1 Setpoint value of dumping angle - Ore: The preset angle for ore dumping in blast furnace charging.
1.28.2 Setpoint value of number of charging rounds - Ore: The preset number of ore charging rounds.
1.28.3 Setpoint value of charging proportion - Ore: The preset proportion of ore in blast furnace charging.
1.28.4 Setpoint value of dumping angle - Coke: The preset angle for coke dumping in blast furnace charging.

1.28.5 Setpoint value of number of charging rounds - Coke: The preset number of coke charging rounds.
1.28.6 Setpoint value of charging proportion - Coke: The preset proportion of coke in blast furnace charging.
1.28.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Production daily report - Charging system configuration”.

1.29 Blast furnace system - Thermocouple - Furnace body cooling stave temperature monitoring

1.29.1 Temperature of section 1 - 001(30°): The temperature at 30° of blast furnace section 1 measuring point 001.
1.29.2 Temperature of section 1 - 002(75°): The temperature at 75° of blast furnace section 1 measuring point 002.
1.29.3 Temperature of section 1 - 003(120°): The temperature at 120° of blast furnace section 1 measuring point 003.
1.29.4 Temperature of section 1 - 004(150°): The temperature at 150° of blast furnace section 1 measuring point 004.
1.29.5 Temperature of section 1 - 005(180°): The temperature at 180° of blast furnace section 1 measuring point 005.
1.29.6 Temperature of section 1 - 006(210°): The temperature at 210° of blast furnace section 1 measuring point 006.
1.29.7 Temperature of section 1 - 007(260°): The temperature at 260° of blast furnace section 1 measuring point 007.
1.29.8 Temperature of section 1 - 008(290°): The temperature at 290° of blast furnace section 1 measuring point 008.
1.29.9 Temperature of section 1 - 009(325°): The temperature at 325° of blast furnace section 1 measuring point 009.
1.29.10 Temperature of section 1 - 010(345°): The temperature at 345° of blast furnace section 1 measuring point 010.
1.29.11 Average temperature of section 1: The average temperature of blast furnace section 1.
1.29.12 Temperature of section 2 - 011(30°): The temperature at 30° of blast furnace section 2 measuring point 011.
1.29.13 Temperature of section 2 - 012(45°): The temperature at 45° of blast furnace section 2 measuring point 012.
1.29.14 Temperature of section 2 - 013(60°): The temperature at 60° of blast furnace section 2 measuring point 013.
1.29.15 Temperature of section 2 - 014(75°): The temperature at 75° of blast furnace section 2 measuring point 014.
1.29.16 Temperature of section 2 - 015(105°): The temperature at 105° of blast furnace section 2 measuring point 015.
1.29.17 Temperature of section 2 - 016(120°): The temperature at 120° of blast furnace section 2 measuring point 016.
1.29.18 Temperature of section 2 - 017(135°): The temperature at 135° of blast furnace section 2 measuring point 017.
1.29.19 Temperature of section 2 - 018(150°): The temperature at 150° of blast furnace section 2 measuring

point 018.
1.29.20 Temperature of section 2 - 019(180°): The temperature at 180° of blast furnace section 2 measuring point 019.
1.29.21 Temperature of section 2 - 020(210°): The temperature at 210° of blast furnace section 2 measuring point 020.
1.29.22 Temperature of section 2 - 021(225°): The temperature at 225° of blast furnace section 2 measuring point 021.
1.29.23 Temperature of section 2 - 022(245°): The temperature at 245° of blast furnace section 2 measuring point 022.
1.29.24 Temperature of section 2 - 023(260°): The temperature at 260° of blast furnace section 2 measuring point 023.
1.29.25 Temperature of section 2 - 024(275°): The temperature at 275° of blast furnace section 2 measuring point 024.
1.29.26 Temperature of section 2 - 025(290°): The temperature at 290° of blast furnace section 2 measuring point 025.
1.29.27 Temperature of section 2 - 026(305°): The temperature at 305° of blast furnace section 2 measuring point 026.
1.29.28 Temperature of section 2 - 027(325°): The temperature at 325° of blast furnace section 2 measuring point 027.
1.29.29 Temperature of section 2 - 028(345°): The temperature at 345° of blast furnace section 2 measuring point 028.
1.29.30 Average temperature of section 2: The average temperature of blast furnace section 2.
1.29.31 Temperature of section 3 - 029(30°): The temperature at 30° of blast furnace section 3 measuring point 029.
1.29.32 Temperature of section 3 - 030(75°): The temperature at 75° of blast furnace section 3 measuring point 030.
1.29.33 Temperature of section 3 - 031(120°): The temperature at 120° of blast furnace section 3 measuring point 031.
1.29.34 Temperature of section 3 - 032(150°): The temperature at 150° of blast furnace section 3 measuring point 032.
1.29.35 Temperature of section 3 - 033(180°): The temperature at 180° of blast furnace section 3 measuring point 033.
1.29.36 Temperature of section 3 - 034(210°): The temperature at 210° of blast furnace section 3 measuring point 034.
1.29.37 Temperature of section 3 - 035(245°): The temperature at 245° of blast furnace section 3 measuring point 035.
1.29.38 Temperature of section 3 - 036(290°): The temperature at 290° of blast furnace section 3 measuring point 036.
1.29.39 Temperature of section 3 - 037(325°): The temperature at 325° of blast furnace section 3 measuring point 037.
1.29.40 Temperature of section 3 - 038(345°): The temperature at 345° of blast furnace section 3 measuring point 038.
1.29.41 Average temperature of section 3: The average temperature of blast furnace section 3.
1.29.42 Temperature of section 6 - 039(30°): The temperature at 30° of blast furnace section 6 measuring

point 039.
1.29.43 Temperature of section 6 - 040(60°): The temperature at 60° of blast furnace section 6 measuring point 040.
1.29.44 Temperature of section 6 - 041(105°): The temperature at 105° of blast furnace section 6 measuring point 041.
1.29.45 Temperature of section 6 - 042(135°): The temperature at 135° of blast furnace section 6 measuring point 042.
1.29.46 Temperature of section 6 - 043(180°): The temperature at 180° of blast furnace section 6 measuring point 043.
1.29.47 Temperature of section 6 - 044(210°): The temperature at 210° of blast furnace section 6 measuring point 044.
1.29.48 Temperature of section 6 - 045(245°): The temperature at 245° of blast furnace section 6 measuring point 045.
1.29.49 Temperature of section 6 - 046(275°): The temperature at 275° of blast furnace section 6 measuring point 046.
1.29.50 Temperature of section 6 - 047(325°): The temperature at 325° of blast furnace section 6 measuring point 047.
1.29.51 Temperature of section 6 - 048(345°): The temperature at 345° of blast furnace section 6 measuring point 048.
1.29.52 Average temperature of section 6: The average temperature of blast furnace section 6.
1.29.53 Temperature of section 7 - 049(45°): The temperature at 45° of blast furnace section 7 measuring point 049.
1.29.54 Temperature of section 7 - 050(75°): The temperature at 75° of blast furnace section 7 measuring point 050.
1.29.55 Temperature of section 7 - 051(105°): The temperature at 105° of blast furnace section 7 measuring point 051.
1.29.56 Temperature of section 7 - 052(150°): The temperature at 150° of blast furnace section 7 measuring point 052.
1.29.57 Temperature of section 7 - 053(180°): The temperature at 180° of blast furnace section 7 measuring point 053.
1.29.58 Temperature of section 7 - 054(210°): The temperature at 210° of blast furnace section 7 measuring point 054.
1.29.59 Temperature of section 7 - 055(245°): The temperature at 245° of blast furnace section 7 measuring point 055.
1.29.60 Temperature of section 7 - 056(290°): The temperature at 290° of blast furnace section 7 measuring point 056.
1.29.61 Temperature of section 7 - 057(325°): The temperature at 325° of blast furnace section 7 measuring point 057.
1.29.62 Temperature of section 7 - 058(345°): The temperature at 345° of blast furnace section 7 measuring point 058.
1.29.63 Average temperature of section 7: The average temperature of blast furnace section 7.
1.29.64 Temperature of section 8 - 059(30°): The temperature at 30° of blast furnace section 8 measuring point 059.
1.29.65 Temperature of section 8 - 060(60°): The temperature at 60° of blast furnace section 8 measuring

point 060.
1.29.66 Temperature of section 8 - 061(105°): The temperature at 105° of blast furnace section 8 measuring point 061.
1.29.67 Temperature of section 8 - 062(135°): The temperature at 135° of blast furnace section 8 measuring point 062.
1.29.68 Temperature of section 8 - 063(180°): The temperature at 180° of blast furnace section 8 measuring point 063.
1.29.69 Temperature of section 8 - 064(225°): The temperature at 225° of blast furnace section 8 measuring point 064.
1.29.70 Temperature of section 8 - 065(245°): The temperature at 245° of blast furnace section 8 measuring point 065.
1.29.71 Temperature of section 8 - 066(275°): The temperature at 275° of blast furnace section 8 measuring point 066.
1.29.72 Temperature of section 8 - 067(325°): The temperature at 325° of blast furnace section 8 measuring point 067.
1.29.73 Temperature of section 8 - 068(345°): The temperature at 345° of blast furnace section 8 measuring point 068.
1.29.74 Average temperature of section 8: The average temperature of blast furnace section 8.
1.29.75 Temperature of section 9 - 069(45°): The temperature at 45° of blast furnace section 9 measuring point 069.
1.29.76 Temperature of section 9 - 070(75°): The temperature at 75° of blast furnace section 9 measuring point 070.
1.29.77 Temperature of section 9 - 071(105°): The temperature at 105° of blast furnace section 9 measuring point 071.
1.29.78 Temperature of section 9 - 072(150°): The temperature at 150° of blast furnace section 9 measuring point 072.
1.29.79 Temperature of section 9 - 073(180°): The temperature at 180° of blast furnace section 9 measuring point 073.
1.29.80 Temperature of section 9 - 074(210°): The temperature at 210° of blast furnace section 9 measuring point 074.
1.29.81 Temperature of section 9 - 075(245°): The temperature at 245° of blast furnace section 9 measuring point 075.
1.29.82 Temperature of section 9 - 076(290°): The temperature at 290° of blast furnace section 9 measuring point 076.
1.29.83 Temperature of section 9 - 077(325°): The temperature at 325° of blast furnace section 9 measuring point 077.
1.29.84 Temperature of section 9 - 078(345°): The temperature at 345° of blast furnace section 9 measuring point 078.
1.29.85 Average temperature of section 9: The average temperature of blast furnace section 9.
1.29.86 Temperature of section 10 - 079(30°): The temperature at 30° of blast furnace section 10 measuring point 079.
1.29.87 Temperature of section 10 - 080(60°): The temperature at 60° of blast furnace section 10 measuring point 080.
1.29.88 Temperature of section 10 - 081(105°): The temperature at 105° of blast furnace section 10

measuring point 081.
1.29.89 Temperature of section 10 - 082(135°): The temperature at 135° of blast furnace section 10 measuring point 082.
1.29.90 Temperature of section 10 - 083(180°): The temperature at 180° of blast furnace section 10 measuring point 083.
1.29.91 Temperature of section 10 - 084(210°): The temperature at 210° of blast furnace section 10 measuring point 084.
1.29.92 Temperature of section 10 - 085(245°): The temperature at 245° of blast furnace section 10 measuring point 085.
1.29.93 Temperature of section 10 - 086(275°): The temperature at 275° of blast furnace section 10 measuring point 086.
1.29.94 Temperature of section 10 - 087(325°): The temperature at 325° of blast furnace section 10 measuring point 087.
1.29.95 Temperature of section 10 - 088(345°): The temperature at 345° of blast furnace section 10 measuring point 088.
1.29.96 Average temperature of section 10: The average temperature of blast furnace section 10.
1.29.97 Temperature of section 11 - 089(45°): The temperature at 45° of blast furnace section 11 measuring point 089.
1.29.98 Temperature of section 11 - 090(75°): The temperature at 75° of blast furnace section 11 measuring point 090.
1.29.99 Temperature of section 11 - 091(105°): The temperature at 105° of blast furnace section 11 measuring point 091.
1.29.100 Temperature of section 11 - 092(150°): The temperature at 150° of blast furnace section 11 measuring point 092.
1.29.101 Temperature of section 11 - 093(180°): The temperature at 180° of blast furnace section 11 measuring point 093.
1.29.102 Temperature of section 11 - 094(210°): The temperature at 210° of blast furnace section 11 measuring point 094.
1.29.103 Temperature of section 11 - 095(245°): The temperature at 245° of blast furnace section 11 measuring point 095.
1.29.104 Temperature of section 11 - 096(290°): The temperature at 290° of blast furnace section 11 measuring point 096.
1.29.105 Temperature of section 11 - 097(325°): The temperature at 325° of blast furnace section 11 measuring point 097.
1.29.106 Temperature of section 11 - 098(345°): The temperature at 345° of blast furnace section 11 measuring point 098.
1.29.107 Average temperature of section 11: The average temperature of blast furnace section 11.
1.29.108 Temperature of section 12 - 099(45°): The temperature at 45° of blast furnace section 12 measuring point 099.
1.29.109 Temperature of section 12 - 100(75°): The temperature at 75° of blast furnace section 12 measuring point 100.
1.29.110 Temperature of section 12 - 101(105°): The temperature at 105° of blast furnace section 12 measuring point 101.
1.29.111 Temperature of section 12 - 102(135°): The temperature at 135° of blast furnace section 12

measuring point 102.
1.29.112 Temperature of section 12 - 103(180°): The temperature at 180° of blast furnace section 12 measuring point 103.
1.29.113 Temperature of section 12 - 104(210°): The temperature at 210° of blast furnace section 12 measuring point 104.
1.29.114 Temperature of section 12 - 105(245°): The temperature at 245° of blast furnace section 12 measuring point 105.
1.29.115 Temperature of section 12 - 106(275°): The temperature at 275° of blast furnace section 12 measuring point 106.
1.29.116 Temperature of section 12 - 107(325°): The temperature at 325° of blast furnace section 12 measuring point 107.
1.29.117 Temperature of section 12 - 108(345°): The temperature at 345° of blast furnace section 12 measuring point 108.
1.29.118 Average temperature of section 12: The average temperature of blast furnace section 12.
1.29.119 Temperature of section 13 - 109(45°): The temperature at 45° of blast furnace section 13 measuring point 109.
1.29.120 Temperature of section 13 - 110(75°): The temperature at 75° of blast furnace section 13 measuring point 110.
1.29.121 Temperature of section 13 - 111(105°): The temperature at 105° of blast furnace section 13 measuring point 111.
1.29.122 Temperature of section 13 - 112(150°): The temperature at 150° of blast furnace section 13 measuring point 112.
1.29.123 Temperature of section 13 - 113(180°): The temperature at 180° of blast furnace section 13 measuring point 113.
1.29.124 Temperature of section 13 - 114(210°): The temperature at 210° of blast furnace section 13 measuring point 114.
1.29.125 Temperature of section 13 - 115(245°): The temperature at 245° of blast furnace section 13 measuring point 115.
1.29.126 Temperature of section 13 - 116(290°): The temperature at 290° of blast furnace section 13 measuring point 116.
1.29.127 Temperature of section 13 - 117(325°): The temperature at 325° of blast furnace section 13 measuring point 117.
1.29.128 Temperature of section 13 - 118(345°): The temperature at 345° of blast furnace section 13 measuring point 118.
1.29.129 Average temperature of section 13: The average temperature of blast furnace section 13.
1.29.130 Temperature of section 14 - 119(45°): The temperature at 45° of blast furnace section 14 measuring point 119.
1.29.131 Temperature of section 14 - 120(75°): The temperature at 75° of blast furnace section 14 measuring point 120.
1.29.132 Temperature of section 14 - 121(105°): The temperature at 105° of blast furnace section 14 measuring point 121.
1.29.133 Temperature of section 14 - 122(135°): The temperature at 135° of blast furnace section 14 measuring point 122.
1.29.134 Temperature of section 14 - 123(180°): The temperature at 180° of blast furnace section 14

measuring point 123.
1.29.135 Temperature of section 14 - 124(210°): The temperature at 210° of blast furnace section 14 measuring point 124.
1.29.136 Temperature of section 14 - 125(245°): The temperature at 245° of blast furnace section 14 measuring point 125.
1.29.137 Temperature of section 14 - 126(275°): The temperature at 275° of blast furnace section 14 measuring point 126.
1.29.138 Temperature of section 14 - 127(325°): The temperature at 325° of blast furnace section 14 measuring point 127.
1.29.139 Temperature of section 14 - 128(345°): The temperature at 345° of blast furnace section 14 measuring point 128.
1.29.140 Average temperature of section 14: The average temperature of blast furnace section 14.
1.29.141 Temperature of section 15 - 129(45°): The temperature at 45° of blast furnace section 15 measuring point 129.
1.29.142 Temperature of section 15 - 130(75°): The temperature at 75° of blast furnace section 15 measuring point 130.
1.29.143 Temperature of section 15 - 131(105°): The temperature at 105° of blast furnace section 15 measuring point 131.
1.29.144 Temperature of section 15 - 132(150°): The temperature at 150° of blast furnace section 15 measuring point 132.
1.29.145 Temperature of section 15 - 133(180°): The temperature at 180° of blast furnace section 15 measuring point 133.
1.29.146 Temperature of section 15 - 134(210°): The temperature at 210° of blast furnace section 15 measuring point 134.
1.29.147 Temperature of section 15 - 135(245°): The temperature at 245° of blast furnace section 15 measuring point 135.
1.29.148 Temperature of section 15 - 136(290°): The temperature at 290° of blast furnace section 15 measuring point 136.
1.29.149 Temperature of section 15 - 137(325°): The temperature at 325° of blast furnace section 15 measuring point 137.
1.29.150 Temperature of section 15 - 138(345°): The temperature at 345° of blast furnace section 15 measuring point 138.
1.29.151 Average temperature of section 15: The average temperature of blast furnace section 15.
1.29.152 Temperature of section 17 - 139(45°): The temperature at 45° of blast furnace section 17 measuring point 139.
1.29.153 Temperature of section 17 - 140(90°): The temperature at 90° of blast furnace section 17 measuring point 140.
1.29.154 Temperature of section 17 - 141(135°): The temperature at 135° of blast furnace section 17 measuring point 141.
1.29.155 Temperature of section 17 - 142(180°): The temperature at 180° of blast furnace section 17 measuring point 142.
1.29.156 Temperature of section 17 - 143(225°): The temperature at 225° of blast furnace section 17 measuring point 143.
1.29.157 Temperature of section 17 - 144(275°): The temperature at 275° of blast furnace section 17

measuring point 144.

1.29.158 Temperature of section 17 - 145(325°): The temperature at 325° of blast furnace section 17 measuring point 145.

1.29.159 Temperature of section 17 - 146(345°): The temperature at 345° of blast furnace section 17 measuring point 146.

1.29.160 Average temperature of section 17: The average temperature of blast furnace section 17.

1.29.161 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Blast furnace system - Thermocouple - Furnace body cooling stave temperature monitoring".

1.30 Blast furnace system - Thermocouple - Furnace hearth side wall temperature monitoring

1.30.1 Temperature of furnace foundation - 001(345°): The temperature at 345° of blast furnace foundation measuring point 001.

1.30.2 Average temperature of furnace foundation: The average temperature of blast furnace foundation.

1.30.3 Temperature of furnace bottom carbon brick - 002(275°): The temperature at 275° of blast furnace bottom carbon brick measuring point 002.

1.30.4 Temperature of furnace bottom carbon brick - 003(290°): The temperature at 290° of blast furnace bottom carbon brick measuring point 003.

1.30.5 Temperature of furnace bottom carbon brick - 004(305°): The temperature at 305° of blast furnace bottom carbon brick measuring point 004.

1.30.6 Temperature of furnace bottom carbon brick - 005(325°): The temperature at 325° of blast furnace bottom carbon brick measuring point 005.

1.30.7 Temperature of furnace bottom carbon brick - 006(345°): The temperature at 345° of blast furnace bottom carbon brick measuring point 006.

1.30.8 Average temperature of furnace bottom carbon brick: The average temperature of blast furnace bottom carbon brick.

1.30.9 Temperature of furnace hearth shell of section 2 - 001(30°): The temperature at 30° of section 2 furnace hearth shell measuring point 001.

1.30.10 Temperature of furnace hearth shell of section 2 - 002(45°): The temperature at 45° of section 2 furnace hearth shell measuring point 002.

1.30.11 Temperature of furnace hearth shell of section 2 - 003(60°): The temperature at 60° of section 2 furnace hearth shell measuring point 003.

1.30.12 Temperature of furnace hearth shell of section 2 - 004(90°): The temperature at 90° of section 2 furnace hearth shell measuring point 004.

1.30.13 Temperature of furnace hearth shell of section 2 - 005(105°): The temperature at 105° of section 2 furnace hearth shell measuring point 005.

1.30.14 Temperature of furnace hearth shell of section 2 - 006(135°): The temperature at 135° of section 2 furnace hearth shell measuring point 006.

1.30.15 Temperature of furnace hearth shell of section 2 - 007(150°): The temperature at 150° of section 2 furnace hearth shell measuring point 007.

1.30.16 Temperature of furnace hearth shell of section 2 - 008(180°): The temperature at 180° of section 2 furnace hearth shell measuring point 008.

1.30.17 Temperature of furnace hearth shell of section 2 - 009(210°): The temperature at 210° of section 2 furnace hearth shell measuring point 009.

1.30.18 Temperature of furnace hearth shell of section 2 - 010(225°): The temperature at 225° of section 2 furnace hearth shell measuring point 010.
1.30.19 Temperature of furnace hearth shell of section 2 - 011(245°): The temperature at 245° of section 2 furnace hearth shell measuring point 011.
1.30.20 Temperature of furnace hearth shell of section 2 - 012(290°): The temperature at 290° of section 2 furnace hearth shell measuring point 012.
1.30.21 Temperature of furnace hearth shell of section 2 - 013(305°): The temperature at 305° of section 2 furnace hearth shell measuring point 013.
1.30.22 Temperature of furnace hearth shell of section 2 - 014(325°): The temperature at 325° of section 2 furnace hearth shell measuring point 014.
1.30.23 Average temperature of furnace hearth shell of section 2: The average temperature of section 2 furnace hearth shell.
1.30.24 Temperature of furnace hearth shell of section 3 - 015(30°): The temperature at 30° of section 3 furnace hearth shell measuring point 015.
1.30.25 Temperature of furnace hearth shell of section 3 - 016(75°): The temperature at 75° of section 3 furnace hearth shell measuring point 016.
1.30.26 Temperature of furnace hearth shell of section 3 - 017(105°): The temperature at 105° of section 3 furnace hearth shell measuring point 017.
1.30.27 Temperature of furnace hearth shell of section 3 - 018(150°): The temperature at 150° of section 3 furnace hearth shell measuring point 018.
1.30.28 Temperature of furnace hearth shell of section 3 - 019(210°): The temperature at 210° of section 3 furnace hearth shell measuring point 019.
1.30.29 Temperature of furnace hearth shell of section 3 - 020(245°): The temperature at 245° of section 3 furnace hearth shell measuring point 020.
1.30.30 Temperature of furnace hearth shell of section 3 - 021(290°): The temperature at 290° of section 3 furnace hearth shell measuring point 021.
1.30.31 Temperature of furnace hearth shell of section 3 - 022(345°): The temperature at 345° of section 3 furnace hearth shell measuring point 022.
1.30.32 Average temperature of furnace hearth shell of section 3: The average temperature of section 3 furnace hearth shell.
1.30.33 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Thermocouple - Furnace hearth side wall temperature monitoring”.

1.31 Blast furnace system - Thermocouple - Furnace lining side wall temperature monitoring

1.31.1 Temperature of furnace lining of section 6 - 001(90°): The temperature at 90° of section 6 furnace lining measuring point 001.
1.31.2 Temperature of furnace lining of section 6 - 002(180°): The temperature at 180° of section 6 furnace lining measuring point 002.
1.31.3 Temperature of furnace lining of section 6 - 003(260°): The temperature at 260° of section 6 furnace lining measuring point 003.
1.31.4 Temperature of furnace lining of section 6 - 004(345°): The temperature at 345° of section 6 furnace lining measuring point 004.
1.31.5 Average temperature of furnace lining of section 6: The average temperature of section 6 furnace

lining.

1.31.6 Temperature of furnace lining of section 7 - 005(90°): The temperature at 90° of section 7 furnace lining measuring point 005.

1.31.7 Temperature of furnace lining of section 7 - 006(180°): The temperature at 180° of section 7 furnace lining measuring point 006.

1.31.8 Temperature of furnace lining of section 7 - 007(275°): The temperature at 275° of section 7 furnace lining measuring point 007.

1.31.9 Temperature of furnace lining of section 7 - 008(345°): The temperature at 345° of section 7 furnace lining measuring point 008.

1.31.10 Average temperature of furnace lining of section 7: The average temperature of section 7 furnace lining.

1.31.11 Temperature of furnace lining of section 8 - 009(90°): The temperature at 90° of section 8 furnace lining measuring point 009.

1.31.12 Temperature of furnace lining of section 8 - 010(180°): The temperature at 180° of section 8 furnace lining measuring point 010.

1.31.13 Temperature of furnace lining of section 8 - 011(260°): The temperature at 260° of section 8 furnace lining measuring point 011.

1.31.14 Temperature of furnace lining of section 8 - 012(345°): The temperature at 345° of section 8 furnace lining measuring point 012.

1.31.15 Average temperature of furnace lining of section 8: The average temperature of section 8 furnace lining.

1.31.16 Temperature of furnace lining of section 9 - 013(90°): The temperature at 90° of section 9 furnace lining measuring point 013.

1.31.17 Temperature of furnace lining of section 9 - 014(180°): The temperature at 180° of section 9 furnace lining measuring point 014.

1.31.18 Temperature of furnace lining of section 9 - 015(275°): The temperature at 275° of section 9 furnace lining measuring point 015.

1.31.19 Temperature of furnace lining of section 9 - 016(345°): The temperature at 345° of section 9 furnace lining measuring point 016.

1.31.20 Average temperature of furnace lining of section 9: The average temperature of section 9 furnace lining.

1.31.21 Temperature of furnace lining of section 10 - 017(60°): The temperature at 60° of section 10 furnace lining measuring point 017.

1.31.22 Temperature of furnace lining of section 10 - 018(120°): The temperature at 120° of section 10 furnace lining measuring point 018.

1.31.23 Temperature of furnace lining of section 10 - 019(180°): The temperature at 180° of section 10 furnace lining measuring point 019.

1.31.24 Temperature of furnace lining of section 10 - 020(245°): The temperature at 245° of section 10 furnace lining measuring point 020.

1.31.25 Temperature of furnace lining of section 10 - 021(290°): The temperature at 290° of section 10 furnace lining measuring point 021.

1.31.26 Temperature of furnace lining of section 10 - 022(345°): The temperature at 345° of section 10 furnace lining measuring point 022.

1.31.27 Average temperature of furnace lining of section 10: The average temperature of section 10 furnace lining.

1.31.28 Temperature of furnace lining of section 11 - 023(60°): The temperature at 60° of section 11 furnace lining measuring point 023.

1.31.29 Temperature of furnace lining of section 11 - 024(120°): The temperature at 120° of section 11 furnace lining measuring point 024.

1.31.30 Temperature of furnace lining of section 11 - 025(180°): The temperature at 180° of section 11 furnace lining measuring point 025.

1.31.31 Temperature of furnace lining of section 11 - 026(245°): The temperature at 245° of section 11 furnace lining measuring point 026.

1.31.32 Temperature of furnace lining of section 11 - 027(305°): The temperature at 305° of section 11 furnace lining measuring point 027.

1.31.33 Temperature of furnace lining of section 11 - 028(345°): The temperature at 345° of section 11 furnace lining measuring point 028.

1.31.34 Average temperature of furnace lining of section 11: The average temperature of section 11 furnace lining.

1.31.35 Temperature of furnace lining of section 12 - 029(60°): The temperature at 60° of section 12 furnace lining measuring point 029.

1.31.36 Temperature of furnace lining of section 12 - 030(120°): The temperature at 120° of section 12 furnace lining measuring point 030.

1.31.37 Temperature of furnace lining of section 12 - 031(180°): The temperature at 180° of section 12 furnace lining measuring point 031.

1.31.38 Temperature of furnace lining of section 12 - 032(245°): The temperature at 245° of section 12 furnace lining measuring point 032.

1.31.39 Temperature of furnace lining of section 12 - 033(290°): The temperature at 290° of section 12 furnace lining measuring point 033.

1.31.40 Temperature of furnace lining of section 12 - 034(345°): The temperature at 345° of section 12 furnace lining measuring point 034.

1.31.41 Average temperature of furnace lining of section 12: The average temperature of section 12 furnace lining.

1.31.42 Temperature of furnace lining of section 13 - 035(60°): The temperature at 60° of section 13 furnace lining measuring point 035.

1.31.43 Temperature of furnace lining of section 13 - 036(120°): The temperature at 120° of section 13 furnace lining measuring point 036.

1.31.44 Temperature of furnace lining of section 13 - 037(180°): The temperature at 180° of section 13 furnace lining measuring point 037.

1.31.45 Temperature of furnace lining of section 13 - 038(245°): The temperature at 245° of section 13 furnace lining measuring point 038.

1.31.46 Temperature of furnace lining of section 13 - 039(305°): The temperature at 305° of section 13 furnace lining measuring point 039.

1.31.47 Temperature of furnace lining of section 13 - 040(345°): The temperature at 345° of section 13 furnace lining measuring point 040.

1.31.48 Average temperature of furnace lining of section 13: The average temperature of section 13 furnace lining.

1.31.49 Temperature of furnace lining of section 14 - 041(60°): The temperature at 60° of section 14 furnace

lining measuring point 041.
1.31.50 Temperature of furnace lining of section 14 - 042(120°): The temperature at 120° of section 14 furnace lining measuring point 042.
1.31.51 Temperature of furnace lining of section 14 - 043(180°): The temperature at 180° of section 14 furnace lining measuring point 043.
1.31.52 Temperature of furnace lining of section 14 - 044(245°): The temperature at 245° of section 14 furnace lining measuring point 044.
1.31.53 Temperature of furnace lining of section 14 - 045(290°): The temperature at 290° of section 14 furnace lining measuring point 045.
1.31.54 Temperature of furnace lining of section 14 - 046(345°): The temperature at 345° of section 14 furnace lining measuring point 046.
1.31.55 Average temperature of furnace lining of section 14: The average temperature of section 14 furnace lining.
1.31.56 Temperature of furnace lining of section 15 - 047(60°): The temperature at 60° of section 15 furnace lining measuring point 047.
1.31.57 Temperature of furnace lining of section 15 - 048(120°): The temperature at 120° of section 15 furnace lining measuring point 048.
1.31.58 Temperature of furnace lining of section 15 - 049(180°): The temperature at 180° of section 15 furnace lining measuring point 049.
1.31.59 Temperature of furnace lining of section 15 - 050(245°): The temperature at 245° of section 15 furnace lining measuring point 050.
1.31.60 Temperature of furnace lining of section 15 - 051(305°): The temperature at 305° of section 15 furnace lining measuring point 051.
1.31.61 Temperature of furnace lining of section 15 - 052(345°): The temperature at 345° of section 15 furnace lining measuring point 052.
1.31.62 Average temperature of furnace lining of section 15: The average temperature of section 15 furnace lining.
1.31.63 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Blast furnace system - Thermocouple - Furnace lining side wall temperature monitoring".

1.32 Blast furnace system - Integrated terminal - Operation monitoring

1.32.1 Start of furnace top shut-off valve: The actuation state of the blast furnace top shut-off valve.
1.32.2 Start of main belt-remote control valve: The actuation state of the main belt remote control valve.
1.32.3 Start of mixed air control valve: The actuation state of the mixed air control valve.
1.32.4 Start of blow-off valve about cold air: The actuation state of the cold air blow-off valve.
1.32.5 Hot blast temperature: The real-time temperature of hot blast supplied to the blast furnace.
1.32.6 Hot blast pressure: The real-time pressure of hot blast supplied to the blast furnace.
1.32.7 Flow rate of cold air: The real-time flow rate of cold air supplied to the blast furnace.
1.32.8 Pressure of cold air: The real-time pressure of cold air supplied to the blast furnace.
1.32.9 Humidity of cold air: The moisture content of cold air supplied to the blast furnace.
1.32.10 Temperature of cold air: The real-time temperature of cold air supplied to the blast furnace.
1.32.11 Pressure of furnace top: The static pressure of gas at the blast furnace top.
1.32.12 Material column difference: The height difference of burden column detected by different probes.

1.32.13 Stockline height of left probe: The burden surface height measured by the left stockline probe.
1.32.14 Stockline height of right probe: The burden surface height measured by the right stockline probe.
1.32.15 Height of radar stock rod: The burden surface height detected by radar stockline gauge.
1.32.16 Temperature difference of whole furnace: The temperature difference between different positions of the blast furnace shaft.
1.32.17 Current of 1# rotating device: The operating current of the No.1 blast furnace rotating charging device.
1.32.18 Current of 2# rotating device: The operating current of the No.2 blast furnace rotating charging device.
1.32.19 Material type of hopper: The type of burden stored in the charging hopper.
1.32.20 Material type of hoisting bucket: The type of burden loaded in the charging hoisting bucket.
1.32.21 Opening value of main belt-remote control valve: The opening degree of the main belt remote control valve.
1.32.22 Opening value of furnace top shut-off valve: The opening degree of the blast furnace top shut-off valve.
1.32.23 Opening value of mixed air control valve: The opening degree of the mixed air control valve.
1.32.24 Opening value of blow-off valve about cold air: The opening degree of the cold air blow-off valve.
1.32.25 Temperature of furnace top - Southeast: The temperature at the southeast part of the blast furnace top.
1.32.26 Temperature of furnace top - Northwest: The temperature at the northwest part of the blast furnace top.
1.32.27 Temperature of furnace top - Northeast: The temperature at the northeast part of the blast furnace top.
1.32.28 Temperature of furnace top - Southwest: The temperature at the southwest part of the blast furnace top.
1.32.29 Temperature of furnace top - Range: The temperature range of the blast furnace top.
1.32.30 Pressure of furnace top - Southeast: The pressure at the southeast part of the blast furnace top.
1.32.31 Pressure of furnace top - Northwest: The pressure at the northwest part of the blast furnace top.
1.32.32 Pressure of furnace top - Northeast: The pressure at the northeast part of the blast furnace top.
1.32.33 Pressure of furnace top - Southwest: The pressure at the southwest part of the blast furnace top.
1.32.34 Pressure of furnace top - Range: The pressure range of the blast furnace top.
1.32.35 Weight of 1# hoisting bucket: The burden weight of the No.1 charging hoisting bucket.
1.32.36 Weight of 2# hoisting bucket: The burden weight of the No.2 charging hoisting bucket.
1.32.37 Center temperature of furnace top cross temperature measurement: The central temperature of furnace top cross temperature measurement.
1.32.38 Sub-center temperature of furnace top cross temperature measurement: The sub-central temperature of furnace top cross temperature measurement.
1.32.39 Dumping angle: The angle of burden dumping during blast furnace charging.
1.32.40 Number of charging rounds: The number of charging cycles in blast furnace operation.
1.32.41 Accumulated volume of cold air: The total accumulated flow of cold air supplied to the blast furnace.
1.32.42 Static pressure of section 15 about furnace body: The static pressure at section 15 of the blast furnace shaft.
1.32.43 Static pressure of section 13 about furnace body: The static pressure at section 13 of the blast furnace shaft.

1.32.44 Static pressure of section 11 about furnace body: The static pressure at section 11 of the blast furnace shaft.
1.32.45 Upper static pressure differential of furnace body: The static pressure difference at the upper part of the blast furnace shaft.
1.32.46 Upper-middle static pressure differential of furnace body: The static pressure difference between upper and middle part of the blast furnace shaft.
1.32.47 Middle static pressure differential of furnace body: The static pressure difference at the middle part of the blast furnace shaft.
1.32.48 Lower static pressure differential of furnace body: The static pressure difference at the lower part of the blast furnace shaft.
1.32.49 Weight of 1# pulverized coal injection tank: The real-time weight of the No.1 pulverized coal injection tank.
1.32.50 Weight of 2# pulverized coal injection tank: The real-time weight of the No.2 pulverized coal injection tank.
1.32.51 Weight of 3# pulverized coal injection tank: The real-time weight of the No.3 pulverized coal injection tank.
1.32.52 Hourly total volume of pulverized coal injection: The total hourly injection volume of pulverized coal into the blast furnace.
1.32.53 CO2 content of coal gas: The volume fraction of carbon dioxide in blast furnace gas.
1.32.54 CO content of coal gas: The volume fraction of carbon monoxide in blast furnace gas.
1.32.55 H2 content of coal gas: The volume fraction of hydrogen in blast furnace gas.
1.32.56 N2 content of coal gas: The volume fraction of nitrogen in blast furnace gas.
1.32.57 CH4 content of coal gas: The volume fraction of methane in blast furnace gas.
1.32.58 Temperature of bag inlet: The gas temperature at the inlet of the blast furnace gas bag filter.
1.32.59 Hourly count of batch: The number of charging batches per hour.
1.32.60 Whole furnace differential pressure of blast furnace: The total pressure difference between furnace top and hearth.
1.32.61 Flow rate of furnace top water injection: The flow rate of cooling water injected into the blast furnace top.
1.32.62 Total count of batch: The total number of charging batches in operation.
1.32.63 Whole furnace temperature difference of blast furnace: The overall temperature difference of the blast furnace.
1.32.64 Pressure of oxygen-enriched: The pressure of oxygen-enriched air supplied to the blast furnace.
1.32.65 Flow rate of oxygen-enriched: The flow rate of oxygen-enriched air supplied to the blast furnace.
1.32.66 Standard blast velocity: The theoretical velocity of hot blast at the blast furnace tuyere.
1.32.67 Actual blast velocity: The real-time velocity of hot blast at the blast furnace tuyere.
1.32.68 Temperature of 1# gas-tight box: The temperature of the No.1 gas-tight box.
1.32.69 Temperature of 2# gas-tight box: The temperature of the No.2 gas-tight box.
1.32.70 Temperature of 3# gas-tight box: The temperature of the No.3 gas-tight box.
1.32.71 Flow rate of nitrogen about gas-tight box: The nitrogen flow rate supplied to the gas-tight box.
1.32.72 Pressure of nitrogen about gas-tight box: The nitrogen pressure supplied to the gas-tight box.
1.32.73 1# Hot stove - Dome temperature: The dome temperature of the No.1 hot blast stove.
1.32.74 2# Hot stove - Dome temperature: The dome temperature of the No.2 hot blast stove.
1.32.75 3# Hot stove - Dome temperature: The dome temperature of the No.3 hot blast stove.

1.32.76 1# Hot stove – Temperature of waste gas: The waste gas temperature of the No.1 hot blast stove.
1.32.77 2# Hot stove – Temperature of waste gas: The waste gas temperature of the No.2 hot blast stove.
1.32.78 3# Hot stove – Temperature of waste gas: The waste gas temperature of the No.3 hot blast stove.
1.32.79 Flow rate of return water about hot blast stove: The return water flow rate of the hot blast stove cooling system.
1.32.80 Flow rate of water supply about hot blast stove: The supply water flow rate of the hot blast stove cooling system.
1.32.81 Pressure of water supply about hot blast stove: The supply water pressure of the hot blast stove cooling system.
1.32.82 Pressure of expansion tank about hot blast stove: The pressure of the hot blast stove expansion tank.
1.32.83 Liquid level of expansion tank about hot blast stove: The liquid level of the hot blast stove expansion tank.
1.32.84 Pressure of main nitrogen pipe: The operating pressure of the main nitrogen pipeline.
1.32.85 Flow rate of main nitrogen pipe: The operating flow rate of the main nitrogen pipeline.
1.32.86 Temperature of saturated steam: The temperature of saturated steam for blast furnace auxiliary use.
1.32.87 Pressure of compressed air: The pressure of compressed air for blast furnace auxiliary use.
1.32.88 Flow rate of 1# water supply pipe: The flow rate of the No.1 water supply pipeline.
1.32.89 Flow rate of 2# water supply pipe: The flow rate of the No.2 water supply pipeline.
1.32.90 Temperature of 1# water supply pipe: The temperature of the No.1 water supply pipeline.
1.32.91 Temperature of 2# water supply pipe: The temperature of the No.2 water supply pipeline.
1.32.92 Pressure of 1# water supply pipe: The pressure of the No.1 water supply pipeline.
1.32.93 Pressure of 2# water supply pipe: The pressure of the No.2 water supply pipeline.
1.32.94 Flow rate of 1# return water pipe: The flow rate of the No.1 return water pipeline.
1.32.95 Flow rate of 2# return water pipe: The flow rate of the No.2 return water pipeline.
1.32.96 Temperature of 1# return water pipe: The temperature of the No.1 return water pipeline.
1.32.97 Temperature of 2# return water pipe: The temperature of the No.2 return water pipeline.
1.32.98 Pressure of 1# return water pipe: The pressure of the No.1 return water pipeline.
1.32.99 Pressure of 2# return water pipe: The pressure of the No.2 return water pipeline.
1.32.100 Pressure of 1# high-pressure water supply pipe: The pressure of the No.1 high-pressure water supply pipeline.
1.32.101 Pressure of 2# high-pressure water supply pipe: The pressure of the No.2 high-pressure water supply pipeline.
1.32.102 Flow rate of 1# high-pressure water supply pipe: The flow rate of the No.1 high-pressure water supply pipeline.
1.32.103 Flow rate of 2# high-pressure water supply pipe: The flow rate of the No.2 high-pressure water supply pipeline.
1.32.104 Temperature of 1# high-pressure water supply pipe: The temperature of the No.1 high-pressure water supply pipeline.
1.32.105 Temperature of 2# high-pressure water supply pipe: The temperature of the No.2 high-pressure water supply pipeline.
1.32.106 Liquid level of main body expansion tank: The liquid level of the blast furnace main expansion tank.
1.32.107 Pressure of main body expansion tank: The pressure of the blast furnace main expansion tank.
1.32.108 Pressure of nitrogen tank: The operating pressure of the blast furnace nitrogen storage tank.

1.32.109 Gas utilization rate: The efficiency of reduction reaction of blast furnace gas components.

1.32.110 K Value (Evaluation index of blast furnace operation status): The comprehensive operation evaluation index of blast furnace.

1.32.111 Permeability index: The index reflecting the ventilation performance of blast furnace burden column.

1.32.112 Bosh gas volume: The volume of gas generated at the blast furnace bosh.

1.32.113 Furnace hearth activity: The index reflecting the reaction activity of blast furnace hearth.

1.32.114 Rate of Oxygen-Enriched: The volume fraction of oxygen in oxygen-enriched blast.

1.32.115 Blast kinetic energy: The kinetic energy of hot blast injected from blast furnace tuyeres.

1.32.116 Theoretical combustion temperature: The theoretical temperature of coal combustion at blast furnace tuyeres.

1.32.117 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast furnace system - Integrated terminal - Operation monitoring”.

1.33 Oxygen enrichment system - Oxygen enrichment pipeline - Operation monitoring

1.33.1 Start of flow control valve: The actuation state of the oxygen-enriched flow control valve.

1.33.2 Flow rate of oxygen-enriched: The real-time flow rate of oxygen-enriched air.

1.33.3 Opening value of flow control valve: The opening degree of the oxygen-enriched flow control valve.

1.33.4 Temperature of oxygen-enriched pipe: The temperature of the oxygen-enriched conveying pipeline.

1.33.5 Pressure of flow control valve inlet: The inlet pressure of the oxygen-enriched flow control valve.

1.33.6 Pressure of flow control valve outlet: The outlet pressure of the oxygen-enriched flow control valve.

1.33.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Oxygen enrichment system - Oxygen enrichment pipeline - Operation monitoring”.

1.34 Oxygen enrichment system - Regulating valve group - Flow control

1.34.1 Start of 1# oxygen-enriched manual shut-off valve: The actuation state of No.1 oxygen-enriched manual shut-off valve.

1.34.2 Start of 2# oxygen-enriched manual shut-off valve: The actuation state of No.2 oxygen-enriched manual shut-off valve.

1.34.3 Start of 1# nitrogen manual shut-off valve: The actuation state of No.1 nitrogen manual shut-off valve.

1.34.4 Start of 2# nitrogen manual shut-off valve: The actuation state of No.2 nitrogen manual shut-off valve.

1.34.5 Start of 3# nitrogen manual shut-off valve: The actuation state of No.3 nitrogen manual shut-off valve.

1.34.6 Opening value of 1# oxygen-enriched manual shut-off valve: The opening degree of No.1 oxygen-enriched manual shut-off valve.

1.34.7 Opening value of 2# oxygen-enriched manual shut-off valve: The opening degree of No.2 oxygen-enriched manual shut-off valve.

1.34.8 Opening value of 1# nitrogen manual shut-off valve: The opening degree of No.1 nitrogen manual shut-off valve.

1.34.9 Opening value of 2# nitrogen manual shut-off valve: The opening degree of No.2 nitrogen manual shut-off valve.

1.34.10 Opening value of 3# nitrogen manual shut-off valve: The opening degree of No.3 nitrogen manual shut-off valve.

1.34.11 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Oxygen enrichment system - Regulating valve group - Flow control”.

1.35 Blast supply system - Industrial camera - Tuyere status monitoring

1.35.1 Additional storage templates: Image data related to data category “Blast supply system - Industrial camera - Tuyere status monitoring”: mainly comes from industrial cameras, image sensors, and other devices.
1.35.2 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast supply system - Industrial camera - Tuyere status monitoring”.

2 Burden feeding

2.1 Raw material system - Laboratory - Sinter performance testing

2.1.1 Tumbler index: A mechanical index representing the mass percentage of material larger than 6.3 mm after tumbling test, reflecting the impact resistance of sinter, pellets or lump ore.
2.1.2 Screening index: An index characterizing the particle size classification efficiency of raw materials, expressed as the mass ratio of material passing or retained on a specified sieve.
2.1.3 Abrasion index: A mechanical index representing the mass percentage of material smaller than 0.5 mm after tumbling test, reflecting the abrasion resistance of ironmaking materials.
2.1.4 Compressive strength: The maximum axial pressure that a single pellet or lump material can withstand without fracture, indicating the mechanical strength of the material.
2.1.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Laboratory - Sinter performance testing”.

2.2 Raw material system - Laboratory - Sinter composition detection

2.2.1 TFe/Iron grade: The mass percentage of total iron in iron-bearing materials, the core index for evaluating the quality of iron ore, sinter and pellets.
2.2.2 FeO content: The mass percentage of ferrous oxide in ironmaking materials, affecting the reducibility and metallurgical properties of the burden.
2.2.3 CaO content: The mass percentage of calcium oxide in blast furnace burden, a main alkaline component for adjusting slag basicity.
2.2.4 MgO content: The mass percentage of magnesium oxide in blast furnace burden, improving slag fluidity and high-temperature stability.
2.2.5 Al2O3 content: The mass percentage of aluminum oxide in blast furnace burden, affecting slag viscosity and melting temperature.
2.2.6 SiO2 content: The mass percentage of silicon dioxide in ironmaking materials, the main acidic component of blast furnace slag.
2.2.7 R2/Binary basicity: The ratio of CaO mass fraction to SiO2 mass fraction in blast furnace burden, a key index for slag property control.
2.2.8 TiO2 content: The mass percentage of titanium dioxide in ironmaking materials, affecting slag fluidity and high-temperature performance.
2.2.9 P content: The mass percentage of phosphorus in raw materials and fuels, a harmful element affecting

hot metal and steel quality.
2.2.10 Moisture content: The mass percentage of free water in raw materials and fuels, affecting charging stability and smelting heat balance.
2.2.11 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Laboratory - Sinter composition detection”.

2.3 Raw material system - Laboratory - Pellet performance testing

2.3.1 Tumbler index: A mechanical index representing the mass percentage of material larger than 6.3 mm after tumbling test, reflecting the impact resistance of sinter, pellets or lump ore.
2.3.2 Abrasion index: A mechanical index representing the mass percentage of material smaller than 0.5 mm after tumbling test, reflecting the abrasion resistance of ironmaking materials.
2.3.3 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Laboratory - Pellet performance testing”.

2.4 Raw material system - Laboratory - Pellet composition detection

2.4.1 TFe/Iron grade: The mass percentage of total iron in iron-bearing materials, the core index for evaluating the quality of iron ore, sinter and pellets.
2.4.2 FeO content: The mass percentage of ferrous oxide in ironmaking materials, affecting the reducibility and metallurgical properties of the burden.
2.4.3 CaO content: The mass percentage of calcium oxide in blast furnace burden, a main alkaline component for adjusting slag basicity.
2.4.4 MgO content: The mass percentage of magnesium oxide in blast furnace burden, improving slag fluidity and high-temperature stability.
2.4.5 Al2O3 content: The mass percentage of aluminum oxide in blast furnace burden, affecting slag viscosity and melting temperature.
2.4.6 SiO2 content: The mass percentage of silicon dioxide in ironmaking materials, the main acidic component of blast furnace slag.
2.4.7 R2/Binary basicity: The ratio of CaO mass fraction to SiO2 mass fraction in blast furnace burden, a key index for slag property control.
2.4.8 TiO2 content: The mass percentage of titanium dioxide in ironmaking materials, affecting slag fluidity and high-temperature performance.
2.4.9 P content: The mass percentage of phosphorus in raw materials and fuels, a harmful element affecting hot metal and steel quality.
2.4.10 Moisture content: The mass percentage of free water in raw materials and fuels, affecting charging stability and smelting heat balance.
2.4.11 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Laboratory - Pellet composition detection”.

2.5 Raw material system - Laboratory - Pulverized coal performance testing

2.5.1 Calorific value: The heat released per unit mass of fuel during complete combustion, a core index for measuring fuel heating capacity.
2.5.2 Mad (Air-dried basis moisture): The mass percentage of moisture in materials calculated on air-dried

basis, a conventional index for fuel analysis.
2.5.3 H_2O content: The total mass percentage of moisture in raw materials and fuels, including free water and bound water.
2.5.4 R200 (Proportion of fine particles): The mass percentage of particles smaller than 200 mesh in materials, representing ultra-fine particle content.
2.5.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Laboratory - Pulverized coal performance testing”.

2.6 Raw material system - Laboratory - Pulverized coal composition detection

2.6.1 Carbon content: The mass percentage of carbon in coke and coal, the main energy-providing component in blast furnace smelting.
2.6.2 A (Ash content): The mass percentage of inorganic residue left after complete combustion of fuel, representing impurity content.
2.6.3 V (Volatile matter content): The mass percentage of gaseous products released from fuel when heated without air, reflecting volatile components.
2.6.4 Sulfur content: The mass percentage of sulfur in raw materials and fuels, a harmful element causing sulfur pickup in hot metal.
2.6.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Laboratory - Pulverized coal composition detection”.

2.7 Raw material system - Laboratory - Coke performance testing

2.7.1 Mad (Air-dried basis moisture): The mass percentage of moisture in materials calculated on air-dried basis, a conventional index for fuel analysis.
2.7.2 H_2O content: The total mass percentage of moisture in raw materials and fuels, including free water and bound water.
2.7.3 M40 (Proportion of fine particles): The mass percentage of coke retained on 40 mm sieve after tumbling test, reflecting coke crushing resistance.
2.7.4 M10 (Proportion of fine particles): The mass percentage of coke passing 10 mm sieve after tumbling test, reflecting coke abrasion resistance.
2.7.5 CRI (Reactivity index): The reaction degree of coke with CO_2 at high temperature, evaluating the reaction performance of blast furnace coke.
2.7.6 CSR (Strength after reaction): The mechanical strength of coke after reaction with CO_2 at high temperature, reflecting strength retention in blast furnace.
2.7.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Laboratory - Coke performance testing”.

2.8 Raw material system - Laboratory - Coke composition detection

2.8.1 Carbon content: The mass percentage of carbon in coke and coal, the main energy-providing component in blast furnace smelting.
2.8.2 A (Ash content): The mass percentage of inorganic residue left after complete combustion of fuel, representing impurity content.

2.8.3 V (Volatile matter content): The mass percentage of gaseous products released from fuel when heated without air, reflecting volatile components.
2.8.4 Sulfur content: The mass percentage of sulfur in raw materials and fuels, a harmful element causing sulfur pickup in hot metal.
2.8.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Laboratory - Coke composition detection”.

2.9 Raw material system - Industrial camera - Pellet particle size recognition

2.9.1 Proportion of <5mm: The mass percentage of particles smaller than 5 mm in raw materials or fuels.
2.9.2 Proportion of 5~8mm: The mass percentage of particles between 5 mm and 8 mm in raw materials or fuels.
2.9.3 Proportion of 8~10mm: The mass percentage of particles between 8 mm and 10 mm in raw materials or fuels.
2.9.4 Proportion of 10~12.5mm: The mass percentage of particles between 10 mm and 12.5 mm in raw materials or fuels.
2.9.5 Proportion of 12.5~16mm: The mass percentage of particles between 12.5 mm and 16 mm in raw materials or fuels.
2.9.6 Proportion of >16mm: The mass percentage of particles larger than 16 mm in raw materials or fuels.
2.9.7 Average particle size: The weighted average diameter of particle size distribution, reflecting the overall particle size level of materials.
2.9.8 Additional storage templates: Image data related to data category “Raw material system - Industrial camera - Pellet particle size recognition”: mainly comes from industrial cameras, image sensors, and other devices.
2.9.9 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Industrial camera - Pellet particle size recognition”.

2.10 Raw material system - Industrial camera - Coke particle size recognition

2.10.1 Proportion of <25mm: The mass percentage of particles smaller than 25 mm in raw materials or fuels.
2.10.2 Proportion of 25~40mm: The mass percentage of particles between 25 mm and 40 mm in raw materials or fuels.
2.10.3 Proportion of 40~60mm: The mass percentage of particles between 40 mm and 60 mm in raw materials or fuels.
2.10.4 Proportion of 60~80mm: The mass percentage of particles between 60 mm and 80 mm in raw materials or fuels.
2.10.5 Proportion of >80mm: The mass percentage of particles larger than 80 mm in raw materials or fuels.
2.10.6 Average particle size: The weighted average diameter of particle size distribution, reflecting the overall particle size level of materials.
2.10.7 Additional storage templates: Image data related to data category “Raw material system - Industrial camera - Coke particle size recognition”: mainly comes from industrial cameras, image sensors, and other devices.
2.10.8 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Industrial camera

- Coke particle size recognition”.

2.11 Raw material system - Industrial camera - Mixed ore particle size recognition

2.11.1 Proportion of <5mm: The mass percentage of particles smaller than 5 mm in raw materials or fuels.

2.11.2 Proportion of 5~10mm: The mass percentage of particles between 5 mm and 10 mm in raw materials or fuels.

2.11.3 Proportion of 10~16mm: The mass percentage of particles between 10 mm and 16 mm in raw materials or fuels.

2.11.4 Proportion of 16~25mm: The mass percentage of particles between 16 mm and 25 mm in raw materials or fuels.

2.11.5 Proportion of 25~40mm: The mass percentage of particles between 25 mm and 40 mm in raw materials or fuels.

2.11.6 Proportion of >40mm: The mass percentage of particles larger than 40 mm in raw materials or fuels.

2.11.7 Average particle size: The weighted average diameter of particle size distribution, reflecting the overall particle size level of materials.

2.11.8 Additional storage templates: Image data related to data category “Raw material system - Industrial camera - Mixed ore particle size recognition”: mainly comes from industrial cameras, image sensors, and other devices.

2.11.9 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Raw material system - Industrial camera - Mixed ore particle size recognition”.

2.12 Ore bin system - Hydraulic station - Operation monitoring

2.12.1 Pressure of hydraulic station at furnace top: The operating pressure of the hydraulic unit at furnace top, providing power for charging equipment.

2.12.2 Pressure of hydraulic station at ore hopper: The operating pressure of the hydraulic system for ore hopper, ensuring normal action of valves and mechanisms.

2.12.3 Temperature of upper seal valve: The real-time operating temperature of the upper seal valve in the charging system.

2.12.4 Temperature of lower seal valve: The real-time operating temperature of the lower seal valve in the charging system.

2.12.5 Flow rate of nitrogen about gas-tight box: The volume flow of nitrogen supplied to the gas-tight box for sealing and purging.

2.12.6 Temperature of valve box: The real-time temperature of the charging valve box, monitoring the thermal environment.

2.12.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Ore bin system - Hydraulic station - Operation monitoring”.

2.13 Ore bin system - Airtight box - Operation monitoring

2.13.1 Liquid level of hydraulic station at furnace top: The hydraulic oil level in the furnace top hydraulic station, ensuring system operation.

2.13.2 Liquid level of hydraulic station at ore hopper: The hydraulic oil level in the ore hopper hydraulic station, maintaining system stability.

2.13.3 Flow rate of circulation water about gas-tight box: The flow rate of cooling circulating water for the gas-tight box.
2.13.4 Temperature A of gas-tight box: One real-time temperature monitoring value of the gas-tight box.
2.13.5 Temperature B of gas-tight box: One real-time temperature monitoring value of the gas-tight box.
2.13.6 Temperature C of gas-tight box: One real-time temperature monitoring value of the gas-tight box.
2.13.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Ore bin system - Airtight box - Operation monitoring”.

2.14 Ore bin system - Ore hopper - Operation monitoring

2.14.1 Weight of 1# feed bin (Silica stone): The real-time weight of silica stone stored in No.1 feed bin.
2.14.2 Weight of 2# feed bin (Manganese ore): The real-time weight of manganese ore stored in No.2 feed bin.
2.14.3 Weight of 3# feed bin (Lump ore): The real-time weight of lump ore stored in No.3 feed bin.
2.14.4 Weight of 4# feed bin (Lump ore): The real-time weight of lump ore stored in No.4 feed bin.
2.14.5 Weight of 5# feed bin (Pellet): The real-time weight of pellets stored in No.5 feed bin.
2.14.6 Weight of 6# feed bin (Pellet): The real-time weight of pellets stored in No.6 feed bin.
2.14.7 Weight of 7# feed bin (Pellet): The real-time weight of pellets stored in No.7 feed bin.
2.14.8 Weight of 8# feed bin (Pellet): The real-time weight of pellets stored in No.8 feed bin.
2.14.9 Weight of 9# feed bin (Pellet): The real-time weight of pellets stored in No.9 feed bin.
2.14.10 Weight of 10# feed bin (Pellet): The real-time weight of pellets stored in No.10 feed bin.
2.14.11 Weight of 11# feed bin (Sinter): The real-time weight of sinter stored in No.11 feed bin.
2.14.12 Weight of 12# feed bin (Sinter): The real-time weight of sinter stored in No.12 feed bin.
2.14.13 Weight of 13# feed bin (Sinter): The real-time weight of sinter stored in No.13 feed bin.
2.14.14 Weight of 14# feed bin (Sinter): The real-time weight of sinter stored in No.14 feed bin.
2.14.15 Weight of 15# feed bin (Sinter): The real-time weight of sinter stored in No.15 feed bin.
2.14.16 Weight of 16# feed bin (Sinter): The real-time weight of sinter stored in No.16 feed bin.
2.14.17 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Ore bin system - Ore hopper - Operation monitoring”.

2.15 Ore bin system - Coke hopper - Operation monitoring

2.15.1 Weight of 1# feed bin: The real-time material weight in No.1 feed bin of the charging system.
2.15.2 Weight of 2# feed bin: The real-time material weight in No.2 feed bin of the charging system.
2.15.3 Weight of 3# feed bin: The real-time material weight in No.3 feed bin of the charging system.
2.15.4 Weight of 4# feed bin: The real-time material weight in No.4 feed bin of the charging system.
2.15.5 Weight of 5# feed bin: The real-time material weight in No.5 feed bin of the charging system.
2.15.6 Weight of 6# feed bin: The real-time material weight in No.6 feed bin of the charging system.
2.15.7 Weight of 7# feed bin (Nut coke): The real-time weight of nut coke stored in No.7 feed bin.
2.15.8 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Ore bin system - Coke hopper - Operation monitoring”.

2.16 Ore bin system - Regulating valve group - Charging mode control

2.16.1 Times of coke insertion per day: The total number of coke supplement operations completed in one day.
2.16.2 Batch loading count of ore: The cumulative number of ore batch charging operations executed.
2.16.3 Batch loading count of coke: The cumulative number of coke batch charging operations executed.
2.16.4 Delay of ore discharge: The time lag of ore discharging action in the charging system.
2.16.5 Delay of nut coke discharge: The time lag of nut coke discharging action in the charging system.
2.16.6 Operation mode of charging: The working control mode of the furnace top charging system, such as automatic or manual.
2.16.7 Type of central charging - coke: The specific pattern of coke centralized charging set in the system.
2.16.8 Prohibition mode of coke charging: The control state that disables coke charging action.
2.16.9 Prohibition mode of coal insertion: The control state that disables coal supplement action.
2.16.10 Type of central charging - ore: The specific pattern of ore centralized charging set in the system.
2.16.11 Prohibition mode of ore charging: The control state that disables ore charging action.
2.16.12 Prohibition Mode of materials discharging: The control state that disables overall material discharging.
2.16.13 Unlock about belt and gate: The control state of releasing interlock for charging belt and material gate.
2.16.14 Furnace top reset: The operation of restoring the furnace top charging system to initial status.
2.16.15 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Ore bin system - Regulating valve group - Charging mode control”.

2.17 Charging system - Rotating chute - Operation monitoring

2.17.1 Material-chasing mode at furnace top: The special control mode for material supplement and tracking at furnace top.
2.17.2 Manual mode of material request: The manual control mode for initiating material demand.
2.17.3 Prohibition Mode of materials discharging: The control state that disables overall material discharging.
2.17.4 Number of charging rounds: The total number of complete charging cycles executed.
2.17.5 Rotation speed of charging rounds: The rotating speed of distribution equipment in charging cycles.
2.17.6 Stepping angle: The fixed rotation angle of the stepping mechanism in distribution equipment.
2.17.7 Reversal batch count: The number of reverse charging batches set or executed.
2.17.8 Daily total batch loading count of ore: The total number of ore batch charging operations completed in one day.
2.17.9 Previous day total batch loading count of ore: The total number of ore batch charging operations completed on the previous day.
2.17.10 Daily total batch loading count of coke: The total number of coke batch charging operations completed in one day.
2.17.11 Previous day total batch loading count of coke: The total number of coke batch charging operations completed on the previous day.
2.17.12 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Charging system - Rotating chute - Operation monitoring”.

2.18 Charging system - Stock rod - Operation monitoring

2.18.1 Height of left probe: The real-time material level detected by the left probe in the blast furnace.
2.18.2 Height of right probe: The real-time material level detected by the right probe in the blast furnace.
2.18.3 Current of left probe: The real-time working current of the left material level probe.
2.18.4 Current of right probe: The real-time working current of the right material level probe.
2.18.5 Upper limit value of height about left probe: The maximum allowable height alarm threshold for the left probe.
2.18.6 Upper limit value of height about right probe: The maximum allowable height alarm threshold for the right probe.
2.18.7 Lower limit value of height about left probe: The minimum allowable height alarm threshold for the left probe.
2.18.8 Lower limit value of height about right probe: The minimum allowable height alarm threshold for the right probe.
2.18.9 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Charging system - Stock rod - Operation monitoring".

2.19 Charging system - Hopper - Ore charging scheme configuration

2.19.1 Gear of α-1: The working gear parameter of No.1 α-axis distribution device.
2.19.2 Gear of α-2: The working gear parameter of No.2 α-axis distribution device.
2.19.3 Gear of α-3: The working gear parameter of No.3 α-axis distribution device.
2.19.4 Gear of α-4: The working gear parameter of No.4 α-axis distribution device.
2.19.5 Gear of α-5: The working gear parameter of No.5 α-axis distribution device.
2.19.6 Gear of α-6: The working gear parameter of No.6 α-axis distribution device.
2.19.7 Gear of α-7: The working gear parameter of No.7 α-axis distribution device.
2.19.8 Gear of α-8: The working gear parameter of No.8 α-axis distribution device.
2.19.9 Gear of α-9: The working gear parameter of No.9 α-axis distribution device.
2.19.10 Gear of α-10: The working gear parameter of No.10 α-axis distribution device.
2.19.11 Gear of α-11: The working gear parameter of No.11 α-axis distribution device.
2.19.12 Gear of α-12: The working gear parameter of No.12 α-axis distribution device.
2.19.13 Gear of β-1: The working gear parameter of No.1 β-axis distribution device.
2.19.14 Gear of β-2: The working gear parameter of No.2 β-axis distribution device.
2.19.15 Gear of β-3: The working gear parameter of No.3 β-axis distribution device.
2.19.16 Gear of β-4: The working gear parameter of No.4 β-axis distribution device.
2.19.17 Gear of β-5: The working gear parameter of No.5 β-axis distribution device.
2.19.18 Gear of β-6: The working gear parameter of No.6 β-axis distribution device.
2.19.19 Gear of β-7: The working gear parameter of No.7 β-axis distribution device.
2.19.20 Gear of β-8: The working gear parameter of No.8 β-axis distribution device.
2.19.21 Gear of β-9: The working gear parameter of No.9 β-axis distribution device.
2.19.22 Gear of β-10: The working gear parameter of No.10 β-axis distribution device.
2.19.23 Gear of β-11: The working gear parameter of No.11 β-axis distribution device.
2.19.24 Gear of β-12: The working gear parameter of No.12 β-axis distribution device.
2.19.25 Weight of gear 1: The material distribution weight corresponding to No.1 gear of distribution device.
2.19.26 Weight of gear 2: The material distribution weight corresponding to No.2 gear of distribution device.
2.19.27 Weight of gear 3: The material distribution weight corresponding to No.3 gear of distribution device.

2.19.28 Weight of gear 4: The material distribution weight corresponding to No.4 gear of distribution device.
2.19.29 Weight of gear 5: The material distribution weight corresponding to No.5 gear of distribution device.
2.19.30 Weight of gear 6: The material distribution weight corresponding to No.6 gear of distribution device.
2.19.31 Weight of gear 7: The material distribution weight corresponding to No.7 gear of distribution device.
2.19.32 Weight of gear 8: The material distribution weight corresponding to No.8 gear of distribution device.
2.19.33 Weight of gear 9: The material distribution weight corresponding to No.9 gear of distribution device.
2.19.34 Weight of gear 10: The material distribution weight corresponding to No.10 gear of distribution device.
2.19.35 Weight of gear 11: The material distribution weight corresponding to No.11 gear of distribution device.
2.19.36 Weight of gear 12: The material distribution weight corresponding to No.12 gear of distribution device.
2.19.37 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Charging system - Hopper - Ore charging scheme configuration".

2.20 Charging system - Hopper - Coke charging scheme configuration

2.20.1 Gear of α-1: The working gear parameter of No.1 α-axis distribution device.
2.20.2 Gear of α-2: The working gear parameter of No.2 α-axis distribution device.
2.20.3 Gear of α-3: The working gear parameter of No.3 α-axis distribution device.
2.20.4 Gear of α-4: The working gear parameter of No.4 α-axis distribution device.
2.20.5 Gear of α-5: The working gear parameter of No.5 α-axis distribution device.
2.20.6 Gear of α-6: The working gear parameter of No.6 α-axis distribution device.
2.20.7 Gear of α-7: The working gear parameter of No.7 α-axis distribution device.
2.20.8 Gear of α-8: The working gear parameter of No.8 α-axis distribution device.
2.20.9 Gear of α-9: The working gear parameter of No.9 α-axis distribution device.
2.20.10 Gear of α-10: The working gear parameter of No.10 α-axis distribution device.
2.20.11 Gear of α-11: The working gear parameter of No.11 α-axis distribution device.
2.20.12 Gear of α-12: The working gear parameter of No.12 α-axis distribution device.
2.20.13 Gear of β-1: The working gear parameter of No.1 β-axis distribution device.
2.20.14 Gear of β-2: The working gear parameter of No.2 β-axis distribution device.
2.20.15 Gear of β-3: The working gear parameter of No.3 β-axis distribution device.
2.20.16 Gear of β-4: The working gear parameter of No.4 β-axis distribution device.
2.20.17 Gear of β-5: The working gear parameter of No.5 β-axis distribution device.
2.20.18 Gear of β-6: The working gear parameter of No.6 β-axis distribution device.
2.20.19 Gear of β-7: The working gear parameter of No.7 β-axis distribution device.
2.20.20 Gear of β-8: The working gear parameter of No.8 β-axis distribution device.
2.20.21 Gear of β-9: The working gear parameter of No.9 β-axis distribution device.
2.20.22 Gear of β-10: The working gear parameter of No.10 β-axis distribution device.
2.20.23 Gear of β-11: The working gear parameter of No.11 β-axis distribution device.
2.20.24 Gear of β-12: The working gear parameter of No.12 β-axis distribution device.
2.20.25 Weight of gear 1: The material distribution weight corresponding to No.1 gear of distribution device.
2.20.26 Weight of gear 2: The material distribution weight corresponding to No.2 gear of distribution device.
2.20.27 Weight of gear 3: The material distribution weight corresponding to No.3 gear of distribution device.

2.20.28 Weight of gear 4: The material distribution weight corresponding to No.4 gear of distribution device.
2.20.29 Weight of gear 5: The material distribution weight corresponding to No.5 gear of distribution device.
2.20.30 Weight of gear 6: The material distribution weight corresponding to No.6 gear of distribution device.
2.20.31 Weight of gear 7: The material distribution weight corresponding to No.7 gear of distribution device.
2.20.32 Weight of gear 8: The material distribution weight corresponding to No.8 gear of distribution device.
2.20.33 Weight of gear 9: The material distribution weight corresponding to No.9 gear of distribution device.
2.20.34 Weight of gear 10: The material distribution weight corresponding to No.10 gear of distribution device.
2.20.35 Weight of gear 11: The material distribution weight corresponding to No.11 gear of distribution device.
2.20.36 Weight of gear 12: The material distribution weight corresponding to No.12 gear of distribution device.
2.20.37 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Charging system - Hopper - Coke charging scheme configuration".

2.21 Charging system - Industrial camera - Stock surface recognition

2.21.1 1# Ring-Temperature monitoring value - A1: Real-time temperature data at point A1 of No.1 ring temperature measuring area.
2.21.2 1# Ring-Temperature monitoring value – A2: Real-time temperature data at point A2 of No.1 ring temperature measuring area.
2.21.3 1# Ring-Temperature monitoring value – A3: Real-time temperature data at point A3 of No.1 ring temperature measuring area.
2.21.4 1# Ring-Temperature monitoring value – A4: Real-time temperature data at point A4 of No.1 ring temperature measuring area.
2.21.5 2# Ring-Temperature monitoring value - B1: Real-time temperature data at point B1 of No.2 ring temperature measuring area.
2.21.6 2# Ring-Temperature monitoring value – B2: Real-time temperature data at point B2 of No.2 ring temperature measuring area.
2.21.7 2# Ring-Temperature monitoring value – B3: Real-time temperature data at point B3 of No.2 ring temperature measuring area.
2.21.8 2# Ring-Temperature monitoring value – B4: Real-time temperature data at point B4 of No.2 ring temperature measuring area.
2.21.9 Additional storage templates: Image data related to data category "Charging system - Industrial camera - Stock surface recognition": mainly comes from industrial cameras, image sensors, and other devices.
2.21.10 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Charging system - Industrial camera - Stock surface recognition".

3 Gas & Dust treating

3.1 Gas system - Pressure equalization recoverer - Operation monitoring

3.1.1 Dust concentration of large dust hopper outlet: The mass of dust per unit volume of gas at the discharge outlet of the large dust hopper in the BFG dry dedusting system. It is the core environmental protection and process monitoring indicator for evaluating whether the dust emission at the system's end meets standards and the system's dedusting efficiency.

3.1.2 Differential pressure of large dust hopper: The pressure difference between the inside of the large dust hopper and the external environment. It reflects the degree of dust accumulation inside the hopper, the smoothness of ash discharge, and the sealing state of the hopper. Abnormal differential pressure indicates problems such as blockage, air leakage, or malfunction of the discharge valve.

3.1.3 Pressure of nitrogen pipe about dust hopper pulse valve: The static pressure of the nitrogen pipeline that provides actuating power for the pulse cleaning valve of the large ash hopper. It is a key pressure parameter to ensure the instantaneous rapid opening of the pulse valve and efficient filter bag cleaning. It must be maintained at a stable value to ensure cleaning effect.

3.1.4 Pressure of pneumatic valve: The pressure of the medium (nitrogen/compressed air) that drives the opening and closing actions of various pneumatic valves in the dedusting, ash conveying, and gas systems. It determines the flexibility, sealing, and response speed of the pneumatic valve and must match the valve specifications.

3.1.5 Flow rate of pneumatic valve: The volume of power medium (nitrogen/compressed air) flowing through the pneumatic valve per unit time. It reflects the flow capacity of the valve and is used to monitor the working status of the pneumatic valve and the stability of the medium pipeline transportation.

3.1.6 Pressure of pulse blowing: The static pressure of the cleaning medium injected by the pulse blowing device into the filter bag in the BFG bag filter. It directly determines the cleaning intensity of the filter bag and must be set reasonably according to dust viscosity and bag material to avoid incomplete cleaning or bag damage.

3.1.7 Flow rate of pulse blowing: The volume of cleaning medium injected by the pulse blowing device into the filter bag per unit time. Together with the blowing pressure, it determines the cleaning effect and reflects the medium supply volume per single injection.

3.1.8 Pressure of pneumatic conveying dust: The static pressure of the carrier gas (nitrogen) in the pipeline when conveying blast furnace ash in the pneumatic ash conveying system. It is a core parameter to ensure that the dust is suspended and stably transported in the pipeline, preventing dust settling and pipe blockage.

3.1.9 Flow rate of pneumatic conveying dust: The total volume of the mixture of carrier gas and dust flowing through the pneumatic ash conveying pipeline per unit time. It reflects the conveying capacity of the ash conveying system and must match the conveying pressure to ensure conveying efficiency.

3.1.10 Pressure of main nitrogen pipe: The static pressure of the main nitrogen pipeline shared by the entire blast furnace dedusting, cleaning, and ash conveying systems. It is the basic parameter for the stable energy supply of the entire nitrogen system and determines the upper limit of pressure supply to each branch nitrogen pipeline.

3.1.11 Flow rate of main nitrogen pipe: The total volume of nitrogen flowing through the main nitrogen pipeline per unit time. It reflects the total gas supply of the nitrogen system and is used to regulate the medium distribution of each branch pipeline to meet the nitrogen demand of each process point.

3.1.12 Pressure of inlet about raw coal gas: The static pressure at the inlet of the dedusting system before the untreated blast furnace raw gas enters the dedusting system. It is a basic input parameter for process control of the dedusting system, reflecting the pressure load generated by the blast furnace gas and determining the operating pressure condition of the dedusting system.

3.1.13 Concentration of inlet about raw coal gas: The mass of dust per unit volume of gas at the inlet of the dedusting system before the untreated blast furnace raw gas enters the dedusting system. It reflects the inlet dust load of the dedusting system and is an important basis for formulating cleaning and ash conveying

process parameters.
3.1.14 Temperature of inlet about raw coal gas: The temperature at the inlet of the dedusting system before the untreated blast furnace raw gas enters the dedusting system. It is a key monitoring indicator for the safe and stable operation of the dedusting system. Too high a temperature can easily damage the filter bag and other equipment, while too low a temperature can easily cause gas condensation.
3.1.15 Pressure of inlet about clean coal gas: The static pressure at the inlet of the subsequent conveying/using system (such as hot blast stove, gas pipeline network) after the cleaned blast furnace gas leaves the dedusting system. It reflects the pressure stability of clean gas transportation and must meet the pressure requirements of gas users.
3.1.16 Concentration of inlet about clean coal gas: The mass of residual dust per unit volume of gas at the inlet of the subsequent conveying/using system after the cleaned blast furnace gas leaves the dedusting system. It is a core indicator for evaluating the dedusting efficiency of the dedusting system and must meet the dust content standard for subsequent gas use.
3.1.17 Temperature of inlet about clean coal gas: The temperature at the inlet of the subsequent conveying/using system after the cleaned blast furnace gas leaves the dedusting system. It reflects the temperature loss during the gas dedusting process and must be kept stable to ensure normal use by gas users.
3.1.18 Pressure of 1# Box: The static pressure inside No. 1 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.
3.1.19 Temperature of 1# Box: The medium temperature inside No. 1 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.
3.1.20 Concentration of 1# Box: The mass of dust per unit volume of gas inside No. 1 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.
3.1.21 Pressure of 2# Box: The static pressure inside No. 2 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.
3.1.22 Temperature of 2# Box: The medium temperature inside No. 2 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.
3.1.23 Concentration of 2# Box: The mass of dust per unit volume of gas inside No. 2 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.
3.1.24 Pressure of 3# Box: The static pressure inside No. 3 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.
3.1.25 Temperature of 3# Box: The medium temperature inside No. 3 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.
3.1.26 Concentration of 3# Box: The mass of dust per unit volume of gas inside No. 3 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.
3.1.27 Pressure of 4# Box: The static pressure inside No. 4 dust removal box of the BFG bag filter. It reflects

the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.

3.1.28 Temperature of 4# Box: The medium temperature inside No. 4 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.

3.1.29 Concentration of 4# Box: The mass of dust per unit volume of gas inside No. 4 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.

3.1.30 Pressure of 5# Box: The static pressure inside No. 5 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.

3.1.31 Temperature of 5# Box: The medium temperature inside No. 5 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.

3.1.32 Concentration of 5# Box: The mass of dust per unit volume of gas inside No. 5 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.

3.1.33 Pressure of 6# Box: The static pressure inside No. 6 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.

3.1.34 Temperature of 6# Box: The medium temperature inside No. 6 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.

3.1.35 Concentration of 6# Box: The mass of dust per unit volume of gas inside No. 6 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.

3.1.36 Pressure of 7# Box: The static pressure inside No. 7 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.

3.1.37 Temperature of 7# Box: The medium temperature inside No. 7 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.

3.1.38 Concentration of 7# Box: The mass of dust per unit volume of gas inside No. 7 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.

3.1.39 Pressure of 8# Box: The static pressure inside No. 8 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.

3.1.40 Temperature of 8# Box: The medium temperature inside No. 8 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.

3.1.41 Concentration of 8# Box: The mass of dust per unit volume of gas inside No. 8 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.

3.1.42 Pressure of 9# Box: The static pressure inside No. 9 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.
3.1.43 Temperature of 9# Box: The medium temperature inside No. 9 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.
3.1.44 Concentration of 9# Box: The mass of dust per unit volume of gas inside No. 9 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.
3.1.45 Pressure of 10# Box: The static pressure inside No. 10 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.
3.1.46 Temperature of 10# Box: The medium temperature inside No. 10 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.
3.1.47 Concentration of 10# Box: The mass of dust per unit volume of gas inside No. 10 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.
3.1.48 Pressure of 11# Box: The static pressure inside No. 11 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.
3.1.49 Temperature of 11# Box: The medium temperature inside No. 11 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.
3.1.50 Concentration of 11# Box: The mass of dust per unit volume of gas inside No. 11 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.
3.1.51 Pressure of 12# Box: The static pressure inside No. 12 dust removal box of the BFG bag filter. It reflects the pressure state of the gas and dust mixture in the box and is used to judge the filtration resistance of the filter bag in the box and whether there is filter bag damage or pipeline blockage.
3.1.52 Temperature of 12# Box: The medium temperature inside No. 12 dust removal box of the BFG bag filter. It monitors whether the temperature inside the box is within a safe range. Abnormal temperature indicates problems such as filter bag damage, gas short circuit, or incomplete cleaning.
3.1.53 Concentration of 12# Box: The mass of dust per unit volume of gas inside No. 12 dust removal box of the BFG bag filter. It reflects the filtration effect of the filter bag in the box. Abnormal concentration in a single box indicates a fault in the filter bag of that box.
3.1.54 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Gas system - Pressure equalization recoverer - Operation monitoring”.

3.2 Gas system - Pressure equalization recoverer - Tank body monitoring

3.2.1 Timed back-blowing mode: A mode in which the blast furnace dedusting system automatically performs pulse cleaning on the filter bags at preset fixed time intervals. It is suitable for normal production conditions where the blast furnace gas load and dust concentration are relatively stable.

3.2.2 Differential pressure back-blowing mode: A mode in which the blast furnace dedusting system automatically triggers cleaning when the pressure difference across the dust removal box/filter bag reaches a preset threshold. It is suitable for conditions where the blast furnace gas load and dust concentration fluctuate greatly.
3.2.3 Manual back-blowing mode: A filter bag cleaning mode triggered manually by the on-site operator through the control terminal. It is suitable for system commissioning, local blockage of a single box filter bag, or temporary cleaning when the automatic mode fails.
3.2.4 Emergency manual back-blowing mode: An emergency filter bag cleaning mode initiated by the operator when a sudden blast furnace fault occurs (such as a sharp rise in box differential pressure or severe filter bag blockage). It has the highest priority and is used for system protection under fault conditions to prevent equipment damage.
3.2.5 System idle mode: A low-load standby mode of the dedusting and ash conveying systems during blast furnace blowing-off, low production, or temporary equipment maintenance. In this mode, only basic parameter monitoring functions are retained, and other process actions (cleaning, ash discharge, ash conveying) are suspended to save energy.
3.2.6 Automatic back-blowing mode: A mode in which the dedusting system automatically determines and initiates filter bag cleaning based on a preset program that integrates multiple triggering conditions such as timed and differential pressure. It is the mainstream cleaning mode during normal blast furnace production and adapts to complex operating condition changes.
3.2.7 Timed dust discharge mode: A mode in which the large ash hopper automatically starts ash discharge at preset fixed time intervals. It is suitable for conditions where the blast furnace dust output is relatively stable and the ash hopper level changes regularly.
3.2.8 Temperature difference dust discharge mode: A mode in which the large ash hopper automatically starts ash discharge when the temperature difference between different positions inside the hopper reaches a preset threshold. It indirectly judges the amount of dust accumulation in the ash hopper through temperature changes and is suitable for conditions where dust accumulation is difficult to monitor directly.
3.2.9 Manual dust discharge mode: An ash hopper discharge mode triggered manually by the on-site operator. It is suitable for system commissioning, automatic ash discharge mode failure, or temporary discharge when the ash hopper level is abnormal.
3.2.10 Emergency manual dust discharge mode: An emergency ash hopper discharge mode initiated by the operator when a sudden blast furnace fault occurs (such as a sudden full ash hopper level or stuck discharge valve). It is used for rapid discharge under fault conditions to prevent ash hopper blockage and overflow.
3.2.11 Automatic dust discharge mode: A mode in which the ash hopper system automatically determines and initiates ash discharge based on a preset program that integrates multiple triggering conditions such as timed, temperature difference, and level. It is the mainstream discharge mode during normal blast furnace production and ensures a stable ash hopper level.
3.2.12 Time of timed back-blowing: The duration of a single pulse back-blowing preset in the timed back-blowing mode. It determines the injection duration of a single cleaning medium and must match the blowing pressure to ensure the cleaning effect.
3.2.13 Pressure of timed back-blowing: The working pressure of the pulse blowing medium preset in the timed back-blowing mode. It is the core parameter of cleaning intensity in this mode, ensuring effective detachment of dust from the filter bag.
3.2.14 Time of timed dust discharge: The duration of a single ash discharge operation preset in the timed ash discharge mode. It determines the amount of dust discharged in a single operation and must match the dust

accumulation rate in the ash hopper.

3.2.15 Temperature of timed dust discharge: The temperature reference value inside the ash hopper monitored at fixed times in the temperature difference ash discharge mode. It serves as a reference for judging temperature changes and assists in triggering ash discharge actions.

3.2.16 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Gas system - Pressure equalization recoverer - Tank body monitoring”.

3.3 Gas system - Pressure equalization recoverer - System configuration

3.3.1 Invoked back-blowing nitrogen consumption per use: The actual amount of nitrogen consumed during a single manual/automatic back-blowing action. It reflects the nitrogen usage efficiency of a single back-blowing and provides data for energy consumption optimization.

3.3.2 Last consumption of invoked back-blowing nitrogen: The amount of nitrogen consumed during the previous back-blowing action. Compared with the current consumption, it helps analyze the stability of nitrogen consumption for back-blowing and determine whether there are problems such as pipeline leakage or valve failure.

3.3.3 Back-blowing nitrogen consumption per use: The actual amount of nitrogen consumed during a single automatic back-blowing cleaning mode. It is a core monitoring indicator for the unit consumption of nitrogen in the back-blowing system.

3.3.4 Last consumption of back-blowing nitrogen: The amount of nitrogen consumed during the previous automatic back-blowing cleaning. It is used for comparative analysis of the consumption trend of nitrogen in the back-blowing system.

3.3.5 Back-blowing nitrogen consumption of current team: The total amount of nitrogen cumulatively consumed by the back-blowing cleaning system during the current shift production cycle of the blast furnace. It is a basic indicator for team nitrogen energy consumption assessment and energy consumption statistics.

3.3.6 Back-blowing nitrogen consumption at 8:00-20:00: The total amount of nitrogen cumulatively consumed by the back-blowing cleaning system during the day shift (8:00-20:00) production cycle of the blast furnace. It is used for shift-based nitrogen consumption statistics to facilitate refined energy consumption management.

3.3.7 Back-blowing nitrogen consumption at 20:00-8:00: The total amount of nitrogen cumulatively consumed by the back-blowing cleaning system during the night shift (20:00-8:00) production cycle of the blast furnace. Compared with the day shift consumption, it helps analyze the energy consumption differences between different shifts under varying operating conditions.

3.3.8 Back-blowing nitrogen consumption of current month: The total amount of nitrogen cumulatively consumed by the back-blowing cleaning system within the current calendar month. It is used for monthly nitrogen energy consumption statistics and energy consumption target assessment.

3.3.9 Back-blowing nitrogen consumption of last month: The total amount of nitrogen cumulatively consumed by the back-blowing cleaning system within the previous calendar month. Compared with the current month's consumption, it helps analyze the monthly trend of nitrogen consumption and optimize energy consumption strategies.

3.3.10 Back-blowing nitrogen consumption of yesterday: The total amount of nitrogen cumulatively consumed by the back-blowing cleaning system within the previous natural day. It is used for daily energy consumption monitoring to timely detect abnormal energy consumption.

3.3.11 Nitrogen consumption per use for dust conveying: The actual amount of nitrogen consumed during a single pneumatic ash conveying operation. It reflects the nitrogen usage efficiency of a single ash conveying and is a core indicator for the unit consumption of the ash conveying system.

3.3.12 Last nitrogen consumption for dust conveying: The amount of nitrogen consumed during the previous pneumatic ash conveying operation. Compared with the current consumption, it helps determine whether there are problems such as pipe blockage or pipeline leakage that cause abnormal gas consumption.

3.3.13 Nitrogen consumption for dust conveying today: The total amount of nitrogen cumulatively consumed by the pneumatic ash conveying system within the current natural day. It is used for daily ash conveying energy consumption monitoring and regulation.

3.3.14 Nitrogen consumption for dust conveying of current month: The total amount of nitrogen cumulatively consumed by the pneumatic ash conveying system within the current calendar month. It is used for monthly ash conveying energy consumption statistics and assessment.

3.3.15 Nitrogen consumption for dust conveying of last month: The total amount of nitrogen cumulatively consumed by the pneumatic ash conveying system within the previous calendar month. Compared with the current month's consumption, it helps analyze the monthly trend of ash conveying energy consumption and optimize ash conveying process parameters.

3.3.16 Upper limit setpoint of temperature about raw coal gas: The preset maximum temperature threshold for raw gas entering the dedusting system to ensure safe and stable operation. When the actual temperature exceeds this value, the system triggers a warning or interlock protection to prevent high-temperature damage to the filter bag and other equipment.

3.3.17 Lower limit setpoint of temperature about raw coal gas: The preset minimum temperature threshold for raw gas entering the dedusting system to ensure safe and stable operation. When the actual temperature falls below this value, the system triggers a warning to prevent gas condensation that may cause bag clogging or dust agglomeration.

3.3.18 Setpoint value of time about timed back-blowing: The preset duration of a single back-blowing in the timed back-blowing mode, set by the operator according to the blast furnace conditions. It is a core control parameter of the timed back-blowing mode and can be adjusted based on dust load.

3.3.19 Setpoint value of pressure about differential pressure back-blowing: The preset pressure difference threshold across the box/filter bag that triggers the cleaning action in the differential pressure back-blowing mode, set by the operator according to the process requirements of the dedusting system. It is a core triggering parameter of the differential pressure back-blowing mode.

3.3.20 Setpoint value of pulse width: The preset duration of a single injection of cleaning medium by the pulse blowing device. It determines the cleaning duration of a single pulse and must be set reasonably according to bag specifications and dust characteristics.

3.3.21 Setpoint value of pulse interval: The preset time interval between two consecutive injections of cleaning medium by the pulse blowing device. It determines the cleaning frequency, avoiding excessive blowing that damages the filter bag or too long an interval that causes bag clogging.

3.3.22 Setpoint value of box interval: In a multi-box dedusting system, the preset time interval for sequentially starting pulse back-blowing in each box. It ensures that most boxes remain in normal filtration mode during the back-blowing process, preventing large fluctuations in dedusting efficiency.

3.3.23 Setpoint value of pressure about high dry nitrogen: The preset pressure threshold for the high-dryness nitrogen supply pipeline supporting the blast furnace raw gas dedusting system. It is used to regulate the nitrogen supplementation pressure during the raw gas dedusting process to ensure dedusting effect and gas dryness.

3.3.24 Setpoint value of pressure about differential pressure back-blowing of dust hopper: The preset pressure difference threshold for triggering back-blowing cleaning of the large ash hopper itself based on differential pressure. When the hopper differential pressure reaches this value, local back-blowing of the hopper is automatically started to prevent dust agglomeration and blockage.

3.3.25 Setpoint value of time about timed dust discharge: The preset duration of a single ash discharge operation in the timed ash discharge mode, set by the operator according to the blast furnace dust output. It is a core control parameter of the timed ash discharge mode and can be adjusted based on the ash hopper level.

3.3.26 Setpoint value of temperature about temperature difference dust discharge: The preset temperature difference threshold that triggers the ash discharge action in the temperature difference ash discharge mode, set by the operator according to the process requirements of the ash hopper. It is a core triggering parameter of the temperature difference ash discharge mode.

3.3.27 Setpoint value of nitrogen cannon width: The preset duration of a single nitrogen injection by the nitrogen cannon (ash hopper anti-blocking device). It determines the injection intensity of the nitrogen cannon and is used to prevent dust adhesion and accumulation on the inner wall of the ash hopper.

3.3.28 Setpoint value of nitrogen cannon interval: The preset time interval between two consecutive nitrogen injections by the nitrogen cannon. It determines the working frequency of the nitrogen cannon and adapts to the dust adhesion characteristics of the ash hopper.

3.3.29 Setpoint value of time about empty hopper: The preset standby interval time after the ash hopper discharge operation is completed. It ensures that residual dust in the ash hopper is completely discharged, avoiding accumulation that may lead to blockage.

3.3.30 Lower limit value of pressure about dust conveying pipe: The preset minimum allowable pressure threshold in the pneumatic ash conveying pipeline to prevent pipe blockage. When the actual pressure falls below this value, the system triggers a warning or stops ash conveying to promptly troubleshoot.

3.3.31 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Gas system - Pressure equalization recoverer - System configuration”.

3.4 Gas system - Purifier - Operation monitoring

3.4.1 Manual valve of main nitrogen pipe: A manually operated valve installed on the main nitrogen pipeline network. It is mainly used for pipeline isolation during equipment maintenance of the nitrogen system, branch pipeline switching, and on-site manual pressure regulation.

3.4.2 Electric ball valve on nitrogen inlet: An electrically controlled ball valve installed at the inlet of the nitrogen system. It has remote automatic control function and is used for quickly switching on/off the nitrogen inlet pipeline, featuring fast response speed and good sealing performance.

3.4.3 Electric ball valve on nitrogen outlet: An electrically controlled ball valve installed at the outlet of each branch of the nitrogen system. It is used for remotely and automatically controlling the on/off of the nitrogen supply pipeline at each process point, enabling precise medium distribution.

3.4.4 Electric blind plate valve of nitrogen: An electrically controlled blind plate valve installed on large-diameter nitrogen pipelines. It has excellent sealing performance and is mainly used for thorough pipeline isolation during major equipment maintenance of the nitrogen system to prevent medium leakage and ensure maintenance safety.

3.4.5 Control valve of nitrogen: An electric/pneumatic control valve installed in the nitrogen pipeline. It has precise pressure and flow regulation functions and is the core valve for precise control of nitrogen system pressure and flow, ensuring stable medium parameters at each process point.

3.4.6 Start ejector valve: A special valve that controls the start of the gas ejector. When opened, it provides power medium for the ejector, triggering the suction function of the ejector to achieve gas pressurization and transportation.
3.4.7 Blind plate valve on coal gas outlet: A blind plate valve installed on the outlet pipeline of the gas dedusting system. It is used for thorough isolation of the gas pipeline during equipment maintenance of the dedusting system to prevent gas leakage and safety accidents.
3.4.8 Butterfly valve on coal gas outlet: A butterfly valve installed on the outlet pipeline of the gas dedusting system. It is used for on/off control and coarse flow regulation of the gas pipeline, suitable for large-diameter gas pipeline conditions, with simple operation.
3.4.9 Butterfly valve on coal gas inlet: A butterfly valve installed on the inlet pipeline of the gas dedusting system. It is used for on/off control and coarse flow regulation of the raw gas pipeline, and can adjust the gas inlet volume according to the dedusting system load.
3.4.10 Electric blind plate valve of ejector: An electric blind plate valve installed in the pipeline of the gas ejector. It is used for thorough pipeline isolation during maintenance of the ejector equipment, achieving medium cut-off and ensuring maintenance safety.
3.4.11 Butterfly valve for reducing pressure of coal gas: A special butterfly valve installed on the clean gas pipeline. It is used to reduce the pressure of the clean gas from the dedusting system outlet to the allowable pressure range for subsequent gas users (such as hot blast stoves).
3.4.12 Temperature of nitrogen outlet: The temperature of nitrogen at the outlet from the main pipe branch to each process point. It reflects the temperature change of nitrogen during pipeline transportation and monitors whether there are problems such as pipeline insulation damage or medium leakage.
3.4.13 Flow rate of nitrogen outlet: The volume of nitrogen flowing from the nitrogen branch pipeline outlet to each process point per unit time. It reflects the actual nitrogen supply to each process point and is used to regulate medium distribution balance.
3.4.14 Pressure of nitrogen outlet: The static pressure of nitrogen at the outlet from the main pipe branch to each process point. It reflects the nitrogen supply pressure status at each process point and must be maintained within the preset range to ensure normal operation of equipment.
3.4.15 Temperature of coal gas before ejector: The temperature of blast furnace gas at the inlet of the ejector. Temperature fluctuation affects the suction capacity, gas mixing effect, and pressurization efficiency of the ejector, and is an important monitoring indicator for ejector operation.
3.4.16 Pressure of main coal gas pipe: The static pressure of the main distribution network for clean blast furnace gas. It reflects the pressure stability of the entire gas transportation system and is a basic parameter to ensure normal gas consumption by all gas users.
3.4.17 Pressure of clean coal gas: The static pressure of clean gas at the outlet of the blast furnace dedusting system. It is the basic source of pressure for the gas pipeline network and directly determines the subsequent gas transportation pressure conditions.
3.4.18 Pressure of ejector: The static pressure inside the cavity of the gas ejector during operation. It reflects the pressurization effect of the ejector and is a core parameter for evaluating the working efficiency of the ejector.
3.4.19 Flow rate of ejector: The volume of gas flowing through the gas ejector per unit time. It reflects the gas conveying capacity of the ejector and must match the blast furnace gas output.
3.4.20 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Gas system - Purifier - Operation monitoring”.

3.5 Furnace top system - Dust recovery tank - Operation monitoring

3.5.1 Blind plate valve for equalizing pressure blow-off: A blind plate valve installed on the equalizing pressure and blow-off pipeline of the blast furnace. It is used for thorough pipeline isolation during maintenance of the equalizing pressure system equipment, preventing medium leakage and ensuring maintenance safety.

3.5.2 Blow-off valve of main equalizing pressure pipe: A blow-off valve installed on the main equalizing pressure pipe. When the pressure in the main equalizing pressure pipe exceeds a preset threshold, it automatically opens to release excess pressure, ensuring stable pressure operation of the equalizing pressure system.

3.5.3 Upper blind plate valve of 1# equalizing pressure pipeline: A blind plate valve installed upstream of No. 1 equalizing pressure pipeline. It is used for upstream pipeline isolation during independent maintenance of No. 1 equalizing pressure pipeline without affecting the normal operation of other equalizing pressure pipelines.

3.5.4 Lower blind plate valve of 1# equalizing pressure pipeline: A blind plate valve installed downstream of No. 1 equalizing pressure pipeline. It is used for downstream pipeline isolation during independent maintenance of No. 1 equalizing pressure pipeline, achieving thorough cut-off of a single pipeline.

3.5.5 Upper blind plate valve of 2# equalizing pressure pipeline: A blind plate valve installed upstream of No. 2 equalizing pressure pipeline. It is used for upstream pipeline isolation during independent maintenance of No. 2 equalizing pressure pipeline without affecting the normal operation of other equalizing pressure pipelines.

3.5.6 Lower blind plate valve of 2# equalizing pressure pipeline: A blind plate valve installed downstream of No. 2 equalizing pressure pipeline. It is used for downstream pipeline isolation during independent maintenance of No. 2 equalizing pressure pipeline, achieving thorough cut-off of a single pipeline.

3.5.7 Upper blind plate valve for recovering and relieving pressure: A blind plate valve installed upstream of the gas recovery and pressure relief pipeline. It is used for upstream pipeline isolation during maintenance of the recovery and pressure relief system to prevent gas leakage.

3.5.8 Lower blind plate valve for recovering and relieving pressure: A blind plate valve installed downstream of the gas recovery and pressure relief pipeline. It is used for downstream pipeline isolation during maintenance of the recovery and pressure relief system, achieving thorough system cut-off.

3.5.9 Pressure control valve of steam: A control valve installed on the steam pipeline of the blast furnace dedusting/equalizing pressure system. It is used for precise regulation of steam pipeline pressure to ensure stable steam medium pressure and adapt to process requirements.

3.5.10 Dust discharge valve: A special valve installed at the discharge outlet of the large ash hopper and dust removal box. It is used to control dust discharge and pipeline on/off. High sealing performance is required to prevent dust leakage and gas escape.

3.5.11 Flow control valve of secondary equalizing pressure: A flow control valve installed on the secondary equalizing pressure pipeline of the blast furnace equalizing pressure system. It is used for precise regulation of the secondary equalizing pressure medium flow to ensure the blast furnace top equalizing effect and stabilize furnace conditions.

3.5.12 Top temperature of box: The medium temperature at the top of each box of the BFG bag filter. It reflects the temperature distribution state in the upper part of the box. Compared with the center temperature of the box, it helps determine whether the temperature inside the box is uniform and indicates problems such as filter bag damage or gas short circuit.

3.5.13 Center temperature of box: The medium temperature at the center of each box of the BFG bag filter. It reflects the core temperature distribution state of the box and is the core reference value for temperature monitoring of the box, ensuring that the equipment operates within a safe temperature range.
3.5.14 Pressure of tank for discharging dust and relieving pressure: The static pressure inside the pressure relief tank supporting the blast furnace ash hopper discharge. It is used to smoothly release the pressure inside the ash hopper during the discharge process, balancing the pressure between the ash hopper and the external environment to prevent dust splashing and gas escape during discharge.
3.5.15 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Dust recovery tank - Operation monitoring”.

3.6 Furnace top system - Ash discharge valve group - Mode control

3.6.1 Timed dust discharge mode: A conventional automatic operation mode for blast furnace ash hopper discharge. The system automatically triggers a series of discharge processes such as opening the discharge valve and interlocking pneumatic ash conveying at preset fixed time intervals, requiring no manual intervention. It is suitable for stable blast furnace production with uniform dust generation and regular ash hopper level changes, ensuring that the ash hopper level remains within the process safety range.
3.6.2 Temperature difference dust discharge mode: An intelligent automatic ash discharge mode based on temperature monitoring. It indirectly determines the amount of dust accumulation in the hopper by detecting the temperature difference between different measurement points (top/bottom, wall/center, etc.) inside the ash hopper (there is a significant temperature difference between the dust accumulation area and the empty area). When the temperature difference reaches the process preset threshold, the system automatically starts the discharge operation. It is suitable for conditions where direct monitoring of the ash hopper level is limited or where strong dust adhesion leads to inaccurate level detection.
3.6.3 Manual dust discharge mode: A manual operation mode for blast furnace ash hopper discharge. The on-site operator manually triggers the discharge process through the central control operation terminal or on-site push button, and can start and stop the discharge as needed. It is mainly used for system commissioning, automatic discharge mode failure, temporary abnormal ash hopper level, and other non-routine scenarios.
3.6.4 Emergency manual dust discharge mode: An emergency priority ash discharge mode for the blast furnace ash hopper, with the highest priority control method. It is suitable for emergency scenarios such as sudden blast furnace faults (e.g., sudden full ash hopper level, stuck discharge valve, interlock failure of the automatic control system). After the operator initiates it, the system skips conventional process interlock checks and directly executes the discharge action to quickly reduce the ash hopper level, preventing process safety accidents such as ash hopper blockage and overflow.
3.6.5 Automatic dust discharge mode: The mainstream intelligent operation mode for the blast furnace ash hopper. The system integrates multiple process triggering conditions such as timed and temperature difference, automatically determines and selects the optimal discharge timing and duration according to the actual blast furnace production conditions (gas load, dust output, ash hopper temperature/level changes). It operates without manual 值守, adapts to complex conditions with fluctuating blast furnace load and dust output, and balances discharge efficiency, system energy consumption, and equipment operation safety. It is the preferred mode for ash hopper discharge during normal blast furnace production.
3.6.6 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace top system - Ash discharge valve group - Mode control”.

3.7 Tapping system - Dust collector - System configuration

3.7.1 Setpoint value for dedusting valve opening of V1-1 side suction main pipe: The preset opening threshold of the dedusting valve for the side suction main pipe in the V1 dedusting branch of the blast furnace casthouse. It is used for precise regulation of the suction volume of the side suction main pipe to match the dust generation in the side suction area of the casthouse.

3.7.2 Setpoint value for dedusting valve opening of V1-2 slag skimmer: The preset opening threshold of the dedusting valve for the slag skimmer station in the V1 dedusting branch of the blast furnace casthouse. It precisely regulates the suction volume of this station to specifically collect dust generated during slag skimmer operation.

3.7.3 Setpoint value for dedusting valve opening of V1-3 slag skimmer at main iron channel: The preset opening threshold of the dedusting valve for the slag skimmer station at the main iron channel in the V1 dedusting branch of the blast furnace casthouse. It matches the dust generation intensity of the main iron channel slag skimmer to ensure dust removal effect at this station.

3.7.4 Setpoint value for dedusting valve opening of V1-4 residual iron channel: The preset opening threshold of the dedusting valve for the residual iron channel station in the V1 dedusting branch of the blast furnace casthouse. It regulates the suction volume of the residual iron channel operation area to collect dust generated at this station.

3.7.5 Setpoint value for dedusting valve opening of V1-5 swinging spout: The preset opening threshold of the dedusting valve for the swinging spout station in the V1 dedusting branch of the blast furnace casthouse. It precisely regulates the suction volume of this station according to the operating state of the swinging spout.

3.7.6 Setpoint value for dedusting valve opening of V1-6 swinging spout at main iron channel: The preset opening threshold of the dedusting valve for the swinging spout station at the main iron channel in the V1 dedusting branch of the blast furnace casthouse. It matches the dust generation of the main iron channel swinging spout to ensure dust removal in the core area of the main iron channel.

3.7.7 Setpoint value for dedusting valve opening of V1-7 swinging spout at main slag channel: The preset opening threshold of the dedusting valve for the swinging spout station at the main slag channel in the V1 dedusting branch of the blast furnace casthouse. It regulates the suction volume of the main slag channel area to collect dust generated during slag trench operation.

3.7.8 Setpoint value for dedusting valve opening of V1-8 top suction swinging spout: The preset opening threshold of the dedusting valve for the top suction swinging spout station in the V1 dedusting branch of the blast furnace casthouse. It collects rising dust generated during swinging spout operation through top suction, adapting to the requirements of the top suction dust removal process.

3.7.9 Setpoint value for dedusting valve opening of V2-1 side suction main pipe: The preset opening threshold of the dedusting valve for the side suction main pipe in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-1, regulating the suction volume of the side suction main pipe of the V2 branch.

3.7.10 Setpoint value for dedusting valve opening of V2-2 slag skimmer: The preset opening threshold of the dedusting valve for the slag skimmer station in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-2, specifically collecting dust from the slag skimmer of the V2 branch.

3.7.11 Setpoint value for dedusting valve opening of V2-3 slag skimmer at main iron channel: The preset opening threshold of the dedusting valve for the slag skimmer station at the main iron channel in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-3, ensuring dust removal effect for the main iron channel slag skimmer of the V2 branch.

3.7.12 Setpoint value for dedusting valve opening of V2-4 residual iron channel: The preset opening threshold of the dedusting valve for the residual iron channel station in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-4, regulating the suction volume of the residual iron channel of the V2 branch.

3.7.13 Setpoint value for dedusting valve opening of V2-5 swinging spout: The preset opening threshold of the dedusting valve for the swinging spout station in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-5, adapting to the operating conditions of the swinging spout of the V2 branch.

3.7.14 Setpoint value for dedusting valve opening of V2-6 swinging spout at main iron channel: The preset opening threshold of the dedusting valve for the swinging spout station at the main iron channel in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-6, ensuring dust removal in the core area of the main iron channel of the V2 branch.

3.7.15 Setpoint value for dedusting valve opening of V2-7 swinging spout at main slag channel: The preset opening threshold of the dedusting valve for the swinging spout station at the main slag channel in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-7, collecting dust from slag trench operation of the V2 branch.

3.7.16 Setpoint value for dedusting valve opening of V2-8 top suction swinging spout: The preset opening threshold of the dedusting valve for the top suction swinging spout station in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-8, collecting rising dust from the swinging spout of the V2 branch through top suction.

3.7.17 Setpoint value for dedusting valve opening of V3-1 side suction main pipe: The preset opening threshold of the dedusting valve for the side suction main pipe in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-1, regulating the suction volume of the side suction main pipe of the V3 branch.

3.7.18 Setpoint value for dedusting valve opening of V3-2 slag skimmer: The preset opening threshold of the dedusting valve for the slag skimmer station in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-2, collecting dust from the slag skimmer of the V3 branch.

3.7.19 Setpoint value for dedusting valve opening of V3-3 slag skimmer at main iron channel: The preset opening threshold of the dedusting valve for the slag skimmer station at the main iron channel in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-3, ensuring dust removal effect for the main iron channel slag skimmer of the V3 branch.

3.7.20 Setpoint value for dedusting valve opening of V3-4 residual iron channel: The preset opening threshold of the dedusting valve for the residual iron channel station in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-4, regulating the suction volume of the residual iron channel of the V3 branch.

3.7.21 Setpoint value for dedusting valve opening of V3-5 swinging spout: The preset opening threshold of the dedusting valve for the swinging spout station in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-5, adapting to the operating state of the swinging spout of the V3 branch.

3.7.22 Setpoint value for dedusting valve opening of V3-6 swinging spout at main iron channel: The preset opening threshold of the dedusting valve for the swinging spout station at the main iron channel in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-6, ensuring dust removal in the core area of the main iron channel of the V3 branch.

3.7.23 Setpoint value for dedusting valve opening of V3-7 swinging spout at main slag channel: The preset opening threshold of the dedusting valve for the swinging spout station at the main slag channel in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-7, collecting dust from slag trench operation of the V3 branch.

3.7.24 Setpoint value for dedusting valve opening of V3-8 top suction swinging spout: The preset opening threshold of the dedusting valve for the top suction swinging spout station in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-8, collecting rising dust from the swinging spout of the V3 branch through top suction.
3.7.25 Setpoint value for dedusting valve opening of V4-1 side suction main pipe: The preset opening threshold of the dedusting valve for the side suction main pipe in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-1, regulating the suction volume of the side suction main pipe of the V4 branch.
3.7.26 Setpoint value for dedusting valve opening of V4-2 slag skimmer: The preset opening threshold of the dedusting valve for the slag skimmer station in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-2, collecting dust from the slag skimmer of the V4 branch.
3.7.27 Setpoint value for dedusting valve opening of V4-3 slag skimmer at main iron channel: The preset opening threshold of the dedusting valve for the slag skimmer station at the main iron channel in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-3, ensuring dust removal effect for the main iron channel slag skimmer of the V4 branch.
3.7.28 Setpoint value for dedusting valve opening of V4-4 residual iron channel: The preset opening threshold of the dedusting valve for the residual iron channel station in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-4, regulating the suction volume of the residual iron channel of the V4 branch.
3.7.29 Setpoint value for dedusting valve opening of V4-5 swinging spout: The preset opening threshold of the dedusting valve for the swinging spout station in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-5, adapting to the operating state of the swinging spout of the V4 branch.
3.7.30 Setpoint value for dedusting valve opening of V4-6 swinging spout at main iron channel: The preset opening threshold of the dedusting valve for the swinging spout station at the main iron channel in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-6, ensuring dust removal in the core area of the main iron channel of the V4 branch.
3.7.31 Setpoint value for dedusting valve opening of V4-7 swinging spout at main slag channel: The preset opening threshold of the dedusting valve for the swinging spout station at the main slag channel in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-7, collecting dust from slag trench operation of the V4 branch.
3.7.32 Setpoint value for dedusting valve opening of V4-8 top suction swinging spout: The preset opening threshold of the dedusting valve for the top suction swinging spout station in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-8, collecting rising dust from the swinging spout of the V4 branch through top suction.
3.7.33 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Tapping system - Dust collector - System configuration”.

3.8 Tapping system - Dust collector - Operation monitoring

3.8.1 Feedback value for dedusting valve opening of V1-1 side suction main pipe: The actual opening feedback value of the dedusting valve for the side suction main pipe in the V1 dedusting branch of the blast furnace casthouse. Compared with the setpoint, it determines whether the valve has moved in place and whether there are problems such as jamming or failure.
3.8.2 Feedback value for dedusting valve opening of V1-2 slag skimmer: The actual opening feedback value

of the dedusting valve for the slag skimmer station in the V1 dedusting branch of the blast furnace casthouse. Compared with the setpoint, it monitors the working status of this valve to ensure the dust removal effect at this station.

3.8.3 Feedback value for dedusting valve opening of V1-3 slag skimmer at main iron channel: The actual opening feedback value of the dedusting valve for the slag skimmer station at the main iron channel in the V1 dedusting branch of the blast furnace casthouse. Compared with the setpoint, it timely detects valve failures to ensure dust removal at the core main iron channel station.

3.8.4 Feedback value for dedusting valve opening of V1-4 residual iron channel: The actual opening feedback value of the dedusting valve for the residual iron channel station in the V1 dedusting branch of the blast furnace casthouse. Compared with the setpoint, it monitors the working status of the valve to ensure that the suction volume at this station meets the standard.

3.8.5 Feedback value for dedusting valve opening of V1-5 swinging spout: The actual opening feedback value of the dedusting valve for the swinging spout station in the V1 dedusting branch of the blast furnace casthouse. Compared with the setpoint, it adapts to the operating state of the swinging spout to ensure the dust removal effect.

3.8.6 Feedback value for dedusting valve opening of V1-6 swinging spout at main iron channel: The actual opening feedback value of the dedusting valve for the swinging spout station at the main iron channel in the V1 dedusting branch of the blast furnace casthouse. Compared with the setpoint, it ensures precise regulation of the suction volume in the core area of the main iron channel.

3.8.7 Feedback value for dedusting valve opening of V1-7 swinging spout at main slag channel: The actual opening feedback value of the dedusting valve for the swinging spout station at the main slag channel in the V1 dedusting branch of the blast furnace casthouse. Compared with the setpoint, it monitors the valve status to collect dust from slag trench operation.

3.8.8 Feedback value for dedusting valve opening of V1-8 top suction swinging spout: The actual opening feedback value of the dedusting valve for the top suction swinging spout station in the V1 dedusting branch of the blast furnace casthouse. Compared with the setpoint, it ensures that the suction volume of the top suction meets the standard to collect rising dust.

3.8.9 Feedback value for dedusting valve opening of V2-1 side suction main pipe: The actual opening feedback value of the dedusting valve for the side suction main pipe in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-1, monitoring the working status of this valve in the V2 branch.

3.8.10 Feedback value for dedusting valve opening of V2-2 slag skimmer: The actual opening feedback value of the dedusting valve for the slag skimmer station in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-2, ensuring the dust removal effect at this station in the V2 branch.

3.8.11 Feedback value for dedusting valve opening of V2-3 slag skimmer at main iron channel: The actual opening feedback value of the dedusting valve for the slag skimmer station at the main iron channel in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-3, ensuring dust removal at the core main iron channel station of the V2 branch.

3.8.12 Feedback value for dedusting valve opening of V2-4 residual iron channel: The actual opening feedback value of the dedusting valve for the residual iron channel station in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-4, monitoring the working status of this valve in the V2 branch.

3.8.13 Feedback value for dedusting valve opening of V2-5 swinging spout: The actual opening feedback value of the dedusting valve for the swinging spout station in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-5, adapting to the operating conditions of the swinging spout in the V2 branch.

3.8.14 Feedback value for dedusting valve opening of V2-6 swinging spout at main iron channel: The actual opening feedback value of the dedusting valve for the swinging spout station at the main iron channel in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-6, ensuring dust removal in the core area of the main iron channel of the V2 branch.

3.8.15 Feedback value for dedusting valve opening of V2-7 swinging spout at main slag channel: The actual opening feedback value of the dedusting valve for the swinging spout station at the main slag channel in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-7, collecting dust from slag trench operation of the V2 branch.

3.8.16 Feedback value for dedusting valve opening of V2-8 top suction swinging spout: The actual opening feedback value of the dedusting valve for the top suction swinging spout station in the V2 dedusting branch of the blast furnace casthouse. Its function is the same as V1-8, ensuring the top suction dust removal effect of the V2 branch.

3.8.17 Feedback value for dedusting valve opening of V3-1 side suction main pipe: The actual opening feedback value of the dedusting valve for the side suction main pipe in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-1, monitoring the working status of this valve in the V3 branch.

3.8.18 Feedback value for dedusting valve opening of V3-2 slag skimmer: The actual opening feedback value of the dedusting valve for the slag skimmer station in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-2, ensuring the dust removal effect at this station in the V3 branch.

3.8.19 Feedback value for dedusting valve opening of V3-3 slag skimmer at main iron channel: The actual opening feedback value of the dedusting valve for the slag skimmer station at the main iron channel in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-3, ensuring dust removal at the core main iron channel station of the V3 branch.

3.8.20 Feedback value for dedusting valve opening of V3-4 residual iron channel: The actual opening feedback value of the dedusting valve for the residual iron channel station in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-4, monitoring the working status of this valve in the V3 branch.

3.8.21 Feedback value for dedusting valve opening of V3-5 swinging spout: The actual opening feedback value of the dedusting valve for the swinging spout station in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-5, adapting to the operating conditions of the swinging spout in the V3 branch.

3.8.22 Feedback value for dedusting valve opening of V3-6 swinging spout at main iron channel: The actual opening feedback value of the dedusting valve for the swinging spout station at the main iron channel in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-6, ensuring dust removal in the core area of the main iron channel of the V3 branch.

3.8.23 Feedback value for dedusting valve opening of V3-7 swinging spout at main slag channel: The actual opening feedback value of the dedusting valve for the swinging spout station at the main slag channel in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-7, collecting dust from slag trench operation of the V3 branch.

3.8.24 Feedback value for dedusting valve opening of V3-8 top suction swinging spout: The actual opening feedback value of the dedusting valve for the top suction swinging spout station in the V3 dedusting branch of the blast furnace casthouse. Its function is the same as V1-8, ensuring the top suction dust removal effect

of the V3 branch.
3.8.25 Feedback value for dedusting valve opening of V4-1 side suction main pipe: The actual opening feedback value of the dedusting valve for the side suction main pipe in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-1, monitoring the working status of this valve in the V4 branch.
3.8.26 Feedback value for dedusting valve opening of V4-2 slag skimmer: The actual opening feedback value of the dedusting valve for the slag skimmer station in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-2, ensuring the dust removal effect at this station in the V4 branch.
3.8.27 Feedback value for dedusting valve opening of V4-3 slag skimmer at main iron channel: The actual opening feedback value of the dedusting valve for the slag skimmer station at the main iron channel in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-3, ensuring dust removal at the core main iron channel station of the V4 branch.
3.8.28 Feedback value for dedusting valve opening of V4-4 residual iron channel: The actual opening feedback value of the dedusting valve for the residual iron channel station in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-4, monitoring the working status of this valve in the V4 branch.
3.8.29 Feedback value for dedusting valve opening of V4-5 swinging spout: The actual opening feedback value of the dedusting valve for the swinging spout station in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-5, adapting to the operating conditions of the swinging spout in the V4 branch.
3.8.30 Feedback value for dedusting valve opening of V4-6 swinging spout at main iron channel: The actual opening feedback value of the dedusting valve for the swinging spout station at the main iron channel in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-6, ensuring dust removal in the core area of the main iron channel of the V4 branch.
3.8.31 Feedback value for dedusting valve opening of V4-7 swinging spout at main slag channel: The actual opening feedback value of the dedusting valve for the swinging spout station at the main slag channel in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-7, collecting dust from slag trench operation of the V4 branch.
3.8.32 Feedback value for dedusting valve opening of V4-8 top suction swinging spout: The actual opening feedback value of the dedusting valve for the top suction swinging spout station in the V4 dedusting branch of the blast furnace casthouse. Its function is the same as V1-8, ensuring the top suction dust removal effect of the V4 branch.
3.8.33 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Tapping system - Dust collector - Operation monitoring”.

4 Cooling monitoring

4.1 Soft water circulation system - Flow monitoring instrument - Flow difference monitoring

4.1.1 Zone 1 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 1 of the blast furnace cooling system (a specific area divided by cooling zone, such as a section of the bosh or belly). It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.

4.1.2 Zone 2 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 2 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.3 Zone 3 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 3 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.4 Zone 4 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 4 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.5 Zone 5 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 5 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.6 Zone 6 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 6 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.7 Zone 7 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 7 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.8 Zone 8 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 8 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.9 Zone 9 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 9 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.10 Zone 10 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 10 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.11 Zone 11 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 11 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.12 Zone 12 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 12 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.

4.1.13 Zone 13 flow difference: The difference between the "design flow rate and actual operating flow rate" or "inlet flow rate and return flow rate" of the cooling medium (water) in Zone 13 of the blast furnace cooling system. It is used to determine whether there is blockage or leakage in the cooling pipeline of this zone, ensuring that the cooling intensity of the zone meets the standard.
4.1.14 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Soft water circulation system - Flow monitoring instrument - Flow difference monitoring".

4.2 Soft water circulation system - Cooling stave - Operation monitoring

4.2.1 Flow rate of return water branch pipe in zone 1: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 1 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 1. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.2 Pressure of return water branch pipe in zone 1: The static pressure of the cooling medium in the return water branch pipe of cooling zone 1 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 1. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.3 Flow rate of return water branch pipe in zone 2: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 2 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 2. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.4 Pressure of return water branch pipe in zone 2: The static pressure of the cooling medium in the return water branch pipe of cooling zone 2 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 2. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.5 Flow rate of return water branch pipe in zone 3: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 3 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 3. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.6 Pressure of return water branch pipe in zone 3: The static pressure of the cooling medium in the return water branch pipe of cooling zone 3 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 3. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.7 Flow rate of return water branch pipe in zone 4: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 4 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 4. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.8 Pressure of return water branch pipe in zone 4: The static pressure of the cooling medium in the return water branch pipe of cooling zone 4 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 4. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.9 Flow rate of return water branch pipe in zone 5: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 5 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 5. Compared with the inlet flow rate, it can determine whether there is

blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.10 Pressure of return water branch pipe in zone 5: The static pressure of the cooling medium in the return water branch pipe of cooling zone 5 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 5. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.11 Flow rate of return water branch pipe in zone 6: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 6 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 6. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.12 Pressure of return water branch pipe in zone 6: The static pressure of the cooling medium in the return water branch pipe of cooling zone 6 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 6. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.13 Flow rate of return water branch pipe in zone 7: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 7 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 7. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.14 Pressure of return water branch pipe in zone 7: The static pressure of the cooling medium in the return water branch pipe of cooling zone 7 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 7. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.15 Flow rate of return water branch pipe in zone 8: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 8 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 8. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.16 Pressure of return water branch pipe in zone 8: The static pressure of the cooling medium in the return water branch pipe of cooling zone 8 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 8. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.17 Flow rate of return water branch pipe in zone 9: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 9 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 9. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.18 Pressure of return water branch pipe in zone 9: The static pressure of the cooling medium in the return water branch pipe of cooling zone 9 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 9. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.
4.2.19 Flow rate of return water branch pipe in zone 10: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 10 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 10. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.
4.2.20 Pressure of return water branch pipe in zone 10: The static pressure of the cooling medium in the return water branch pipe of cooling zone 10 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 10. Abnormal pressure indicates pipeline blockage or return system failure, which

requires timely troubleshooting to avoid affecting the cooling medium circulation.

4.2.21 Flow rate of return water branch pipe in zone 11: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 11 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 11. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.

4.2.22 Pressure of return water branch pipe in zone 11: The static pressure of the cooling medium in the return water branch pipe of cooling zone 11 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 11. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.

4.2.23 Flow rate of return water branch pipe in zone 12: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 12 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 12. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.

4.2.24 Pressure of return water branch pipe in zone 12: The static pressure of the cooling medium in the return water branch pipe of cooling zone 12 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 12. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.

4.2.25 Flow rate of return water branch pipe in zone 13: The volume of cooling medium (water) flowing per unit time in the return water branch pipe of cooling zone 13 of the blast furnace. It reflects the actual return flow of the cooling medium in zone 13. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the pipeline, ensuring cooling balance in the zone.

4.2.26 Pressure of return water branch pipe in zone 13: The static pressure of the cooling medium in the return water branch pipe of cooling zone 13 of the blast furnace. It reflects the resistance state of the return water pipeline in zone 13. Abnormal pressure indicates pipeline blockage or return system failure, which requires timely troubleshooting to avoid affecting the cooling medium circulation.

4.2.27 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Soft water circulation system - Cooling stave - Operation monitoring”.

4.3 Soft water circulation system - Return water pipeline - Operation monitoring

4.3.1 returning per unit time in the No. 1 main return water pipe of the blast furnace. It is a core flow parameter that collects the return medium from multiple cooling zones, reflecting the overall return load of the No. 1 return water system of the blast furnace.

4.3.2 Pressure of 1# main return water pipe: The static pressure of the cooling medium in the No. 1 main return water pipe of the blast furnace. It is the overall pressure reference of the No. 1 return water system. Stable pressure directly determines the medium return efficiency of each branch return pipe. Abnormality indicates blockage in the main pipe or failure of the return pump station.

4.3.3 Temperature of 1# main return water pipe: The temperature of the cooling medium in the No. 1 main return water pipe of the blast furnace. It reflects the overall cooling heat absorption effect of the area covered by the No. 1 return water system. An excessive temperature rise indicates that the cooling load in this area is too high or the cooling medium supply is insufficient.

4.3.4 Flow rate of 2# main return water pipe: The total volume of cooling medium returning per unit time in the No. 2 main return water pipe of the blast furnace. It is a core flow parameter that collects the return medium from multiple cooling zones, reflecting the overall return load of the No. 2 return water system of

the blast furnace.
4.3.5 Pressure of 2# main return water pipe: The static pressure of the cooling medium in the No. 2 main return water pipe of the blast furnace. It is the overall pressure reference of the No. 2 return water system. Stable pressure directly determines the medium return efficiency of each branch return pipe. Abnormality indicates blockage in the main pipe or failure of the return pump station.
4.3.6 Temperature of 2# main return water pipe: The temperature of the cooling medium in the No. 2 main return water pipe of the blast furnace. It reflects the overall cooling heat absorption effect of the area covered by the No. 2 return water system. An excessive temperature rise indicates that the cooling load in this area is too high or the cooling medium supply is insufficient.
4.3.7 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Soft water circulation system - Return water pipeline - Operation monitoring”.

4.4 Soft water circulation system - Water supply pipeline - Flow monitoring

4.4.1 Flow rate of supply water branch pipe in zone 1: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 1 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 1 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.
4.4.2 Flow rate of supply water branch pipe in zone 2: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 2 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 2 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.
4.4.3 Flow rate of supply water branch pipe in zone 3: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 3 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 3 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.
4.4.4 Flow rate of supply water branch pipe in zone 4: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 4 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 4 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.
4.4.5 Flow rate of supply water branch pipe in zone 5: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 5 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 5 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.
4.4.6 Flow rate of supply water branch pipe in zone 6: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 6 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 6 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.
4.4.7 Flow rate of supply water branch pipe in zone 7: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 7 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 7 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.
4.4.8 Flow rate of supply water branch pipe in zone 8: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 8 of the blast furnace. It reflects the actual supply

flow of the cooling medium to zone 8 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.

4.4.9 Flow rate of supply water branch pipe in zone 9: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 9 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 9 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.

4.4.10 Flow rate of supply water branch pipe in zone 10: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 10 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 10 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.

4.4.11 Flow rate of supply water branch pipe in zone 11: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 11 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 11 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.

4.4.12 Flow rate of supply water branch pipe in zone 12: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 12 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 12 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.

4.4.13 Flow rate of supply water branch pipe in zone 13: The volume of cooling medium (water) flowing per unit time in the supply water branch pipe of cooling zone 13 of the blast furnace. It reflects the actual supply flow of the cooling medium to zone 13 and must match the heat dissipation load of this zone. Insufficient flow will lead to a decrease in cooling intensity in this zone, causing local overheating.

4.4.14 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Soft water circulation system - Water supply pipeline - Flow monitoring”.

4.5 Hot blast valve cooling system - Cooling pipeline - Operation monitoring

4.5.1 Water flow rate of inlet about stop-wind valve: The inlet water flow rate of the cooling water circuit of the blast furnace stop-wind valve (a key valve that cuts off the gas path during blast furnace blowing-off). It is a core parameter to ensure the cooling of the valve body. Insufficient flow will cause the valve body to deform due to high temperature and seal failure.

4.5.2 Water flow rate of outlet about stop-wind valve: The return water flow rate of the cooling water circuit of the blast furnace stop-wind valve. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the cooling pipeline of the stop-wind valve, ensuring the integrity of the valve cooling cycle.

4.5.3 Water flow rate of inlet about 1# hot blast valve: The inlet water flow rate of the cooling water circuit of the No. 1 hot blast valve of the blast furnace (a key valve that controls the hot blast from the hot blast stove into the blast furnace). It provides cooling medium for the valve body. The flow rate must meet the heat dissipation requirements of the hot blast valve under high-temperature hot blast environment to prevent valve body damage.

4.5.4 Water flow rate of outlet about 1# hot blast valve: The return water flow rate of the cooling water circuit of the No. 1 hot blast valve of the blast furnace. It reflects the return state of the cooling medium of the hot blast valve. The matching degree with the inlet flow rate directly determines the efficiency of the valve cooling cycle.

4.5.5 Water temperature of outlet about 1# hot blast valve: The return water temperature of the cooling water circuit of the No. 1 hot blast valve of the blast furnace. It reflects the heat absorption and cooling effect of the valve body. An excessive temperature rise indicates insufficient valve cooling, which may be caused by pipeline blockage or insufficient flow.

4.5.6 Water flow rate of inlet about 2# hot blast valve: The inlet water flow rate of the cooling water circuit of the No. 2 hot blast valve of the blast furnace. It provides cooling medium for the valve body. The flow rate must meet the heat dissipation requirements of the hot blast valve under high-temperature hot blast environment to prevent valve body damage.

4.5.7 Water flow rate of outlet about 2# hot blast valve: The return water flow rate of the cooling water circuit of the No. 2 hot blast valve of the blast furnace. It reflects the return state of the cooling medium of the hot blast valve. The matching degree with the inlet flow rate directly determines the efficiency of the valve cooling cycle.

4.5.8 Water temperature of outlet about 2# hot blast valve: The return water temperature of the cooling water circuit of the No. 2 hot blast valve of the blast furnace. It reflects the heat absorption and cooling effect of the valve body. An excessive temperature rise indicates insufficient valve cooling, which may be caused by pipeline blockage or insufficient flow.

4.5.9 Water flow rate of inlet about 3# hot blast valve: The inlet water flow rate of the cooling water circuit of the No. 3 hot blast valve of the blast furnace. It provides cooling medium for the valve body. The flow rate must meet the heat dissipation requirements of the hot blast valve under high-temperature hot blast environment to prevent valve body damage.

4.5.10 Water flow rate of outlet about 3# hot blast valve: The return water flow rate of the cooling water circuit of the No. 3 hot blast valve of the blast furnace. It reflects the return state of the cooling medium of the hot blast valve. The matching degree with the inlet flow rate directly determines the efficiency of the valve cooling cycle.

4.5.11 Water temperature of outlet about 3# hot blast valve: The return water temperature of the cooling water circuit of the No. 3 hot blast valve of the blast furnace. It reflects the heat absorption and cooling effect of the valve body. An excessive temperature rise indicates insufficient valve cooling, which may be caused by pipeline blockage or insufficient flow.

4.5.12 Temperature of main return water pipe: The temperature of the cooling medium in the main return water pipe of the blast furnace cooling system. It is the comprehensive heat absorption effect feedback of the entire blast furnace cooling system. An excessive temperature rise indicates that the overall cooling load of the blast furnace is too high or the total supply of cooling medium is insufficient.

4.5.13 Flow rate of main return water pipe: The total volume of cooling medium returning per unit time in the main return water pipe of the blast furnace cooling system. It reflects the total return load of the entire cooling system. Matching with the total supply water flow determines the efficiency of the overall cooling cycle.

4.5.14 Temperature of main supply water pipe: The temperature of the cooling medium in the main supply water pipe of the blast furnace cooling system. It is the initial temperature reference of the entire cooling system. Too high a temperature will reduce the heat absorption capacity of the cooling medium, directly affecting the cooling effect of the entire furnace.

4.5.15 Pressure of main supply water pipe: The static pressure of the cooling medium in the main supply water pipe of the blast furnace cooling system. It is the pressure core of the cooling medium supply for the entire furnace. Stable pressure directly determines the medium supply efficiency of each branch pipeline and each cooling zone, and is the basic parameter for the normal operation of the cooling system.

4.5.16 Flow rate of main supply water pipe: The total volume of cooling medium delivered per unit time in the main supply water pipe of the blast furnace cooling system. It is the total supply of cooling medium for the entire furnace and must meet the heat dissipation needs of all cooling parts such as the furnace body, tuyeres, and valves.
4.5.17 Pressure of booster pump: The pressure of the cooling medium at the outlet of the booster pump of the blast furnace cooling system. It is a core parameter that provides power for the cooling medium transportation, ensuring that the cooling medium can be smoothly delivered to all cooling parts of the blast furnace. Insufficient pressure will lead to a shortage of cooling medium supply in remote areas.
4.5.18 Flow rate of booster pump: The volume of cooling medium delivered per unit time by the booster pump of the blast furnace cooling system. It reflects the actual liquid supply capacity of the booster pump and must match the total demand for cooling medium of the entire furnace to ensure the medium supply of the cooling system.
4.5.19 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast valve cooling system - Cooling pipeline - Operation monitoring”.

4.6 Hot blast valve cooling system - Cooling pipeline - System configuration

4.6.1 Upper limit setpoint of water flow about main pipe inlet: The process threshold upper limit for the inlet water flow of the main supply water pipe of the blast furnace cooling system. It is a protection parameter to prevent excessively high flow from causing excessive pipeline pressure and valve damage. When the actual flow exceeds the upper limit, the system triggers a warning or interlock regulation.
4.6.2 Lower limit setpoint of water flow about main pipe inlet: The process threshold lower limit for the inlet water flow of the main supply water pipe of the blast furnace cooling system. It is a protection parameter to ensure the basic supply of cooling medium for the entire furnace. When the actual flow falls below the lower limit, a warning is triggered to prevent a decrease in the cooling intensity of the entire furnace due to insufficient total flow.
4.6.3 Upper limit setpoint of return water flow about main pipe: The process threshold upper limit for the return water flow of the main return water pipe of the blast furnace cooling system. It is used to prevent the return water system from operating overload due to excessive return flow. Exceeding the upper limit triggers system interlock to ensure the safety of the return pump station and pipeline.
4.6.4 Lower limit setpoint of return water flow about main pipe: The process threshold lower limit for the return water flow of the main return water pipe of the blast furnace cooling system. Combined with the inlet water flow, it determines whether there is a serious leakage in the entire furnace cooling pipeline. When the actual flow falls below the lower limit, a leakage warning is triggered for timely troubleshooting.
4.6.5 Upper limit setpoint of water pressure about main pipe inlet: The process threshold upper limit for the inlet water pressure of the main supply water pipe of the blast furnace cooling system. It prevents overpressure load from damaging cooling pipelines, valves, and equipment. Exceeding the upper limit triggers booster pump pressure reduction or interlock shutdown.
4.6.6 Lower limit setpoint of water pressure about main pipe inlet: The process threshold lower limit for the inlet water pressure of the main supply water pipe of the blast furnace cooling system. It ensures that the cooling medium can be smoothly delivered to all cooling parts of the blast furnace (especially remote and high-altitude parts). When the pressure falls below the lower limit, it triggers booster pump pressurization or a warning.
4.6.7 Upper limit setpoint of water temperature about main pipe inlet: The process threshold upper limit for

the inlet water temperature of the main supply water pipe of the blast furnace cooling system. It prevents the heat absorption capacity of the cooling medium from decreasing due to excessively high initial temperature, affecting the cooling effect of the entire furnace. Exceeding the upper limit triggers an interlock of the cooling medium cooling system.

4.6.8 Lower limit setpoint of water temperature about main pipe inlet: The process threshold lower limit for the inlet water temperature of the main supply water pipe of the blast furnace cooling system. It avoids condensation or thermal stress on pipelines and equipment due to excessively low cooling medium temperature. Falling below the lower limit triggers medium preheating or a warning.

4.6.9 Lower limit setpoint of water supply flow about stop-wind valve: The process threshold lower limit for the inlet water flow of the cooling water circuit of the blast furnace stop-wind valve. It is a protection parameter to ensure basic cooling of the stop-wind valve body. When the actual flow falls below the lower limit, a warning is triggered to prevent the valve body from overheating and damage due to insufficient cooling.

4.6.10 Lower limit setpoint of return water flow about stop-wind valve: The process threshold lower limit for the return water flow of the cooling water circuit of the blast furnace stop-wind valve. Combined with the inlet water flow, it determines whether the cooling pipeline of the stop-wind valve is blocked. When the actual flow falls below the lower limit, a pipeline fault warning is triggered for timely troubleshooting.

4.6.11 Lower limit setpoint of water supply flow about 1# hot blast valve: The process threshold lower limit for the inlet water flow of the cooling water circuit of the No. 1 hot blast valve of the blast furnace. It ensures the basic cooling supply for the valve body of the No. 1 hot blast valve, preventing deformation and seal failure due to insufficient flow under high-temperature hot blast environment.

4.6.12 Lower limit setpoint of water supply flow about 2# hot blast valve: The process threshold lower limit for the inlet water flow of the cooling water circuit of the No. 2 hot blast valve of the blast furnace. It ensures the basic cooling supply for the valve body of the No. 2 hot blast valve, preventing deformation and seal failure due to insufficient flow under high-temperature hot blast environment.

4.6.13 Lower limit setpoint of water supply flow about 3# hot blast valve: The process threshold lower limit for the inlet water flow of the cooling water circuit of the No. 3 hot blast valve of the blast furnace. It ensures the basic cooling supply for the valve body of the No. 3 hot blast valve, preventing deformation and seal failure due to insufficient flow under high-temperature hot blast environment.

4.6.14 Lower limit setpoint of return water flow about 1# hot blast valve: The process threshold lower limit for the return water flow of the cooling water circuit of the No. 1 hot blast valve of the blast furnace. Combined with the inlet water flow, it determines whether the cooling pipeline of the No. 1 hot blast valve is blocked or leaking. When the flow falls below the lower limit, a pipeline fault warning is triggered.

4.6.15 Lower limit setpoint of return water flow about 2# hot blast valve: The process threshold lower limit for the return water flow of the cooling water circuit of the No. 2 hot blast valve of the blast furnace. Combined with the inlet water flow, it determines whether the cooling pipeline of the No. 2 hot blast valve is blocked or leaking. When the flow falls below the lower limit, a pipeline fault warning is triggered.

4.6.16 Lower limit setpoint of return water flow about 3# hot blast valve: The process threshold lower limit for the return water flow of the cooling water circuit of the No. 3 hot blast valve of the blast furnace. Combined with the inlet water flow, it determines whether the cooling pipeline of the No. 3 hot blast valve is blocked or leaking. When the flow falls below the lower limit, a pipeline fault warning is triggered.

4.6.17 Upper limit setpoint of return water temperature about 1# hot blast valve: The process threshold upper limit for the return water temperature of the cooling water circuit of the No. 1 hot blast valve of the blast furnace. It is a core protection parameter to prevent overheating of the No. 1 hot blast valve body. When the

return water temperature exceeds the upper limit, it triggers inlet water flow regulation or a fault warning.
4.6.18 Upper limit setpoint of return water temperature about 2# hot blast valve: The process threshold upper limit for the return water temperature of the cooling water circuit of the No. 2 hot blast valve of the blast furnace. It is a core protection parameter to prevent overheating of the No. 2 hot blast valve body. When the return water temperature exceeds the upper limit, it triggers inlet water flow regulation or a fault warning.
4.6.19 Upper limit setpoint of return water temperature about 3# hot blast valve: The process threshold upper limit for the return water temperature of the cooling water circuit of the No. 3 hot blast valve of the blast furnace. It is a core protection parameter to prevent overheating of the No. 3 hot blast valve body. When the return water temperature exceeds the upper limit, it triggers inlet water flow regulation or a fault warning.
4.6.20 Upper limit setpoint of pressure about expansion tank: The pressure process threshold upper limit for the expansion tank (which balances cooling medium volume changes and stabilizes system pressure) of the blast furnace cooling system. It prevents overpressure damage to the expansion tank and cooling pipeline. Exceeding the upper limit triggers pressure relief interlock.
4.6.21 Lower limit setpoint of pressure about expansion tank: The pressure process threshold lower limit for the expansion tank of the blast furnace cooling system. It ensures that the expansion tank can normally perform its pressure balancing and replenishment functions. When the pressure falls below the lower limit, it triggers the pressure replenishment system interlock to maintain the cooling system pressure stability.
4.6.22 Upper limit setpoint of temperature about expansion tank: The process threshold upper limit for the temperature of the cooling medium in the expansion tank of the blast furnace cooling system. It prevents an abnormal pressure increase in the expansion tank due to excessive temperature, and also avoids vaporization of the cooling medium due to high temperature, affecting the cooling cycle.
4.6.23 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast valve cooling system - Cooling pipeline - System configuration”.

4.7 Cooling system - Soft water return pipeline - Operation monitoring

4.7.1 Temperature of 1# main pipe: The temperature of the cooling medium in the No. 1 process main pipe of the blast furnace cooling system. It reflects the overall heat absorption state of the cooling area covered by this main pipe and is an important parameter for evaluating the cooling effect of this area.
4.7.2 Flow rate of 1# main pipe: The volume of cooling medium delivered per unit time in the No. 1 process main pipe of the blast furnace cooling system. It is the total supply of cooling medium for the area covered by this main pipe and must meet the heat dissipation needs of all cooling parts in this area.
4.7.3 Pressure of 1# main pipe: The static pressure of the cooling medium in the No. 1 process main pipe of the blast furnace cooling system. It ensures the cooling medium supply efficiency of each branch pipeline in this main pipe. Stable pressure is the basis for the normal operation of the cooling system in this area.
4.7.4 Temperature of 2# main pipe: The temperature of the cooling medium in the No. 2 process main pipe of the blast furnace cooling system. It reflects the overall heat absorption state of the cooling area covered by this main pipe and is an important parameter for evaluating the cooling effect of this area.
4.7.5 Flow rate of 2# main pipe: The volume of cooling medium delivered per unit time in the No. 2 process main pipe of the blast furnace cooling system. It is the total supply of cooling medium for the area covered by this main pipe and must meet the heat dissipation needs of all cooling parts in this area.
4.7.6 Pressure of 2# main pipe: The static pressure of the cooling medium in the No. 2 process main pipe of the blast furnace cooling system. It ensures the cooling medium supply efficiency of each branch pipeline in this main pipe. Stable pressure is the basis for the normal operation of the cooling system in this area.

4.7.7 Pressure of return water ring pipe: The static pressure of the cooling medium in the return water ring pipe (which achieves uniform collection and circulation of return medium) of the blast furnace cooling system. It maintains the return pressure balance of each branch of the return water ring pipe, preventing poor return in some areas due to uneven pressure.

4.7.8 Temperature of return water ring pipe: The temperature of the cooling medium in the return water ring pipe of the blast furnace cooling system. It reflects the overall cooling heat absorption effect of the area covered by the ring pipe and is a comprehensive parameter for evaluating the cooling state of the furnace body in this area.

4.7.9 Pressure of return water pipe about 1# “tuyere – outer”: The static pressure of the cooling medium in the return water pipe of the No. 1 tuyere outer cover (the outermost protective sleeve of the tuyere, resisting high temperature inside the furnace). It reflects the resistance state of the return water pipeline of the No. 1 tuyere outer cover. Abnormal pressure indicates pipeline blockage or return failure.

4.7.10 Flow rate of return water pipe about 1# “tuyere – outer”: The volume of cooling medium returning per unit time in the return water pipe of the No. 1 tuyere outer cover. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the cooling pipeline of the No. 1 tuyere outer cover, ensuring the stability of the outer cover cooling cycle.

4.7.11 Temperature of return water pipe about 1# “tuyere – outer”: The temperature of the cooling medium in the return water pipe of the No. 1 tuyere outer cover. It reflects the heat absorption and cooling effect of the No. 1 tuyere outer cover. An excessive temperature rise indicates possible wear or insufficient cooling of the outer cover, requiring timely troubleshooting.

4.7.12 Pressure of return water pipe about 2# “tuyere – outer”: The static pressure of the cooling medium in the return water pipe of the No. 2 tuyere outer cover. It reflects the resistance state of the return water pipeline of the No. 2 tuyere outer cover. Abnormal pressure indicates pipeline blockage or return failure.

4.7.13 Flow rate of return water pipe about 2# “tuyere – outer”: The volume of cooling medium returning per unit time in the return water pipe of the No. 2 tuyere outer cover. Compared with the inlet flow rate, it can determine whether there is blockage or leakage in the cooling pipeline of the No. 2 tuyere outer cover, ensuring the stability of the outer cover cooling cycle.

4.7.14 Temperature of return water pipe about 2# “tuyere – outer”: The temperature of the cooling medium in the return water pipe of the No. 2 tuyere outer cover. It reflects the heat absorption and cooling effect of the No. 2 tuyere outer cover. An excessive temperature rise indicates possible wear or insufficient cooling of the outer cover, requiring timely troubleshooting.

4.7.15 Pressure of return water pipe about 1# “tuyere – midrange”: The static pressure of the cooling medium in the return water pipe of the No. 1 tuyere midrange (the middle layer structure of the tuyere, assisting cooling and fixing the tuyere). It reflects the resistance state of the return water pipeline of the No. 1 tuyere midrange. Abnormal pressure indicates pipeline blockage or return failure.

4.7.16 Flow rate of return water pipe about 1# “tuyere – midrange”: The volume of cooling medium returning per unit time in the return water pipe of the No. 1 tuyere midrange. Compared with the inlet flow rate, it can determine whether there is a fault in the cooling pipeline of the No. 1 tuyere midrange, ensuring the stability of the midrange cooling cycle.

4.7.17 Temperature of return water pipe about 1# “tuyere – midrange”: The temperature of the cooling medium in the return water pipe of the No. 1 tuyere midrange. It reflects the heat absorption and cooling state of the No. 1 tuyere midrange. An excessive temperature rise indicates insufficient cooling of the midrange, which may lead to structural damage of the entire tuyere.

4.7.18 Pressure of return water pipe about 2# “tuyere – midrange”: The static pressure of the cooling medium in the return water pipe of the No. 2 tuyere midrange. It reflects the resistance state of the return water pipeline of the No. 2 tuyere midrange. Abnormal pressure indicates pipeline blockage or return failure.

4.7.19 Flow rate of return water pipe about 2# “tuyere – midrange”: The volume of cooling medium returning per unit time in the return water pipe of the No. 2 tuyere midrange. Compared with the inlet flow rate, it can determine whether there is a fault in the cooling pipeline of the No. 2 tuyere midrange, ensuring the stability of the midrange cooling cycle.

4.7.20 Temperature of return water pipe about 2# “tuyere – midrange”: The temperature of the cooling medium in the return water pipe of the No. 2 tuyere midrange. It reflects the heat absorption and cooling state of the No. 2 tuyere midrange. An excessive temperature rise indicates insufficient cooling of the midrange, which may lead to structural damage of the entire tuyere.

4.7.21 Pressure of return water pipe about 3# “tuyere – midrange”: The static pressure of the cooling medium in the return water pipe of the No. 3 tuyere midrange. It reflects the resistance state of the return water pipeline of the No. 3 tuyere midrange. Abnormal pressure indicates pipeline blockage or return failure.

4.7.22 Flow rate of return water pipe about 3# “tuyere – midrange”: The volume of cooling medium returning per unit time in the return water pipe of the No. 3 tuyere midrange. Compared with the inlet flow rate, it can determine whether there is a fault in the cooling pipeline of the No. 3 tuyere midrange, ensuring the stability of the midrange cooling cycle.

4.7.23 Temperature of return water pipe about 3# “tuyere – midrange”: The temperature of the cooling medium in the return water pipe of the No. 3 tuyere midrange. It reflects the heat absorption and cooling state of the No. 3 tuyere midrange. An excessive temperature rise indicates insufficient cooling of the midrange, which may lead to structural damage of the entire tuyere.

4.7.24 Pressure of return water pipe about 4# “tuyere – midrange”: The static pressure of the cooling medium in the return water pipe of the No. 4 tuyere midrange. It reflects the resistance state of the return water pipeline of the No. 4 tuyere midrange. Abnormal pressure indicates pipeline blockage or return failure.

4.7.25 Flow rate of return water pipe about 4# “tuyere – midrange”: The volume of cooling medium returning per unit time in the return water pipe of the No. 4 tuyere midrange. Compared with the inlet flow rate, it can determine whether there is a fault in the cooling pipeline of the No. 4 tuyere midrange, ensuring the stability of the midrange cooling cycle.

4.7.26 Temperature of return water pipe about 4# “tuyere – midrange”: The temperature of the cooling medium in the return water pipe of the No. 4 tuyere midrange. It reflects the heat absorption and cooling state of the No. 4 tuyere midrange. An excessive temperature rise indicates insufficient cooling of the midrange, which may lead to structural damage of the entire tuyere.

4.7.27 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Cooling system - Soft water return pipeline - Operation monitoring”.

4.8 Cooling system - Soft water supply pipeline - Operation monitoring

4.8.1 Temperature of 1# main pipe: The real-time temperature of the cooling medium (water) in the core process No. 1 main pipe of the blast furnace cooling system. It is the initial temperature reference of the cooling medium for the cooling area covered by this main pipe (such as a part of the furnace body or a tuyere group), directly affecting the cooling heat absorption efficiency of the corresponding area. Too high a temperature will reduce the cooling effect and must be controlled within the process preset range.

4.8.2 Flow rate of 1# main pipe: The volume of cooling medium delivered per unit time in the No. 1 main pipe of the blast furnace cooling system. It is the total supply of cooling medium for the cooling area covered

by this main pipe and must precisely match the total heat dissipation load of the furnace body and equipment in this area. Insufficient flow will lead to a decrease in the overall cooling intensity of the corresponding area, causing local overheating.
4.8.3 Temperature of 2# main pipe: The real-time temperature of the cooling medium (water) in the core process No. 2 main pipe of the blast furnace cooling system. It is the initial temperature reference of the cooling medium for the cooling area covered by this main pipe (such as a part of the furnace body or a tuyere group), directly affecting the cooling heat absorption efficiency of the corresponding area. Too high a temperature will reduce the cooling effect and must be controlled within the process preset range.
4.8.4 Flow rate of 2# main pipe: The volume of cooling medium delivered per unit time in the No. 2 main pipe of the blast furnace cooling system. It is the total supply of cooling medium for the cooling area covered by this main pipe and must precisely match the total heat dissipation load of the furnace body and equipment in this area. Insufficient flow will lead to a decrease in the overall cooling intensity of the corresponding area, causing local overheating.
4.8.5 Pressure of furnace front water supply pipe: The static pressure of the cooling/replenishment medium in the furnace front water supply pipeline of the blast furnace casthouse. It is the pressure basis for ensuring furnace front water replenishment (iron trench/slag trench cooling, slag flushing, emergency cooling). It must be maintained at the process set value. Insufficient pressure will lead to poor water replenishment transportation, affecting the cooling and slag flushing effect of the furnace front operation and increasing safety risks.
4.8.6 Flow rate of furnace front water supply pipe: The volume of water replenishment per unit time in the furnace front water supply pipeline of the blast furnace casthouse. It must precisely match the tapping rhythm, the heating state of the iron trench/slag trench, and the slag flushing demand. It is a key parameter to ensure the normal progress of furnace front iron trench/slag trench cooling and slag flushing operations. Insufficient flow can easily cause iron trench crusting and slag trench blockage.
4.8.7 Pressure of supply water ring pipe: The static pressure of the cooling medium in the supply water ring pipe (the core pipeline for uniform distribution of cooling medium to the entire furnace) of the blast furnace cooling system. It is the pressure reference for all cooling branch pipelines of the entire furnace, directly determining the pressure balance of cooling medium supply to various areas and equipment of the furnace body. Pressure abnormality will lead to uneven flow distribution, causing local insufficient cooling.
4.8.8 Temperature of supply water ring pipe: The real-time temperature of the cooling medium in the supply water ring pipe of the blast furnace cooling system. It is the unified initial temperature reference of the cooling medium for the entire furnace. Stable temperature is a prerequisite for ensuring consistent heat absorption efficiency of all cooling areas of the furnace. Too high a temperature will overall reduce the heat dissipation capacity of the cooling system, while too low a temperature is prone to pipeline thermal stress.
4.8.9 Pressure of furnace bottom water-cooled pipe: The static pressure of the cooling medium in the furnace bottom water-cooled pipe (special cooling pipeline for the furnace bottom, preventing high-temperature erosion of the furnace bottom and extending the furnace life) of the blast furnace. It reflects the resistance state of the furnace bottom water-cooled pipeline. Too high a pressure indicates pipeline blockage, and too low a pressure indicates leakage. It must be maintained within the process range to ensure smooth circulation of the cooling medium at the furnace bottom.
4.8.10 Flow rate of furnace bottom water-cooled pipe: The volume of cooling medium delivered per unit time in the furnace bottom water-cooled pipe of the blast furnace. It is a core parameter to meet the heat dissipation needs of the furnace bottom refractory material. The furnace bottom is one of the high-temperature core areas of the blast furnace. Insufficient flow will cause the furnace bottom temperature to

rise, accelerating the erosion of refractory materials and seriously affecting the structural safety and service life of the blast furnace body.

4.8.11 Pressure of supply water pipe about 1# “tuyere – outer”: The static pressure of the cooling medium in the supply water pipe of the No. 1 tuyere outer cover. It is the pressure guarantee for the smooth flow of the cooling medium into the outer cover and must match the resistance of the outer cover cooling water circuit. Insufficient pressure will cause poor circulation of the cooling medium, leading to overheating deformation of the outer cover.

4.8.12 Flow rate of supply water pipe about 1# “tuyere – outer”: The volume of cooling medium delivered per unit time in the supply water pipe of the No. 1 tuyere outer cover. It directly determines the cooling intensity of the No. 1 tuyere outer cover and must match the heat load of the outer cover. Insufficient flow will cause the outer cover temperature to exceed the standard, damage the tuyere sealing structure, and affect the stability of the blast furnace air supply.

4.8.13 Temperature of supply water pipe about 1# “tuyere – outer”: The initial temperature of the cooling medium in the supply water pipe of the No. 1 tuyere outer cover. It is the temperature reference for outer cover cooling. Too high a temperature will reduce the heat absorption capacity of the cooling medium, making it unable to effectively 带走 the heat absorbed by the outer cover from the furnace. It must be strictly controlled within the low temperature range of the process to ensure the cooling effect.

4.8.14 Pressure of supply water pipe about 2# “tuyere – outer”: The static pressure of the cooling medium in the supply water pipe of the No. 2 tuyere outer cover. It is the pressure guarantee for the smooth flow of the cooling medium into the outer cover and must match the resistance of the outer cover cooling water circuit. Insufficient pressure will cause poor circulation of the cooling medium, leading to overheating deformation of the outer cover.

4.8.15 Flow rate of supply water pipe about 2# “tuyere – outer”: The volume of cooling medium delivered per unit time in the supply water pipe of the No. 2 tuyere outer cover. It directly determines the cooling intensity of the No. 2 tuyere outer cover and must match the heat load of the outer cover. Insufficient flow will cause the outer cover temperature to exceed the standard, damage the tuyere sealing structure, and affect the stability of the blast furnace air supply.

4.8.16 Temperature of supply water pipe about 2# “tuyere – outer”: The initial temperature of the cooling medium in the supply water pipe of the No. 2 tuyere outer cover. It is the temperature reference for outer cover cooling. Too high a temperature will reduce the heat absorption capacity of the cooling medium, making it unable to effectively 带走 the heat absorbed by the outer cover from the furnace. It must be strictly controlled within the low temperature range of the process to ensure the cooling effect.

4.8.17 Flow rate of 1# bypass pipe: The volume of cooling medium per unit time in the No. 1 bypass pipe (a regulating pipeline supporting the main pipeline) of the blast furnace cooling system. It is mainly used to regulate the pressure and flow balance of the No. 1 main pipe, diverting flow when the main pipe flow is too high or the pressure is too high, or temporarily supplying liquid to key cooling areas when the main pipe fails, ensuring the stability of the cooling system.

4.8.18 Flow rate of 2# bypass pipe: The volume of cooling medium per unit time in the No. 2 bypass pipe of the blast furnace cooling system. It is mainly used to regulate the pressure and flow balance of the No. 2 main pipe, diverting flow when the main pipe flow is too high or the pressure is too high, or temporarily supplying liquid to key cooling areas when the main pipe fails, ensuring the stability of the cooling system.

4.8.19 Flow rate of supply water pipe about 1# “tuyere – midrange”: The volume of cooling medium delivered per unit time in the supply water pipe of the No. 1 tuyere midrange. It is a core parameter for midrange cooling. The midrange connects the outer cover and the small sleeve. Insufficient flow will cause the midrange to overheat, leading to structural loosening and seal failure of the entire tuyere.

4.8.20 Flow rate of supply water pipe about 2# “tuyere – midrange”: The volume of cooling medium delivered per unit time in the supply water pipe of the No. 2 tuyere midrange. It is a core parameter for midrange cooling. The midrange connects the outer cover and the small sleeve. Insufficient flow will cause the midrange to overheat, leading to structural loosening and seal failure of the entire tuyere.

4.8.21 Flow rate of supply water pipe about 3# “tuyere – midrange”: The volume of cooling medium delivered per unit time in the supply water pipe of the No. 3 tuyere midrange. It is a core parameter for midrange cooling. The midrange connects the outer cover and the small sleeve. Insufficient flow will cause the midrange to overheat, leading to structural loosening and seal failure of the entire tuyere.

4.8.22 Flow rate of supply water pipe about 4# “tuyere – midrange”: The volume of cooling medium delivered per unit time in the supply water pipe of the No. 4 tuyere midrange. It is a core parameter for midrange cooling. The midrange connects the outer cover and the small sleeve. Insufficient flow will cause the midrange to overheat, leading to structural loosening and seal failure of the entire tuyere.

4.8.23 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Cooling system - Soft water supply pipeline - Operation monitoring”.

4.9 Cooling system - High-pressure industrial water pipeline - Operation monitoring

4.9.1 Water flow rate of inlet about “1#→8#” tuyere: The total volume of cooling medium per unit time in the main inlet pipe for the small sleeves of tuyeres No. 1 to No. 8. It is the total cooling supply for this group of tuyere small sleeves. The tuyere small sleeve is a key component for blast furnace air supply and must have sufficient flow to prevent the small sleeve from burning out and causing blast furnace blow-off.

4.9.2 Water flow rate of inlet about “9#→16#” tuyere: The total volume of cooling medium per unit time in the main inlet pipe for the small sleeves of tuyeres No. 9 to No. 16. It is the total cooling supply for this group of tuyere small sleeves. The tuyere small sleeve is a key component for blast furnace air supply and must have sufficient flow to prevent the small sleeve from burning out and causing blast furnace blow-off.

4.9.3 Water flow rate of inlet about “17#→24#” tuyere: The total volume of cooling medium per unit time in the main inlet pipe for the small sleeves of tuyeres No. 17 to No. 24. It is the total cooling supply for this group of tuyere small sleeves. The tuyere small sleeve is a key component for blast furnace air supply and must have sufficient flow to prevent the small sleeve from burning out and causing blast furnace blow-off.

4.9.4 Water flow rate of inlet about “25#→32#” tuyere: The total volume of cooling medium per unit time in the main inlet pipe for the small sleeves of tuyeres No. 25 to No. 32. It is the total cooling supply for this group of tuyere small sleeves. The tuyere small sleeve is a key component for blast furnace air supply and must have sufficient flow to prevent the small sleeve from burning out and causing blast furnace blow-off.

4.9.5 Pressure of 1# main water supply pipe: The static pressure of the cooling medium in the No. 1 main water supply pipe (the main supply pipeline for the core cooling area of the entire furnace, such as the lower part of the furnace body and the core tuyere group) of the blast furnace cooling system. It is the pressure core for all cooling branches covered by this main pipe, directly determining the medium supply pressure of the branch pipelines. Stable pressure is the basis for normal cooling of the core area.

4.9.6 Flow rate of 1# main water supply pipe: The total volume of cooling medium delivered per unit time in the No. 1 main water supply pipe of the blast furnace cooling system. It is the total liquid supply for the core cooling area of the entire furnace and must meet the total heat dissipation needs of high-temperature areas such as the lower part of the furnace body and the core tuyere group. It is a key parameter for ensuring the cooling of the core parts of the blast furnace.

4.9.7 Temperature of 1# main water supply pipe: The real-time temperature of the cooling medium in the No. 1 main water supply pipe of the blast furnace cooling system. It is the initial temperature of the cooling medium for the core cooling area of the entire furnace and must be strictly controlled within the low temperature range of the process. Too high a temperature will greatly reduce the cooling heat absorption efficiency of the core area, causing high-temperature failures of the furnace body and tuyeres.
4.9.8 Pressure of 2# main water supply pipe: The static pressure of the cooling medium in the No. 2 main water supply pipe of the blast furnace cooling system. It is the pressure core for all cooling branches covered by this main pipe, directly determining the medium supply pressure of the branch pipelines. Stable pressure is the basis for normal cooling of the core area.
4.9.9 Flow rate of 2# main water supply pipe: The total volume of cooling medium delivered per unit time in the No. 2 main water supply pipe of the blast furnace cooling system. It is the total liquid supply for the core cooling area of the entire furnace and must meet the total heat dissipation needs of high-temperature areas such as the lower part of the furnace body and the core tuyere group. It is a key parameter for ensuring the cooling of the core parts of the blast furnace.
4.9.10 Temperature of 2# main water supply pipe: The real-time temperature of the cooling medium in the No. 2 main water supply pipe of the blast furnace cooling system. It is the initial temperature of the cooling medium for the core cooling area of the entire furnace and must be strictly controlled within the low temperature range of the process. Too high a temperature will greatly reduce the cooling heat absorption efficiency of the core area, causing high-temperature failures of the furnace body and tuyeres.
4.9.11 Temperature of 1# main return water pipe: The real-time temperature in the No. 1 main return water pipe (the main pipe that collects the return cooling medium from the area covered by the No. 1 supply water main pipe) of the blast furnace cooling system. It is the overall cooling effect feedback parameter for the core cooling area. An excessive temperature rise in the return water indicates that the cooling load in the core area is too high or the cooling medium supply is insufficient, requiring timely adjustment of the water supply parameters.
4.9.12 Temperature of 2# main return water pipe: The real-time temperature in the No. 2 main return water pipe of the blast furnace cooling system. It is the overall cooling effect feedback parameter for the core cooling area. An excessive temperature rise in the return water indicates that the cooling load in the core area is too high or the cooling medium supply is insufficient, requiring timely adjustment of the water supply parameters.
4.9.13 Flow rate of cooling water supply pipe about 1# furnace throat steel brick: The inlet water flow rate of the cooling water circuit for the No. 1 furnace throat steel brick (special steel brick that resists furnace charge impact and high-temperature gas erosion at the furnace throat) of the blast furnace. The furnace throat is the intersection area where furnace charge falls and gas rises, and the steel brick is subject to high heat and impact load. Sufficient flow must be ensured to prevent overheating deformation and damage of the steel brick, which would affect the stability of the furnace throat material flow.
4.9.14 Flow rate of cooling water supply pipe about 2# furnace throat steel brick: The inlet water flow rate of the cooling water circuit for the No. 2 furnace throat steel brick of the blast furnace. The furnace throat is the intersection area where furnace charge falls and gas rises, and the steel brick is subject to high heat and impact load. Sufficient flow must be ensured to prevent overheating deformation and damage of the steel brick, which would affect the stability of the furnace throat material flow.
4.9.15 Flow rate of furnace top water injection: The volume of water injected per unit time by the water spray cooling system at the blast furnace top area (furnace top cover, material flow chute, furnace top gas pipeline, etc.). It is a core parameter for precise control of the furnace top temperature. Too high a furnace

top temperature will damage furnace top equipment and affect gas quality. The water injection flow rate must be dynamically adjusted according to the actual furnace top temperature, while avoiding excessive water injection that may cause gas carryover and charge sticking, affecting the smooth operation of the blast furnace.
4.9.16 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Cooling system - High-pressure industrial water pipeline - Operation monitoring”.

5 BF tapping

5.1 Tapping system - Hot metal ladle - Operation monitoring

5.1.1 Temperature of hot metal about 1# taphole: The real-time measured temperature of hot metal during tapping from No. 1 taphole of the blast furnace. It is a core thermal parameter reflecting the thermal state of the blast furnace hearth and the smelting process, directly determining the quality of hot metal and the process suitability for subsequent steelmaking. It must be maintained within a reasonable preset process temperature range.
5.1.2 Temperature of hot metal about 2# taphole: The real-time measured temperature of hot metal during tapping from No. 2 taphole of the blast furnace. It is a core thermal parameter reflecting the thermal state of the blast furnace hearth and the smelting process, directly determining the quality of hot metal and the process suitability for subsequent steelmaking. It must be maintained within a reasonable preset process temperature range.
5.1.3 Temperature of hot metal about 3# taphole: The real-time measured temperature of hot metal during tapping from No. 3 taphole of the blast furnace. It is a core thermal parameter reflecting the thermal state of the blast furnace hearth and the smelting process, directly determining the quality of hot metal and the process suitability for subsequent steelmaking. It must be maintained within a reasonable preset process temperature range.
5.1.4 Temperature of hot metal about 4# taphole: The real-time measured temperature of hot metal during tapping from No. 4 taphole of the blast furnace. It is a core thermal parameter reflecting the thermal state of the blast furnace hearth and the smelting process, directly determining the quality of hot metal and the process suitability for subsequent steelmaking. It must be maintained within a reasonable preset process temperature range.
5.1.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Tapping system - Hot metal ladle - Operation monitoring”.

5.2 Tapping system - Taphole - Operation monitoring

5.2.1 Temperature of 1# taphole: The real-time temperature of the No. 1 taphole body (the refractory material around the taphole clay pad and taphole channel) of the blast furnace. It reflects the heat state of the refractory material in the taphole area. Too high a temperature indicates serious erosion of the taphole clay pad and degraded sealing performance, which can easily cause taphole splashing and iron leakage.
5.2.2 Temperature of 2# taphole: The real-time temperature of the No. 2 taphole body of the blast furnace. It reflects the heat state of the refractory material in the taphole area. Too high a temperature indicates serious erosion of the taphole clay pad and degraded sealing performance, which can easily cause taphole splashing and iron leakage.
5.2.3 Temperature of 3# taphole: The real-time temperature of the No. 3 taphole body of the blast furnace.

It reflects the heat state of the refractory material in the taphole area. Too high a temperature indicates serious erosion of the taphole clay pad and degraded sealing performance, which can easily cause taphole splashing and iron leakage.

5.2.4 Temperature of 4# taphole: The real-time temperature of the No. 4 taphole body of the blast furnace. It reflects the heat state of the refractory material in the taphole area. Too high a temperature indicates serious erosion of the taphole clay pad and degraded sealing performance, which can easily cause taphole splashing and iron leakage.

5.2.5 Opening time of 1# taphole: The total time from starting the taphole drill to drill through the taphole and the smooth flow of hot metal for No. 1 taphole of the blast furnace. It reflects the firmness of the taphole clay pad. Too long an opening time indicates that the clay pad is too hard, while too short indicates that the clay pad is soft and easily eroded.

5.2.6 Slag arrival time of 1# taphole: The time interval from the start of hot metal flow to the first appearance of slag from the taphole after opening No. 1 taphole of the blast furnace. It reflects the iron-slag separation state and slag fluidity in the blast furnace hearth. Abnormal slag arrival time indicates deviations in hearth conditions and slag formation practice.

5.2.7 Blockage time of 1# taphole: The total time from starting the mud gun to completely sealing the taphole with no leakage of hot metal/slag after completing the tapping operation for No. 1 taphole of the blast furnace. It is an indicator of the efficiency of the plugging operation. Too long a plugging time can easily cause taphole splashing and abnormal hearth pressure.

5.2.8 Slag appearance rate of 1# taphole: The ratio of the duration of slag flow to the total tapping duration in a single tapping operation of No. 1 taphole of the blast furnace. It is a core indicator reflecting whether the slag discharge from the blast furnace is complete. A too low slag appearance rate indicates a high slag level in the hearth, which can easily cause hearth accumulation and smelting difficulties.

5.2.9 Physical heat of 1# taphole: The physical sensible heat of the hot metal from No. 1 taphole of the blast furnace, calculated based on the hot metal temperature. It is a direct reflection of the heat content of the hot metal. Insufficient physical heat will lead to poor fluidity of the hot metal and increase the heating load of the subsequent steelmaking process.

5.2.10 Total tapping time of 1# taphole: The total duration from the start of tapping to the completion of plugging for No. 1 taphole of the blast furnace. It reflects the tapping capacity of a single taphole and must match the hot metal generation rate in the blast furnace hearth. Abnormal total tapping time will affect the tapping rhythm and hearth level stability.

5.2.11 Diameter of 1# taphole: The actual inner diameter of the taphole channel after drilling for No. 1 taphole of the blast furnace. It is a key parameter determining the tapping flow rate of a single taphole. The diameter must be reasonably controlled according to the blast furnace tapping demand. Too large a diameter can easily cause excessively fast tapping and a sudden drop in hearth pressure, while too small a diameter leads to slow tapping and affects production efficiency.

5.2.12 Angle of 1# taphole: The angle between the No. 1 taphole channel and the horizontal direction of the blast furnace hearth. The taphole angle directly affects the insertion depth of the taphole channel into the hearth. A reasonable angle ensures thorough discharge of hot metal/slag from the hearth and avoids dead zone accumulation.

5.2.13 Depth of 1# taphole: The actual length from the outside of the furnace shell to the end of the taphole channel (the clay pad in the hearth) for No. 1 taphole of the blast furnace. It is a core indicator for taphole maintenance. Sufficient taphole depth can form a solid clay pad, preventing taphole diameter expansion and iron leakage during tapping.

5.2.14 Mud amount injected of 1# taphole: The total weight of mud injected into the taphole channel by the mud gun during plugging of No. 1 taphole of the blast furnace. It is a key parameter for maintaining the taphole clay pad. The mud amount must be dynamically adjusted according to the degree of taphole erosion and tapping conditions to ensure good clay pad formation.

5.2.15 Drill bit diameter of 1# taphole: The nominal diameter of the taphole drill bit used during the opening operation of No. 1 taphole of the blast furnace. It determines the initial aperture of the taphole opening. The appropriate drill bit diameter must be selected according to the tapping flow demand, and it is a basic parameter for controlling the tapping rhythm.

5.2.16 Temperature of hot metal at 1# taphole: The on-site real-time temperature measurement data of hot metal during the tapping process of No. 1 taphole of the blast furnace. It is the basis for measuring the hot metal temperature of No. 1 taphole, providing direct on-site data for the regulation of the blast furnace thermal system and the judgment of hot metal quality.

5.2.17 Previous temperature of hot metal at 1# taphole: The measured temperature of hot metal in the previous tapping operation of No. 1 taphole of the blast furnace. Compared with the current temperature measurement data, it helps analyze the change trend of the hot metal temperature at No. 1 taphole, judge the stability of the blast furnace thermal state, and provide a reference for process adjustment.

5.2.18 Estimated yield of hot metal about 1# taphole: The pre-estimated hot metal output of a single tapping operation of No. 1 taphole, based on parameters such as the blast furnace smelting rhythm, hearth level, and tapping flow rate of No. 1 taphole. It is an important basis for formulating the blast furnace tapping plan and allocating hot metal ladles.

5.2.19 Total weight through the pump at 1# taphole: The actual total weight of hot metal flowing out measured by the hot metal delivery pump (or weighing system) during a single tapping operation of No. 1 taphole of the blast furnace. It is the precise measured value of the single tapping output of No. 1 taphole, used to verify the estimated output and calculate the actual production of the blast furnace.

5.2.20 Yield deviation of hot metal at 1# taphole: The difference between the estimated hot metal output and the total weight through the pump for No. 1 taphole of the blast furnace. It reflects the accuracy of the tapping volume estimation. Too large a deviation indicates deviations in the monitoring of the blast furnace hearth level and the prediction of the smelting rhythm, requiring optimization of the estimation model and operating parameters.

5.2.21 Estimated slag volume about 1# taphole: The estimated slag output of a single tapping operation of No. 1 taphole, based on the blast furnace slag formation practice, the tapping volume of No. 1 taphole, and the slag-to-hot metal ratio. It is the planning basis for the blast furnace slag treatment system (such as slag flushing, slag ladle allocation).

5.2.22 Opening time of 2# taphole: The total time from starting the taphole drill to drill through the taphole and the smooth flow of hot metal for No. 2 taphole of the blast furnace. It reflects the firmness of the taphole clay pad. Too long an opening time indicates that the clay pad is too hard, while too short indicates that the clay pad is soft and easily eroded.

5.2.23 Slag arrival time of 2# taphole: The time interval from the start of hot metal flow to the first appearance of slag from the taphole after opening No. 2 taphole of the blast furnace. It reflects the iron-slag separation state and slag fluidity in the blast furnace hearth. Abnormal slag arrival time indicates deviations in hearth conditions and slag formation practice.

5.2.24 Blockage time of 2# taphole: The total time from starting the mud gun to completely sealing the taphole with no leakage of hot metal/slag after completing the tapping operation for No. 2 taphole of the blast furnace. It is an indicator of the efficiency of the plugging operation. Too long a plugging time can easily

cause taphole splashing and abnormal hearth pressure.
5.2.25 Slag appearance rate of 2# taphole: The ratio of the duration of slag flow to the total tapping duration in a single tapping operation of No. 2 taphole of the blast furnace. It is a core indicator reflecting whether the slag discharge from the blast furnace is complete. A too low slag appearance rate indicates a high slag level in the hearth, which can easily cause hearth accumulation and smelting difficulties.
5.2.26 Physical heat of 2# taphole: The physical sensible heat of the hot metal from No. 2 taphole of the blast furnace, calculated based on the hot metal temperature. It is a direct reflection of the heat content of the hot metal. Insufficient physical heat will lead to poor fluidity of the hot metal and increase the heating load of the subsequent steelmaking process.
5.2.27 Total tapping time of 2# taphole: The total duration from the start of tapping to the completion of plugging for No. 2 taphole of the blast furnace. It reflects the tapping capacity of a single taphole and must match the hot metal generation rate in the blast furnace hearth. Abnormal total tapping time will affect the tapping rhythm and hearth level stability.
5.2.28 Diameter of 2# taphole: The actual inner diameter of the taphole channel after drilling for No. 2 taphole of the blast furnace. It is a key parameter determining the tapping flow rate of a single taphole. The diameter must be reasonably controlled according to the blast furnace tapping demand. Too large a diameter can easily cause excessively fast tapping and a sudden drop in hearth pressure, while too small a diameter leads to slow tapping and affects production efficiency.
5.2.29 Angle of 2# taphole: The angle between the No. 2 taphole channel and the horizontal direction of the blast furnace hearth. The taphole angle directly affects the insertion depth of the taphole channel into the hearth. A reasonable angle ensures thorough discharge of hot metal/slag from the hearth and avoids dead zone accumulation.
5.2.30 Depth of 2# taphole: The actual length from the outside of the furnace shell to the end of the taphole channel (the clay pad in the hearth) for No. 2 taphole of the blast furnace. It is a core indicator for taphole maintenance. Sufficient taphole depth can form a solid clay pad, preventing taphole diameter expansion and iron leakage during tapping.
5.2.31 Mud amount injected of 2# taphole: The total weight of mud injected into the taphole channel by the mud gun during plugging of No. 2 taphole of the blast furnace. It is a key parameter for maintaining the taphole clay pad. The mud amount must be dynamically adjusted according to the degree of taphole erosion and tapping conditions to ensure good clay pad formation.
5.2.32 Drill bit diameter of 2# taphole: The nominal diameter of the taphole drill bit used during the opening operation of No. 2 taphole of the blast furnace. It determines the initial aperture of the taphole opening. The appropriate drill bit diameter must be selected according to the tapping flow demand, and it is a basic parameter for controlling the tapping rhythm.
5.2.33 Temperature of hot metal at 2# taphole: The on-site real-time temperature measurement data of hot metal during the tapping process of No. 2 taphole of the blast furnace. It is the basis for measuring the hot metal temperature of No. 2 taphole, providing direct on-site data for the regulation of the blast furnace thermal system and the judgment of hot metal quality.
5.2.34 Previous temperature of hot metal at 2# taphole: The measured temperature of hot metal in the previous tapping operation of No. 2 taphole of the blast furnace. Compared with the current temperature measurement data, it helps analyze the change trend of the hot metal temperature at No. 2 taphole, judge the stability of the blast furnace thermal state, and provide a reference for process adjustment.
5.2.35 Estimated yield of hot metal about 2# taphole: The pre-estimated hot metal output of a single tapping operation of No. 2 taphole, based on parameters such as the blast furnace smelting rhythm, hearth level, and

tapping flow rate of No. 2 taphole. It is an important basis for formulating the blast furnace tapping plan and allocating hot metal ladles.

5.2.36 Total weight through the pump at 2# taphole: The actual total weight of hot metal flowing out measured by the hot metal delivery pump (or weighing system) during a single tapping operation of No. 2 taphole of the blast furnace. It is the precise measured value of the single tapping output of No. 2 taphole, used to verify the estimated output and calculate the actual production of the blast furnace.

5.2.37 Yield deviation of hot metal at 2# taphole: The difference between the estimated hot metal output and the total weight through the pump for No. 2 taphole of the blast furnace. It reflects the accuracy of the tapping volume estimation. Too large a deviation indicates deviations in the monitoring of the blast furnace hearth level and the prediction of the smelting rhythm, requiring optimization of the estimation model and operating parameters.

5.2.38 Estimated slag volume about 2# taphole: The estimated slag output of a single tapping operation of No. 2 taphole, based on the blast furnace slag formation practice, the tapping volume of No. 2 taphole, and the slag-to-hot metal ratio. It is the planning basis for the blast furnace slag treatment system (such as slag flushing, slag ladle allocation).

5.2.39 Opening time of 3# taphole: The total time from starting the taphole drill to drill through the taphole and the smooth flow of hot metal for No. 3 taphole of the blast furnace. It reflects the firmness of the taphole clay pad. Too long an opening time indicates that the clay pad is too hard, while too short indicates that the clay pad is soft and easily eroded.

5.2.40 Slag arrival time of 3# taphole: The time interval from the start of hot metal flow to the first appearance of slag from the taphole after opening No. 3 taphole of the blast furnace. It reflects the iron-slag separation state and slag fluidity in the blast furnace hearth. Abnormal slag arrival time indicates deviations in hearth conditions and slag formation practice.

5.2.41 Blockage time of 3# taphole: The total time from starting the mud gun to completely sealing the taphole with no leakage of hot metal/slag after completing the tapping operation for No. 3 taphole of the blast furnace. It is an indicator of the efficiency of the plugging operation. Too long a plugging time can easily cause taphole splashing and abnormal hearth pressure.

5.2.42 Slag appearance rate of 3# taphole: The ratio of the duration of slag flow to the total tapping duration in a single tapping operation of No. 3 taphole of the blast furnace. It is a core indicator reflecting whether the slag discharge from the blast furnace is complete. A too low slag appearance rate indicates a high slag level in the hearth, which can easily cause hearth accumulation and smelting difficulties.

5.2.43 Physical heat of 3# taphole: The physical sensible heat of the hot metal from No. 3 taphole of the blast furnace, calculated based on the hot metal temperature. It is a direct reflection of the heat content of the hot metal. Insufficient physical heat will lead to poor fluidity of the hot metal and increase the heating load of the subsequent steelmaking process.

5.2.44 Total tapping time of 3# taphole: The total duration from the start of tapping to the completion of plugging for No. 3 taphole of the blast furnace. It reflects the tapping capacity of a single taphole and must match the hot metal generation rate in the blast furnace hearth. Abnormal total tapping time will affect the tapping rhythm and hearth level stability.

5.2.45 Diameter of 3# taphole: The actual inner diameter of the taphole channel after drilling for No. 3 taphole of the blast furnace. It is a key parameter determining the tapping flow rate of a single taphole. The diameter must be reasonably controlled according to the blast furnace tapping demand. Too large a diameter can easily cause excessively fast tapping and a sudden drop in hearth pressure, while too small a diameter leads to slow tapping and affects production efficiency.

5.2.46 Angle of 3# taphole: The angle between the No. 3 taphole channel and the horizontal direction of the blast furnace hearth. The taphole angle directly affects the insertion depth of the taphole channel into the hearth. A reasonable angle ensures thorough discharge of hot metal/slag from the hearth and avoids dead zone accumulation.
5.2.47 Depth of 3# taphole: The actual length from the outside of the furnace shell to the end of the taphole channel (the clay pad in the hearth) for No. 3 taphole of the blast furnace. It is a core indicator for taphole maintenance. Sufficient taphole depth can form a solid clay pad, preventing taphole diameter expansion and iron leakage during tapping.
5.2.48 Mud amount injected of 3# taphole: The total weight of mud injected into the taphole channel by the mud gun during plugging of No. 3 taphole of the blast furnace. It is a key parameter for maintaining the taphole clay pad. The mud amount must be dynamically adjusted according to the degree of taphole erosion and tapping conditions to ensure good clay pad formation.
5.2.49 Drill bit diameter of 3# taphole: The nominal diameter of the taphole drill bit used during the opening operation of No. 3 taphole of the blast furnace. It determines the initial aperture of the taphole opening. The appropriate drill bit diameter must be selected according to the tapping flow demand, and it is a basic parameter for controlling the tapping rhythm.
5.2.50 Temperature of hot metal at 3# taphole: The on-site real-time temperature measurement data of hot metal during the tapping process of No. 3 taphole of the blast furnace. It is the basis for measuring the hot metal temperature of No. 3 taphole, providing direct on-site data for the regulation of the blast furnace thermal system and the judgment of hot metal quality.
5.2.51 Previous temperature of hot metal at 3# taphole: The measured temperature of hot metal in the previous tapping operation of No. 3 taphole of the blast furnace. Compared with the current temperature measurement data, it helps analyze the change trend of the hot metal temperature at No. 3 taphole, judge the stability of the blast furnace thermal state, and provide a reference for process adjustment.
5.2.52 Estimated yield of hot metal about 3# taphole: The pre-estimated hot metal output of a single tapping operation of No. 3 taphole, based on parameters such as the blast furnace smelting rhythm, hearth level, and tapping flow rate of No. 3 taphole. It is an important basis for formulating the blast furnace tapping plan and allocating hot metal ladles.
5.2.53 Total weight through the pump at 3# taphole: The actual total weight of hot metal flowing out measured by the hot metal delivery pump (or weighing system) during a single tapping operation of No. 3 taphole of the blast furnace. It is the precise measured value of the single tapping output of No. 3 taphole, used to verify the estimated output and calculate the actual production of the blast furnace.
5.2.54 Yield deviation of hot metal at 3# taphole: The difference between the estimated hot metal output and the total weight through the pump for No. 3 taphole of the blast furnace. It reflects the accuracy of the tapping volume estimation. Too large a deviation indicates deviations in the monitoring of the blast furnace hearth level and the prediction of the smelting rhythm, requiring optimization of the estimation model and operating parameters.
5.2.55 Estimated slag volume about 3# taphole: The estimated slag output of a single tapping operation of No. 3 taphole, based on the blast furnace slag formation practice, the tapping volume of No. 3 taphole, and the slag-to-hot metal ratio. It is the planning basis for the blast furnace slag treatment system (such as slag flushing, slag ladle allocation).
5.2.56 Opening time of 4# taphole: The total time from starting the taphole drill to drill through the taphole and the smooth flow of hot metal for No. 4 taphole of the blast furnace. It reflects the firmness of the taphole clay pad. Too long an opening time indicates that the clay pad is too hard, while too short indicates that the

clay pad is soft and easily eroded.

5.2.57 Slag arrival time of 4# taphole: The time interval from the start of hot metal flow to the first appearance of slag from the taphole after opening No. 4 taphole of the blast furnace. It reflects the iron-slag separation state and slag fluidity in the blast furnace hearth. Abnormal slag arrival time indicates deviations in hearth conditions and slag formation practice.

5.2.58 Blockage time of 4# taphole: The total time from starting the mud gun to completely sealing the taphole with no leakage of hot metal/slag after completing the tapping operation for No. 4 taphole of the blast furnace. It is an indicator of the efficiency of the plugging operation. Too long a plugging time can easily cause taphole splashing and abnormal hearth pressure.

5.2.59 Slag appearance rate of 4# taphole: The ratio of the duration of slag flow to the total tapping duration in a single tapping operation of No. 4 taphole of the blast furnace. It is a core indicator reflecting whether the slag discharge from the blast furnace is complete. A too low slag appearance rate indicates a high slag level in the hearth, which can easily cause hearth accumulation and smelting difficulties.

5.2.60 Physical heat of 4# taphole: The physical sensible heat of the hot metal from No. 4 taphole of the blast furnace, calculated based on the hot metal temperature. It is a direct reflection of the heat content of the hot metal. Insufficient physical heat will lead to poor fluidity of the hot metal and increase the heating load of the subsequent steelmaking process.

5.2.61 Total tapping time of 4# taphole: The total duration from the start of tapping to the completion of plugging for No. 4 taphole of the blast furnace. It reflects the tapping capacity of a single taphole and must match the hot metal generation rate in the blast furnace hearth. Abnormal total tapping time will affect the tapping rhythm and hearth level stability.

5.2.62 Diameter of 4# taphole: The actual inner diameter of the taphole channel after drilling for No. 4 taphole of the blast furnace. It is a key parameter determining the tapping flow rate of a single taphole. The diameter must be reasonably controlled according to the blast furnace tapping demand. Too large a diameter can easily cause excessively fast tapping and a sudden drop in hearth pressure, while too small a diameter leads to slow tapping and affects production efficiency.

5.2.63 Angle of 4# taphole: The angle between the No. 4 taphole channel and the horizontal direction of the blast furnace hearth. The taphole angle directly affects the insertion depth of the taphole channel into the hearth. A reasonable angle ensures thorough discharge of hot metal/slag from the hearth and avoids dead zone accumulation.

5.2.64 Depth of 4# taphole: The actual length from the outside of the furnace shell to the end of the taphole channel (the clay pad in the hearth) for No. 4 taphole of the blast furnace. It is a core indicator for taphole maintenance. Sufficient taphole depth can form a solid clay pad, preventing taphole diameter expansion and iron leakage during tapping.

5.2.65 Mud amount injected of 4# taphole: The total weight of mud injected into the taphole channel by the mud gun during plugging of No. 4 taphole of the blast furnace. It is a key parameter for maintaining the taphole clay pad. The mud amount must be dynamically adjusted according to the degree of taphole erosion and tapping conditions to ensure good clay pad formation.

5.2.66 Drill bit diameter of 4# taphole: The nominal diameter of the taphole drill bit used during the opening operation of No. 4 taphole of the blast furnace. It determines the initial aperture of the taphole opening. The appropriate drill bit diameter must be selected according to the tapping flow demand, and it is a basic parameter for controlling the tapping rhythm.

5.2.67 Temperature of hot metal at 4# taphole: The on-site real-time temperature measurement data of hot metal during the tapping process of No. 4 taphole of the blast furnace. It is the basis for measuring the hot metal temperature of No. 4 taphole, providing direct on-site data for the regulation of the blast furnace thermal system and the judgment of hot metal quality.

5.2.68 Previous temperature of hot metal at 4# taphole: The measured temperature of hot metal in the previous tapping operation of No. 4 taphole of the blast furnace. Compared with the current temperature measurement data, it helps analyze the change trend of the hot metal temperature at No. 4 taphole, judge the stability of the blast furnace thermal state, and provide a reference for process adjustment.

5.2.69 Estimated yield of hot metal about 4# taphole: The pre-estimated hot metal output of a single tapping operation of No. 4 taphole, based on parameters such as the blast furnace smelting rhythm, hearth level, and tapping flow rate of No. 4 taphole. It is an important basis for formulating the blast furnace tapping plan and allocating hot metal ladles.

5.2.70 Total weight through the pump at 4# taphole: The actual total weight of hot metal flowing out measured by the hot metal delivery pump (or weighing system) during a single tapping operation of No. 4 taphole of the blast furnace. It is the precise measured value of the single tapping output of No. 4 taphole, used to verify the estimated output and calculate the actual production of the blast furnace.

5.2.71 Yield deviation of hot metal at 4# taphole: The difference between the estimated hot metal output and the total weight through the pump for No. 4 taphole of the blast furnace. It reflects the accuracy of the tapping volume estimation. Too large a deviation indicates deviations in the monitoring of the blast furnace hearth level and the prediction of the smelting rhythm, requiring optimization of the estimation model and operating parameters.

5.2.72 Estimated slag volume about 4# taphole: The estimated slag output of a single tapping operation of No. 4 taphole, based on the blast furnace slag formation practice, the tapping volume of No. 4 taphole, and the slag-to-hot metal ratio. It is the planning basis for the blast furnace slag treatment system (such as slag flushing, slag ladle allocation).

5.2.73 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Tapping system - Taphole - Operation monitoring”.

5.3 Tapping system - Laboratory - Hot metal composition detection

5.3.1 Si content: The mass percentage of silicon in the blast furnace hot metal. It is a core chemical indicator for judging hot metal quality and blast furnace hearth temperature. Too high a Si content indicates a high furnace temperature, and too low indicates a low furnace temperature. It must be precisely controlled according to steelmaking requirements and blast furnace smelting smoothness.

5.3.2 S content: The mass percentage of sulfur in the blast furnace hot metal. It is a key harmful element indicator for hot metal quality. Sulfur reduces the plasticity and toughness of steel products. It must be controlled within the low range allowed by the process through blast furnace slag formation and desulfurization operations.

5.3.3 P content: The mass percentage of phosphorus in the blast furnace hot metal. Phosphorus is one of the harmful elements in hot metal, causing "cold shortness" in steel products and reducing low-temperature toughness. Phosphorus is difficult to remove in the steelmaking process, so the phosphorus content of the raw materials must be strictly controlled in blast furnace smelting to reduce the P content in hot metal.

5.3.4 Ti content: The mass percentage of titanium in the blast furnace hot metal. Titanium can form compounds with nitrogen and sulfur in hot metal, improving steel properties. At the same time, an appropriate amount of Ti can protect the blast furnace hearth refractory material, but too high a content will increase slag viscosity and affect smelting smoothness. It must be controlled as needed.

5.3.5 C content: The mass percentage of carbon in the blast furnace hot metal. Carbon is one of the main components of hot metal, directly determining the hardness, fluidity, and decarburization load of hot metal in subsequent steelmaking. Blast furnace hot metal is high-carbon hot metal, and the carbon content must be maintained within the preset process range.

5.3.6 V content: The mass percentage of vanadium in the blast furnace hot metal. Vanadium is a precious alloying element that significantly improves the strength, toughness, and wear resistance of steel. If the blast furnace raw materials contain vanadium, the V content in hot metal should be maintained through smelting control to achieve comprehensive utilization of vanadium.

5.3.7 Cr content: The mass percentage of chromium in the blast furnace hot metal. Chromium improves the corrosion resistance, wear resistance, and high-temperature strength of steel. It is an important alloying element. The Cr content in hot metal can be controlled by regulating the raw materials according to steel product requirements.

5.3.8 Mn content: The mass percentage of manganese in the blast furnace hot metal. Manganese is a beneficial alloying element that reduces the harm of sulfur in hot metal, improves slag fluidity, and enhances the strength and toughness of steel. It must be reasonably controlled according to slag formation practice and steel product requirements.

5.3.9 Fe content: The mass percentage of iron in the blast furnace hot metal. Iron is the core effective component of hot metal. The Fe content directly reflects the smelting efficiency of the blast furnace and the utilization efficiency of raw materials, and is an important chemical indicator for measuring the production efficiency of the blast furnace.

5.3.10 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Tapping system - Laboratory - Hot metal composition detection”.

5.4 Tapping system - Laboratory - Slag composition detection

5.4.1 FeO content: The mass percentage of ferrous oxide in the blast furnace slag. It reflects the oxidizing property of the slag. Too high a FeO content will cause sulfur return to the blast furnace hot metal and a drop in hearth temperature, and will also accelerate furnace lining erosion. The oxidizing property of the slag must be controlled through slag formation practice to reduce the FeO content.

5.4.2 SiO2 content: The mass percentage of silicon dioxide in the blast furnace slag. It is the main acidic oxide component of the slag and is one of the core indicators for determining the slag basicity. Too high a SiO2 content will increase slag viscosity and worsen fluidity. The content must be balanced by adding basic fluxes.

5.4.3 CaO content: The mass percentage of calcium oxide in the blast furnace slag. It is the main basic oxide component of the slag and is a key indicator for regulating the slag basicity. CaO can combine with acidic oxides in the slag to reduce slag viscosity, improve fluidity, and enhance the desulfurization capacity of the slag.

5.4.4 MgO content: The mass percentage of magnesium oxide in the blast furnace slag. It is one of the basic oxide components of the slag. An appropriate amount of MgO can lower the slag melting point, improve slag fluidity, and protect the blast furnace lining refractory material, preventing slag accretion. It must be maintained at a reasonable content.

5.4.5 TiO2 content: The mass percentage of titanium dioxide in the blast furnace slag. It is introduced with raw materials (such as ilmenite). An appropriate amount of TiO2 can improve slag fluidity, but too high a content will generate high-melting-point compounds, increasing slag viscosity and affecting smelting smoothness.

5.4.6 Al_2O_3 content: The mass percentage of aluminum oxide in the blast furnace slag. It is one of the acidic oxide components of the slag. Too high an Al_2O_3 content will increase the slag melting point and viscosity, reducing the desulfurization capacity of the slag. Its content must be regulated within a reasonable range through slag formation practice.

5.4.7 MnO content: The mass percentage of manganese oxide in the blast furnace slag. It is a beneficial component of the slag. MnO can improve slag fluidity and desulfurization capacity, and part of the MnO will be reduced and enter the hot metal, increasing the Mn content in hot metal.

5.4.8 S content: The mass percentage of sulfur in the blast furnace slag. It reflects the desulfurization effect of the slag. The blast furnace achieves hot metal desulfurization through slag. The higher the S content in the slag, the more thorough the desulfurization of the hot metal. It is an important indicator for evaluating the rationality of the blast furnace slag formation practice.

5.4.9 R2/Binary basicity: The ratio of the mass percentage of CaO to SiO_2 (CaO/SiO_2) in the blast furnace slag. It is the most core control indicator for the blast furnace slag formation practice, directly determining the alkalinity, melting point, fluidity, and desulfurization capacity of the slag. It must be precisely set according to the blast furnace smelting conditions and hot metal quality requirements.

5.4.10 R3/Ternary basicity: The ratio of the mass percentage of CaO to (SiO_2+Al_2O_3) [$CaO/(SiO_2+Al_2O_3)$] in the blast furnace slag. It takes into account the influence of Al_2O_3 on the acidity of the slag. It is a supplementary indicator to the binary basicity, more accurately reflecting the alkalinity and fluidity of the slag, and is suitable for slag regulation with high Al_2O_3 content.

5.4.11 R4/Quaternary basicity: The ratio of (CaO+MgO) to (SiO_2+Al_2O_3) [$(CaO+MgO)/(SiO_2+Al_2O_3)$] in the blast furnace slag. It comprehensively considers the combined action of basic oxides CaO and MgO and acidic oxides SiO_2 and Al_2O_3, most comprehensively reflecting the alkalinity, melting point, and fluidity of the slag. It is an important indicator for fine control of the blast furnace slag formation practice.

5.4.12 Ls (Sulfur partition coefficient): The ratio of the mass percentage of sulfur in the slag to the mass percentage of sulfur in the hot metal during the blast furnace smelting process. It is a core indicator for measuring the desulfurization capacity of the blast furnace slag. The larger the Ls value, the better the desulfurization effect of the slag, and the more thorough the desulfurization of the hot metal. Its magnitude is closely related to the slag basicity, temperature, and fluidity.

5.4.13 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Tapping system - Laboratory - Slag composition detection”.

6 Slag treating

6.1 Slag granulation system - Circulating water pipeline - Operation monitoring

6.1.1 1# Slag yard - Pressure of 1# filtering branch pipe: The liquid pressure in the first filtering branch pipe of No. 1 slag yard, reflecting the smoothness of the filtering pipeline.

6.1.2 1# Slag yard - Pressure of 2# filtering branch pipe: The liquid pressure in the second filtering branch pipe of No. 1 slag yard, reflecting the smoothness of the filtering pipeline.

6.1.3 1# Slag yard - Pressure of 1# slag flushing branch pipe: The water supply pressure in the first slag flushing branch pipe of No. 1 slag yard, monitoring the slag flushing water pressure.

6.1.4 1# Slag yard - Pressure of 2# slag flushing branch pipe: The water supply pressure in the second slag flushing branch pipe of No. 1 slag yard, monitoring the slag flushing water pressure.

6.1.5 1# Slag yard - Temperature of outlet pipe in hot water pump group: The medium temperature in the outlet pipe of the hot water pump group of No. 1 slag yard, monitoring the return water temperature.
6.1.6 1# Slag yard - Pressure of outlet pipe in hot water pump group: The liquid pressure in the outlet pipe of the hot water pump group of No. 1 slag yard, monitoring the pump outlet pressure.
6.1.7 1# Slag yard - Flow rate of outlet pipe in hot water pump group: The instantaneous flow rate in the outlet pipe of the hot water pump group of No. 1 slag yard, monitoring the return water flow rate.
6.1.8 1# Slag yard - Temperature of outlet pipe in slag flushing pump group: The medium temperature in the outlet pipe of the slag flushing pump group of No. 1 slag yard, monitoring the slag flushing water temperature.
6.1.9 1# Slag yard - Pressure of outlet pipe in slag flushing pump group: The pressure in the outlet pipe of the slag flushing pump group of No. 1 slag yard, the core working pressure of the slag flushing system.
6.1.10 1# Slag yard - Flow rate of outlet pipe in slag flushing pump group: The instantaneous flow rate in the outlet pipe of the slag flushing pump group of No. 1 slag yard, monitoring the slag flushing water supply.
6.1.11 2# Slag yard - Pressure of 1# filtering branch pipe: The liquid pressure in the first filtering branch pipe of No. 2 slag yard, reflecting the smoothness of the filtering pipeline.
6.1.12 2# Slag yard - Pressure of 2# filtering branch pipe: The liquid pressure in the second filtering branch pipe of No. 2 slag yard, reflecting the smoothness of the filtering pipeline.
6.1.13 2# Slag yard - Pressure of 1# slag flushing branch pipe: The water supply pressure in the first slag flushing branch pipe of No. 2 slag yard, monitoring the slag flushing water pressure.
6.1.14 2# Slag yard - Pressure of 2# slag flushing branch pipe: The water supply pressure in the second slag flushing branch pipe of No. 2 slag yard, monitoring the slag flushing water pressure.
6.1.15 2# Slag yard - Temperature of outlet pipe in hot water pump group: The medium temperature in the outlet pipe of the hot water pump group of No. 2 slag yard, monitoring the return water temperature.
6.1.16 2# Slag yard - Pressure of outlet pipe in hot water pump group: The pressure in the outlet pipe of the hot water pump group of No. 2 slag yard, monitoring the pump outlet pressure.
6.1.17 2# Slag yard - Flow rate of outlet pipe in hot water pump group: The instantaneous flow rate in the outlet pipe of the hot water pump group of No. 2 slag yard, monitoring the return water flow rate.
6.1.18 2# Slag yard - Temperature of outlet pipe in slag flushing pump group: The medium temperature in the outlet pipe of the slag flushing pump group of No. 2 slag yard, monitoring the slag flushing water temperature.
6.1.19 2# Slag yard - Pressure of outlet pipe in slag flushing pump group: The pressure in the outlet pipe of the slag flushing pump group of No. 2 slag yard, the core working pressure of the slag flushing system.
6.1.20 2# Slag yard - Flow rate of outlet pipe in slag flushing pump group: The instantaneous flow rate in the outlet pipe of the slag flushing pump group of No. 2 slag yard, monitoring the slag flushing water supply.
6.1.21 1# Slag yard - Flow rate of production water: The instantaneous flow rate of production fresh water/new water replenishment for No. 1 slag yard.
6.1.22 1# Slag yard - Flow rate of concentrated brine: The instantaneous flow rate of concentrated brine/sewage discharged from the No. 1 slag yard system.
6.1.23 1# Slag yard - Daily accumulated value of flow rate of production water: The cumulative total of production water replenishment for No. 1 slag yard during the current day.
6.1.24 1# Slag yard - Daily accumulated value of flow rate of concentrated brine: The cumulative total of concentrated brine discharge for No. 1 slag yard during the current day.
6.1.25 2# Slag yard - Flow rate of production water: The instantaneous flow rate of production fresh water/new water replenishment for No. 2 slag yard.
6.1.26 2# Slag yard - Flow rate of concentrated brine: The instantaneous flow rate of concentrated

brine/sewage discharged from the No. 2 slag yard system.
6.1.27 2# Slag yard - Daily accumulated value of flow rate of production water: The cumulative total of production water replenishment for No. 2 slag yard during the current day.
6.1.28 2# Slag yard - Daily accumulated value of flow rate of concentrated brine: The cumulative total of concentrated brine discharge for No. 2 slag yard during the current day.
6.1.29 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Slag granulation system - Circulating water pipeline - Operation monitoring”.

6.2 Slag granulation system - Water pump motor - Operation monitoring

6.2.1 1# Slag yard - 1# Pump motor - Temperature of front shaft: The front bearing temperature of the No. 1 pump motor in No. 1 slag yard.
6.2.2 1# Slag yard - 1# Pump motor - Temperature of rear shaft: The rear bearing temperature of the No. 1 pump motor in No. 1 slag yard.
6.2.3 1# Slag yard - 1# Pump motor - Temperature of stator A: The temperature of stator phase A winding of the No. 1 pump motor in No. 1 slag yard.
6.2.4 1# Slag yard - 1# Pump motor - Temperature of stator B: The temperature of stator phase B winding of the No. 1 pump motor in No. 1 slag yard.
6.2.5 1# Slag yard - 1# Pump motor - Temperature of stator C: The temperature of stator phase C winding of the No. 1 pump motor in No. 1 slag yard.
6.2.6 1# Slag yard - 2# Pump motor - Temperature of front shaft: The front bearing temperature of the No. 2 pump motor in No. 1 slag yard.
6.2.7 1# Slag yard - 2# Pump motor - Temperature of rear shaft: The rear bearing temperature of the No. 2 pump motor in No. 1 slag yard.
6.2.8 1# Slag yard - 2# Pump motor - Temperature of stator A: The temperature of stator phase A winding of the No. 2 pump motor in No. 1 slag yard.
6.2.9 1# Slag yard - 2# Pump motor - Temperature of stator B: The temperature of stator phase B winding of the No. 2 pump motor in No. 1 slag yard.
6.2.10 1# Slag yard - 2# Pump motor - Temperature of stator C: The temperature of stator phase C winding of the No. 2 pump motor in No. 1 slag yard.
6.2.11 1# Slag yard - 3# Pump motor - Temperature of front shaft: The front bearing temperature of the No. 3 pump motor in No. 1 slag yard.
6.2.12 1# Slag yard - 3# Pump motor - Temperature of rear shaft: The rear bearing temperature of the No. 3 pump motor in No. 1 slag yard.
6.2.13 1# Slag yard - 3# Pump motor - Temperature of stator A: The temperature of stator phase A winding of the No. 3 pump motor in No. 1 slag yard.
6.2.14 1# Slag yard - 3# Pump motor - Temperature of stator B: The temperature of stator phase B winding of the No. 3 pump motor in No. 1 slag yard.
6.2.15 1# Slag yard - 3# Pump motor - Temperature of stator C: The temperature of stator phase C winding of the No. 3 pump motor in No. 1 slag yard.
6.2.16 1# Slag yard - 4# Pump motor - Temperature of front shaft: The front bearing temperature of the No. 4 pump motor in No. 1 slag yard.
6.2.17 1# Slag yard - 4# Pump motor - Temperature of rear shaft: The rear bearing temperature of the No. 4 pump motor in No. 1 slag yard.

6.2.18 1# Slag yard - 4# Pump motor - Temperature of stator A: The temperature of stator phase A winding of the No. 4 pump motor in No. 1 slag yard.
6.2.19 1# Slag yard - 4# Pump motor - Temperature of stator B: The temperature of stator phase B winding of the No. 4 pump motor in No. 1 slag yard.
6.2.20 1# Slag yard - 4# Pump motor - Temperature of stator C: The temperature of stator phase C winding of the No. 4 pump motor in No. 1 slag yard.
6.2.21 2# Slag yard - 1# Pump motor - Temperature of front shaft: The front bearing temperature of the No. 1 pump motor in No. 2 slag yard.
6.2.22 2# Slag yard - 1# Pump motor - Temperature of rear shaft: The rear bearing temperature of the No. 1 pump motor in No. 2 slag yard.
6.2.23 2# Slag yard - 1# Pump motor - Temperature of stator A: The temperature of stator phase A winding of the No. 1 pump motor in No. 2 slag yard.
6.2.24 2# Slag yard - 1# Pump motor - Temperature of stator B: The temperature of stator phase B winding of the No. 1 pump motor in No. 2 slag yard.
6.2.25 2# Slag yard - 1# Pump motor - Temperature of stator C: The temperature of stator phase C winding of the No. 1 pump motor in No. 2 slag yard.
6.2.26 2# Slag yard - 2# Pump motor - Temperature of front shaft: The front bearing temperature of the No. 2 pump motor in No. 2 slag yard.
6.2.27 2# Slag yard - 2# Pump motor - Temperature of rear shaft: The rear bearing temperature of the No. 2 pump motor in No. 2 slag yard.
6.2.28 2# Slag yard - 2# Pump motor - Temperature of stator A: The temperature of stator phase A winding of the No. 2 pump motor in No. 2 slag yard.
6.2.29 2# Slag yard - 2# Pump motor - Temperature of stator B: The temperature of stator phase B winding of the No. 2 pump motor in No. 2 slag yard.
6.2.30 2# Slag yard - 2# Pump motor - Temperature of stator C: The temperature of stator phase C winding of the No. 2 pump motor in No. 2 slag yard.
6.2.31 2# Slag yard - 3# Pump motor - Temperature of front shaft: The front bearing temperature of the No. 3 pump motor in No. 2 slag yard.
6.2.32 2# Slag yard - 3# Pump motor - Temperature of rear shaft: The rear bearing temperature of the No. 3 pump motor in No. 2 slag yard.
6.2.33 2# Slag yard - 3# Pump motor - Temperature of stator A: The temperature of stator phase A winding of the No. 3 pump motor in No. 2 slag yard.
6.2.34 2# Slag yard - 3# Pump motor - Temperature of stator B: The temperature of stator phase B winding of the No. 3 pump motor in No. 2 slag yard.
6.2.35 2# Slag yard - 3# Pump motor - Temperature of stator C: The temperature of stator phase C winding of the No. 3 pump motor in No. 2 slag yard.
6.2.36 2# Slag yard - 4# Pump motor - Temperature of front shaft: The front bearing temperature of the No. 4 pump motor in No. 2 slag yard.
6.2.37 2# Slag yard - 4# Pump motor - Temperature of rear shaft: The rear bearing temperature of the No. 4 pump motor in No. 2 slag yard.
6.2.38 2# Slag yard - 4# Pump motor - Temperature of stator A: The temperature of stator phase A winding of the No. 4 pump motor in No. 2 slag yard.
6.2.39 2# Slag yard - 4# Pump motor - Temperature of stator B: The temperature of stator phase B winding of the No. 4 pump motor in No. 2 slag yard.

6.2.40 2# Slag yard - 4# Pump motor - Temperature of stator C: The temperature of stator phase C winding of the No. 4 pump motor in No. 2 slag yard.
6.2.41 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Slag granulation system - Water pump motor - Operation monitoring”.

6.3 Slag granulation system - Cooling tower - Operation monitoring

6.3.1 1# Slag yard - Temperature of 1# cooling tower: The temperature inside/outlet water temperature of No. 1 cooling tower in No. 1 slag yard.
6.3.2 1# Slag yard - Temperature of 2# cooling tower: The temperature inside/outlet water temperature of No. 2 cooling tower in No. 1 slag yard.
6.3.3 1# Slag yard - Temperature of 3# cooling tower: The temperature inside/outlet water temperature of No. 3 cooling tower in No. 1 slag yard.
6.3.4 1# Slag yard - Temperature of 4# cooling tower: The temperature inside/outlet water temperature of No. 4 cooling tower in No. 1 slag yard.
6.3.5 1# Slag yard - Liquid level of 1# cooling tower: The liquid level in the sump of No. 1 cooling tower in No. 1 slag yard.
6.3.6 1# Slag yard - Liquid level of 2# cooling tower: The liquid level in the sump of No. 2 cooling tower in No. 1 slag yard.
6.3.7 1# Slag yard - Liquid level of 3# cooling tower: The liquid level in the sump of No. 3 cooling tower in No. 1 slag yard.
6.3.8 1# Slag yard - Liquid level of 4# cooling tower: The liquid level in the sump of No. 4 cooling tower in No. 1 slag yard.
6.3.9 1# Slag yard - Flow rate of 1# cooling tower: The cooling circulating water flow rate of No. 1 cooling tower in No. 1 slag yard.
6.3.10 1# Slag yard - Flow rate of 2# cooling tower: The cooling circulating water flow rate of No. 2 cooling tower in No. 1 slag yard.
6.3.11 1# Slag yard - Flow rate of 3# cooling tower: The cooling circulating water flow rate of No. 3 cooling tower in No. 1 slag yard.
6.3.12 1# Slag yard - Flow rate of 4# cooling tower: The cooling circulating water flow rate of No. 4 cooling tower in No. 1 slag yard.
6.3.13 1# Slag yard - Current of 1# cooling tower: The working current of the fan motor of No. 1 cooling tower in No. 1 slag yard.
6.3.14 1# Slag yard - Current of 2# cooling tower: The working current of the fan motor of No. 2 cooling tower in No. 1 slag yard.
6.3.15 1# Slag yard - Current of 3# cooling tower: The working current of the fan motor of No. 3 cooling tower in No. 1 slag yard.
6.3.16 1# Slag yard - Current of 4# cooling tower: The working current of the fan motor of No. 4 cooling tower in No. 1 slag yard.
6.3.17 2# Slag yard - Temperature of 1# cooling tower: The temperature inside/outlet water temperature of No. 1 cooling tower in No. 2 slag yard.
6.3.18 2# Slag yard - Temperature of 2# cooling tower: The temperature inside/outlet water temperature of No. 2 cooling tower in No. 2 slag yard.
6.3.19 2# Slag yard - Temperature of 3# cooling tower: The temperature inside/outlet water temperature of

No. 3 cooling tower in No. 2 slag yard.

6.3.20 2# Slag yard - Temperature of 4# cooling tower: The temperature inside/outlet water temperature of No. 4 cooling tower in No. 2 slag yard.

6.3.21 2# Slag yard - Liquid level of 1# cooling tower: The liquid level in the sump of No. 1 cooling tower in No. 2 slag yard.

6.3.22 2# Slag yard - Liquid level of 2# cooling tower: The liquid level in the sump of No. 2 cooling tower in No. 2 slag yard.

6.3.23 2# Slag yard - Liquid level of 3# cooling tower: The liquid level in the sump of No. 3 cooling tower in No. 2 slag yard.

6.3.24 2# Slag yard - Liquid level of 4# cooling tower: The liquid level in the sump of No. 4 cooling tower in No. 2 slag yard.

6.3.25 2# Slag yard - Flow rate of 1# cooling tower: The cooling circulating water flow rate of No. 1 cooling tower in No. 2 slag yard.

6.3.26 2# Slag yard - Flow rate of 2# cooling tower: The cooling circulating water flow rate of No. 2 cooling tower in No. 2 slag yard.

6.3.27 2# Slag yard - Flow rate of 3# cooling tower: The cooling circulating water flow rate of No. 3 cooling tower in No. 2 slag yard.

6.3.28 2# Slag yard - Flow rate of 4# cooling tower: The cooling circulating water flow rate of No. 4 cooling tower in No. 2 slag yard.

6.3.29 2# Slag yard - Current of 1# cooling tower: The working current of the fan motor of No. 1 cooling tower in No. 2 slag yard.

6.3.30 2# Slag yard - Current of 2# cooling tower: The working current of the fan motor of No. 2 cooling tower in No. 2 slag yard.

6.3.31 2# Slag yard - Current of 3# cooling tower: The working current of the fan motor of No. 3 cooling tower in No. 2 slag yard.

6.3.32 2# Slag yard - Current of 4# cooling tower: The working current of the fan motor of No. 4 cooling tower in No. 2 slag yard.

6.3.33 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Slag granulation system - Cooling tower - Operation monitoring”.

7 Hot blast supplying

7.1 Blast supply system - Expansion tank - Pressure control

7.1.1 Blow-off valve: A valve installed on the gas or pressure system pipeline, used to automatically or manually open and discharge the medium to the atmosphere or a designated area when the system is overpressurized or needs to relieve pressure, ensuring system pressure safety.

7.1.2 Pressurization valve: A valve used to supplement medium (such as nitrogen, gas) into a closed container or system to increase pressure, commonly used in process links such as blast furnace top equalizing pressure and hot blast stove pressurization.

7.1.3 Emergency water supply valve: A valve that automatically or manually opens to replenish cooling water to the system when the cooling system is short of water or in an emergency, used to prevent equipment from overheating and damage due to water shortage.

7.1.4 Level of expansion tank: The liquid level height of the cooling medium (such as water) in the expansion tank, reflecting the volume change of the medium in the system and the buffering capacity of the expansion tank. It is an important monitoring parameter for the stable operation of the cooling system.

7.1.5 Pressure of expansion tank: The static pressure of the medium in the expansion tank, used to balance pressure fluctuations in the cooling system caused by temperature changes and maintain system pressure stability.

7.1.6 Pressure of nitrogen tank: The gas pressure in the nitrogen storage tank, reflecting the nitrogen reserve and gas supply capacity, and is a key parameter for ensuring the stable supply of the nitrogen system.

7.1.7 Pressure of supply ring pipe about soft water: The static pressure of the medium in the soft water supply ring pipe, which is the basis for the soft water system to evenly distribute pressure to each water consumption point, ensuring stable water supply to each cooling point.

7.1.8 Flow rate of emergency water supply pipe: The volume of cooling water flowing through the emergency water supply pipe per unit time, reflecting the actual water supply capacity of the emergency water supply system, used to determine whether the water replenishment is timely and sufficient.

7.1.9 Pressure of emergency water supply pipe: The static pressure of the cooling water in the emergency water supply pipe, ensuring that the cooling water can be smoothly injected into the system in an emergency. Insufficient pressure will cause the water replenishment to fail.

7.1.10 Ultimate upper limit setpoint of level of expansion tank: The absolute maximum threshold allowed for the liquid level of the expansion tank. When the liquid level exceeds this value, the system will trigger an emergency interlock protection to prevent medium overflow or equipment damage.

7.1.11 Upper limit setpoint of level of expansion tank: The upper limit of the normal operation range for the liquid level of the expansion tank. When the liquid level reaches this value, the system issues a warning or automatically adjusts water replenishment/drainage to maintain the liquid level within a reasonable range.

7.1.12 Lower limit setpoint of level of expansion tank: The lower limit of the normal operation range for the liquid level of the expansion tank. When the liquid level falls below this value, the system issues a warning and automatically replenishes water to prevent system pressure fluctuations due to too low a liquid level.

7.1.13 Ultimate lower limit setpoint of level of expansion tank: The absolute minimum threshold allowed for the liquid level of the expansion tank. When the liquid level falls below this value, the system triggers an emergency interlock to stop the operation of related equipment, avoiding equipment damage due to water shortage.

7.1.14 Pressure setpoint of supply ring pipe about soft water: The target pressure value of the soft water supply ring pipe set by the operator according to process requirements, used to automatically adjust the water supply system and maintain the ring pipe pressure stability.

7.1.15 Upper limit setpoint of pressure of blow-off valve: The upper limit threshold for the action of the blow-off valve. When the system pressure exceeds this value, the blow-off valve automatically opens to relieve pressure, preventing overpressure accidents.

7.1.16 Lower limit setpoint of pressure of blow-off valve: The lower limit threshold for closing the blow-off valve. When the system pressure falls below this value, the blow-off valve automatically closes to ensure the normal operating pressure of the system.

7.1.17 Upper limit setpoint of pressure of pressurization valve: The upper limit threshold for stopping pressurization of the pressurization valve. When the system pressure reaches this value, the pressurization valve closes to prevent overpressure.

7.1.18 Lower limit setpoint of pressure of pressurization valve: The lower limit threshold for starting pressurization of the pressurization valve. When the system pressure falls below this value, the pressurization valve opens to supplement medium to the system to increase pressure.

7.1.19 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast supply system - Expansion tank - Pressure control”.

7.2 Blast supply system - Valve group - Hot blast control

7.2.1 Hot blast temperature - Hot blast stove side: The measured temperature of the hot blast at the outlet of the hot blast stove, reflecting the heat supply capacity of the hot blast stove, and is an important feedback parameter for combustion control of the hot blast stove.

7.2.2 Hot blast temperature - Blast furnace side: The measured temperature of the hot blast before entering the blast furnace, which is the actual control value of the blast furnace hot blast temperature, directly affecting the smelting intensity and furnace condition of the blast furnace.

7.2.3 Hot blast pressure - Hot blast stove side: The static pressure of the hot blast at the outlet of the hot blast stove, reflecting the blast pressure of the hot blast stove, and is a basic parameter for pressure control of the hot blast system.

7.2.4 Hot blast pressure - Blast furnace side: The static pressure of the hot blast before entering the blast furnace, which is the actual hot blast pressure received by the blast furnace, and has an important impact on the permeability of the blast furnace and the smelting process.

7.2.5 Lower limit setpoint of opening about cold air flow control valve: The minimum allowable opening threshold of the cold air flow control valve, preventing the valve from closing too much, which would cause too low a cold air flow and affect normal blast furnace production.

7.2.6 Feedback value of opening about cold air flow control valve: The actual opening feedback signal of the cold air flow control valve. Compared with the setpoint, it is used to determine whether the valve has moved in place and whether there is jamming or failure.

7.2.7 Temperature setpoint of mixed air: The target temperature of the mixed air (after mixing cold air and hot air) set by the operator, used to automatically adjust the opening of the mixing valve to precisely control the hot blast temperature entering the furnace.

7.2.8 Feedback value of opening about mixed air valve: The actual opening feedback signal of the mixing valve, used to monitor whether the valve is acting as required to ensure stable mixed air temperature.

7.2.9 Setpoint of opening about mixed air valve: The target opening of the mixing valve set by the operator, used to manually or automatically adjust the mixing ratio of cold and hot air to control the hot blast temperature entering the furnace.

7.2.10 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Blast supply system - Valve group - Hot blast control”.

7.3 Hot blast supply system - Combustion air fan - Operation monitoring

7.3.1 Flow rate of carbon monoxide in 1# heat exchanger outlet: The instantaneous flow rate of carbon monoxide in the flue gas at the outlet of No. 1 heat exchanger, reflecting the combustion efficiency and the degree of complete gas combustion, and is an environmental protection and energy saving monitoring indicator.

7.3.2 Flow rate of carbon monoxide in 2# heat exchanger outlet: The instantaneous flow rate of carbon monoxide in the flue gas at the outlet of No. 2 heat exchanger, reflecting the combustion efficiency and the degree of complete gas combustion, and is an environmental protection and energy saving monitoring indicator.

7.3.3 Air main pipe pressure in 1# heat exchanger outlet: The static pressure of the air main pipe at the outlet of No. 1 heat exchanger, reflecting the pressure state of air delivery after the heat exchanger, and is a pressure monitoring point for the combustion air system.
7.3.4 Air main pipe pressure in 2# heat exchanger outlet: The static pressure of the air main pipe at the outlet of No. 2 heat exchanger, reflecting the pressure state of air delivery after the heat exchanger, and is a pressure monitoring point for the combustion air system.
7.3.5 Pressure of cold air main pipe: The static pressure of air in the cold air main pipe, which is a core parameter of the cold air system supply pressure, directly determining whether cold air can be smoothly delivered to the hot blast stove and blast furnace.
7.3.6 Flow rate of cold air main pipe: The volume of air flowing through the cold air main pipe per unit time, reflecting the total air supply of the cold air system, which must meet the blast furnace smelting demand.
7.3.7 Temperature of cold air main pipe: The temperature of air in the cold air main pipe, which is a basic temperature parameter of the cold air system, affecting the combustion efficiency of the hot blast stove and the hot blast temperature of the blast furnace.
7.3.8 Pressure of hot blast main pipe: The static pressure of hot blast in the hot blast main pipe, which is the final reflection of the blast furnace blast pressure, and has a direct impact on the stability of the blast furnace condition and smelting intensity.
7.3.9 Pressure of nitrogen main pipe: The static pressure of gas in the nitrogen main pipe, which is the pressure reference of the entire nitrogen system, determining the pressure supply capacity of each branch nitrogen pipeline.
7.3.10 Temperature of nitrogen main pipe: The temperature of gas in the nitrogen main pipe, reflecting the temperature change of nitrogen during transportation, used to monitor pipeline insulation and medium state.
7.3.11 Flow rate of nitrogen main pipe: The volume of nitrogen flowing through the nitrogen main pipe per unit time, reflecting the total gas supply of the nitrogen system, used to monitor nitrogen consumption and distribution.
7.3.12 Bearing temperature of support end in 1# combustion air fan: The measured temperature of the support end (non-drive end) bearing of No. 1 combustion air fan, used to monitor the bearing operation state, preventing bearing overheating damage due to poor lubrication or overload.
7.3.13 Bearing temperature of thrust end in 1# combustion air fan: The measured temperature of the thrust end (bearing axial thrust) bearing of No. 1 combustion air fan, reflecting the operating state of the bearing under axial load, and is an important parameter for judging bearing wear and lubrication condition.
7.3.14 Bearing temperature of extension end in 1# combustion air fan: The measured temperature of the extension end (shaft extension end) bearing of No. 1 combustion air fan, used to monitor the thermal state of this end bearing. Compared with the support end temperature, it can determine whether the bearing temperature difference is normal.
7.3.15 Bearing temperature of non-shaft extension end in 1# combustion air fan: The measured temperature of the non-shaft extension end (the end opposite the extension end) bearing of No. 1 combustion air fan, used to comprehensively evaluate the overall thermal balance state of the fan bearings.
7.3.16 Temperature of motor stator U in 1# combustion air fan: The measured temperature of the U-phase stator winding of the motor of No. 1 combustion air fan, used to monitor the insulation state of the motor winding, preventing winding overheating and burnout due to overload or poor heat dissipation.
7.3.17 Temperature of motor stator V in 1# combustion air fan: The measured temperature of the V-phase stator winding of the motor of No. 1 combustion air fan, together with the U-phase and W-phase temperatures, determines whether the three-phase winding temperatures of the motor are balanced, and timely detects motor

abnormalities.

7.3.18 Temperature of motor stator W in 1# combustion air fan: The measured temperature of the W-phase stator winding of the motor of No. 1 combustion air fan, used to monitor the thermal state of this phase winding, preventing insulation damage due to single-phase overheating.

7.3.19 X-Direction bearing amplitude of support end in 1# combustion air fan: The vibration amplitude of the support end bearing of No. 1 combustion air fan in the X direction (horizontal direction), used to judge the rotor dynamic balance and bearing operation state of the fan. Excessive X-direction amplitude may indicate rotor imbalance or abnormal bearing clearance.

7.3.20 Y-Direction bearing amplitude of support end in 1# combustion air fan: The vibration amplitude of the support end bearing of No. 1 combustion air fan in the Y direction (vertical direction). Combined with the X-direction amplitude, it comprehensively evaluates the fan vibration condition. Excessive Y-direction amplitude may indicate foundation loosening or bearing wear.

7.3.21 X-Direction bearing amplitude of thrust end in 1# combustion air fan: The vibration amplitude of the thrust end bearing of No. 1 combustion air fan in the X direction, reflecting the vibration state of the thrust end bearing in the horizontal direction, used to judge the operational stability of this end bearing.

7.3.22 Y-Direction bearing amplitude of thrust end in 1# combustion air fan: The vibration amplitude of the thrust end bearing of No. 1 combustion air fan in the Y direction, reflecting the vibration state of the thrust end bearing in the vertical direction. Combined with the X-direction amplitude, it comprehensively judges whether the thrust end bearing operation is normal.

7.3.23 Current of 1# combustion air fan: The working current of the motor of No. 1 combustion air fan, reflecting the actual load of the fan, used to determine whether the fan is overloaded or operating abnormally. Sudden current changes may indicate mechanical failure.

7.3.24 Bearing temperature of support end in 2# combustion air fan: The measured temperature of the support end (non-drive end) bearing of No. 2 combustion air fan, used to monitor the bearing operation state, preventing bearing overheating damage due to poor lubrication or overload.

7.3.25 Bearing temperature of thrust end in 2# combustion air fan: The measured temperature of the thrust end bearing of No. 2 combustion air fan, reflecting the operating state of the bearing under axial load, and is an important parameter for judging bearing wear and lubrication condition.

7.3.26 Bearing temperature of extension end in 2# combustion air fan: The measured temperature of the extension end bearing of No. 2 combustion air fan, used to monitor the thermal state of this end bearing. Compared with the support end temperature, it can determine whether the bearing temperature difference is normal.

7.3.27 Bearing temperature of non-shaft extension end in 2# combustion air fan: The measured temperature of the non-shaft extension end bearing of No. 2 combustion air fan, used to comprehensively evaluate the overall thermal balance state of the fan bearings.

7.3.28 Temperature of motor stator U in 2# combustion air fan: The measured temperature of the U-phase stator winding of the motor of No. 2 combustion air fan, used to monitor the insulation state of the motor winding, preventing winding overheating and burnout due to overload or poor heat dissipation.

7.3.29 Temperature of motor stator V in 2# combustion air fan: The measured temperature of the V-phase stator winding of the motor of No. 2 combustion air fan, together with the U-phase and W-phase temperatures, determines whether the three-phase winding temperatures of the motor are balanced, and timely detects motor abnormalities.

7.3.30 Temperature of motor stator W in 2# combustion air fan: The measured temperature of the W-phase stator winding of the motor of No. 2 combustion air fan, used to monitor the thermal state of this phase winding, preventing insulation damage due to single-phase overheating.

7.3.31 X-Direction bearing amplitude of support end in 2# combustion air fan: The vibration amplitude of the support end bearing of No. 2 combustion air fan in the X direction, used to judge the rotor dynamic balance and bearing operation state of the fan. Excessive X-direction amplitude may indicate rotor imbalance or abnormal bearing clearance.

7.3.32 Y-Direction bearing amplitude of support end in 2# combustion air fan: The vibration amplitude of the support end bearing of No. 2 combustion air fan in the Y direction. Combined with the X-direction amplitude, it comprehensively evaluates the fan vibration condition. Excessive Y-direction amplitude may indicate foundation loosening or bearing wear.

7.3.33 X-Direction bearing amplitude of thrust end in 2# combustion air fan: The vibration amplitude of the thrust end bearing of No. 2 combustion air fan in the X direction, reflecting the vibration state of the thrust end bearing in the horizontal direction, used to judge the operational stability of this end bearing.

7.3.34 Y-Direction bearing amplitude of thrust end in 2# combustion air fan: The vibration amplitude of the thrust end bearing of No. 2 combustion air fan in the Y direction, reflecting the vibration state of the thrust end bearing in the vertical direction. Combined with the X-direction amplitude, it comprehensively judges whether the thrust end bearing operation is normal.

7.3.35 Current of 2# combustion air fan: The working current of the motor of No. 2 combustion air fan, reflecting the actual load of the fan, used to determine whether the fan is overloaded or operating abnormally. Sudden current changes may indicate mechanical failure.

7.3.36 Pressure of preheated combustion air: The static pressure of the combustion air pipeline after preheating by the heat exchanger, reflecting the delivery capacity of the preheating system to the combustion air and the pressure loss.

7.3.37 Temperature of preheated combustion air: The temperature of the combustion air after preheating by the heat exchanger. It is a key parameter for improving the combustion efficiency of the hot blast stove and reducing gas consumption. The higher the temperature, the more favorable the combustion.

7.3.38 Pressure of preheated coal gas: The static pressure of the gas pipeline after preheating by the heat exchanger, reflecting the delivery capacity of the preheating system to the gas and the pressure loss, and is an important parameter for ensuring stable combustion.

7.3.39 Temperature of preheated coal gas: The temperature of the gas after preheating by the heat exchanger. Increasing the gas temperature helps stabilize combustion and reduce heat loss, and is an important indicator for the energy-saving operation of the hot blast stove.

7.3.40 PV value of 1# combustion air fan: The process variable of No. 1 combustion air fan, usually referring to real-time measured values such as fan outlet pressure, flow rate, or vibration, used to monitor the fan operation state and participate in automatic regulation.

7.3.41 VI value of 1# combustion air fan: The vibration intensity value of No. 1 combustion air fan, comprehensively reflecting the vibration severity of the fan, used to judge whether the mechanical state of the fan is good. An excessive VI value indicates possible imbalance, misalignment, or bearing failure.

7.3.42 SP value of 1# combustion air fan: The setpoint value of No. 1 combustion air fan, usually referring to target control values such as pressure or flow rate, used for the automatic regulation system to make the process variable approach this setpoint.

7.3.43 CV value of 1# combustion air fan: The control output value of No. 1 combustion air fan, such as the inverter frequency or valve opening, is the output quantity of the control system execution instruction, used to regulate the fan operating condition.

7.3.44 PV value of 2# combustion air fan: The process variable of No. 2 combustion air fan, usually referring to real-time measured values such as fan outlet pressure, flow rate, or vibration, used to monitor the fan operation state and participate in automatic regulation.

7.3.45 VI value of 2# combustion air fan: The vibration intensity value of No. 2 combustion air fan, comprehensively reflecting the vibration severity of the fan, used to judge whether the mechanical state of the fan is good. An excessive VI value indicates possible imbalance, misalignment, or bearing failure.

7.3.46 SP value of 2# combustion air fan: The setpoint value of No. 2 combustion air fan, usually referring to target control values such as pressure or flow rate, used for the automatic regulation system to make the process variable approach this setpoint.

7.3.47 CV value of 2# combustion air fan: The control output value of No. 2 combustion air fan, such as the inverter frequency or valve opening, is the output quantity of the control system execution instruction, used to regulate the fan operating condition.

7.3.48 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast supply system - Combustion air fan - Operation monitoring”.

7.4 Hot blast supply system - Combustion air fan - System configuration

7.4.1 Upper limit setpoint of bearing temperature of support end: The maximum allowable temperature threshold for the support end bearing of the fan set by the operator. Exceeding this value triggers an alarm or interlock shutdown to prevent the bearing from burning out due to overheating.

7.4.2 Upper limit setpoint of bearing temperature of thrust end: The maximum allowable temperature threshold for the thrust end bearing of the fan set by the operator. Exceeding this value triggers an alarm or interlock shutdown to protect the thrust bearing from high-temperature damage.

7.4.3 Upper limit setpoint of bearing temperature of extension end: The maximum allowable temperature threshold for the extension end bearing of the fan set by the operator. Exceeding this value triggers an alarm or interlock shutdown to prevent this end bearing from failing due to overheating.

7.4.4 Upper limit setpoint of bearing temperature of non-shaft extension end: The maximum allowable temperature threshold for the non-shaft extension end bearing of the fan set by the operator. Exceeding this value triggers an alarm, indicating a risk of overheating for this end bearing.

7.4.5 Upper limit setpoint of temperature of motor stator U: The maximum allowable temperature threshold for the U-phase stator winding of the motor set by the operator. Exceeding this value triggers an alarm or shutdown to protect the motor winding insulation from damage.

7.4.6 Upper limit setpoint of temperature of motor stator V: The maximum allowable temperature threshold for the V-phase stator winding of the motor set by the operator. Exceeding this value triggers an alarm or shutdown to prevent this phase winding from overheating and damage.

7.4.7 Upper limit setpoint of temperature of motor stator W: The maximum allowable temperature threshold for the W-phase stator winding of the motor set by the operator. Exceeding this value triggers an alarm or shutdown to ensure the safe operation of the three-phase windings of the motor.

7.4.8 Upper limit setpoint of X-Direction bearing amplitude of support end: The maximum allowable vibration amplitude in the X direction for the support end bearing set by the operator. Exceeding this value triggers an alarm to prevent equipment damage due to excessive vibration.

7.4.9 Upper limit setpoint of Y-Direction bearing amplitude of support end: The maximum allowable vibration amplitude in the Y direction for the support end bearing set by the operator. Exceeding this value triggers an alarm, indicating abnormal vertical vibration of the fan.

7.4.10 Upper limit setpoint of X-Direction bearing amplitude of thrust end: The maximum allowable vibration amplitude in the X direction for the thrust end bearing set by the operator. Exceeding this value triggers an alarm, reflecting excessive horizontal vibration of the thrust end.

7.4.11 Upper limit setpoint of Y-Direction bearing amplitude of thrust end: The maximum allowable vibration amplitude in the Y direction for the thrust end bearing set by the operator. Exceeding this value triggers an alarm, indicating abnormal vertical vibration of the thrust end.

7.4.12 Upper limit setpoint of current of combustion air fan: The maximum allowable working current of the motor of the combustion air fan set by the operator. Exceeding this value triggers an alarm or interlock to prevent motor overload burnout.

7.4.13 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Hot blast supply system - Combustion air fan - System configuration".

7.5 Hot blast supply system - Hydraulic station - Operation monitoring

7.5.1 Fault about hydraulic station system: A comprehensive fault signal of the hydraulic station system. When the hydraulic station experiences problems such as abnormal pressure, excessively high oil temperature, low liquid level, or motor overload, an alarm is triggered to prompt the operator to troubleshoot.

7.5.2 Start about 1# pump oil pump: The operation status feedback signal of No. 1 hydraulic oil pump, indicating whether the oil pump is started and running, used to monitor the working status of the oil pump.

7.5.3 Start about 2# pump oil pump: The operation status feedback signal of No. 2 hydraulic oil pump, indicating whether the oil pump is started and running, used to monitor the working status of the oil pump.

7.5.4 Low oil temperature of hydraulic station: An alarm signal when the oil temperature of the hydraulic station falls below the lower limit setpoint. Too low an oil temperature will cause the hydraulic oil viscosity to be too high, affecting the system response speed and actuator action.

7.5.5 High oil temperature of hydraulic station: An alarm signal when the oil temperature of the hydraulic station exceeds the upper limit setpoint. Too high an oil temperature will accelerate oil oxidation, damage seals, and affect system stability and service life.

7.5.6 Motor overload of hydraulic station: An action signal of the overload protection of the hydraulic station motor. When the motor current exceeds the set value, it is triggered, indicating possible motor jamming, excessive load, or power supply failure.

7.5.7 High liquid level of hydraulic station: An alarm signal when the oil level in the hydraulic station tank exceeds the upper limit setpoint. It may be caused by incorrect water addition, system leakage, or cooling water entering the oil, requiring timely troubleshooting.

7.5.8 Low liquid level of hydraulic station: An alarm signal when the oil level in the hydraulic station tank falls below the lower limit setpoint, indicating insufficient oil, requiring timely replenishment to prevent the oil pump from sucking air and being damaged.

7.5.9 Extremely low liquid level of hydraulic station: An emergency alarm signal when the oil level in the hydraulic station tank reaches an extremely low value. At this time, the oil pump may no longer work properly, requiring immediate shutdown and oil replenishment.

7.5.10 Extremely high pressure of hydraulic station: An emergency alarm signal when the system pressure of the hydraulic station reaches an extremely high value. It may be caused by a malfunctioning relief valve or pipeline blockage, requiring emergency treatment to prevent system damage.

7.5.11 Extremely high oil temperature of hydraulic station: An emergency alarm signal when the oil temperature of the hydraulic station reaches an extremely high value. The oil temperature is too high and has seriously threatened the system safety, requiring immediate shutdown for cooling and cause investigation.

7.5.12 Circulation oil filter clogged: A blockage alarm signal for the circulation filter of the hydraulic station, indicating that the filter element is clogged with contaminants, requiring cleaning or replacement of the filter element to ensure oil cleanliness.

7.5.13 1# return oil filter clogged: A blockage alarm signal for the No. 1 return oil filter, indicating that the return oil filter element is clogged with contaminants, requiring timely cleaning or replacement to prevent poor return flow.

7.5.14 2# return oil filter clogged: A blockage alarm signal for the No. 2 return oil filter, indicating that the return oil filter element is clogged with contaminants, requiring timely cleaning or replacement to prevent poor return flow.

7.5.15 Current of 1# pump: The working current of the motor of No. 1 hydraulic oil pump, used to monitor the oil pump load and operation status. An abnormally high current indicates pump wear or increased system resistance.

7.5.16 Temperature of 1# pump: The measured temperature of the No. 1 hydraulic oil pump body, reflecting the thermal state of the pump body, preventing seal failure or pump damage due to overheating.

7.5.17 Current of 2# pump: The working current of the motor of No. 2 hydraulic oil pump, used to monitor the oil pump load and operation status. An abnormally high current indicates pump wear or increased system resistance.

7.5.18 Temperature of 2# pump: The measured temperature of the No. 2 hydraulic oil pump body, reflecting the thermal state of the pump body, preventing seal failure or pump damage due to overheating.

7.5.19 Oil pressure of hydraulic station: The pressure of the main oil circuit of the hydraulic station system. It is a core parameter for the operation of the hydraulic system, reflecting the system output force and the action capability of the actuators. Abnormal pressure directly affects equipment operation.

7.5.20 Oil temperature of hydraulic station: The measured temperature of the hydraulic oil in the tank of the hydraulic station. It is an important parameter for monitoring the thermal balance of the hydraulic system. Too high or too low an oil temperature affects system performance.

7.5.21 Liquid level of hydraulic station: The liquid level height of the hydraulic oil in the tank of the hydraulic station. It is used to determine whether the oil volume is sufficient and is a guarantee for the normal operation of the system. Too low a liquid level requires timely oil replenishment.

7.5.22 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast supply system - Hydraulic station - Operation monitoring”.

7.6 Hot blast supply system - Hydraulic station - System configuration

7.6.1 1# Oil pump interlock start: The status signal of No. 1 hydraulic oil pump automatically starting due to interlock conditions (such as main pump failure, low pressure, etc.), used to monitor whether the interlock control logic is executed normally.

7.6.2 2# Oil pump interlock start: The status signal of No. 2 hydraulic oil pump automatically starting due to interlock conditions (such as main pump failure, low pressure, etc.), used to monitor whether the interlock control logic is executed normally.

7.6.3 1# Oil pump start: The feedback signal of the manual or automatic start command for No. 1 hydraulic oil pump, used to confirm whether the oil pump has been successfully started.

7.6.4 2# Oil pump start: The feedback signal of the manual or automatic start command for No. 2 hydraulic oil pump, used to confirm whether the oil pump has been successfully started.

7.6.5 Circulation pump start: The operation status feedback signal of the hydraulic station circulation pump.

The circulation pump is used for oil cooling and filtration. After starting, it ensures stable oil temperature and cleanliness.

7.6.6 Upper limit setpoint of oil temperature about hydraulic station: The maximum allowable oil temperature of the hydraulic station set by the operator. Exceeding this value triggers an alarm to prevent oil degradation and system failure.

7.6.7 Lower limit setpoint of oil temperature about hydraulic station: The minimum allowable oil temperature of the hydraulic station set by the operator. Falling below this value triggers an alarm or starts heating to ensure oil fluidity and system response speed.

7.6.8 Upper limit setpoint of oil pressure about hydraulic station: The maximum allowable system pressure of the hydraulic station set by the operator. Exceeding this value triggers an alarm to prevent pipeline rupture or component damage.

7.6.9 Lower limit setpoint of oil pressure about hydraulic station: The minimum allowable system pressure of the hydraulic station set by the operator. Falling below this value triggers an alarm, indicating insufficient pressure, which may affect the action of the actuators.

7.6.10 Upper limit setpoint of Liquid Level about hydraulic station: The maximum allowable oil level in the tank of the hydraulic station set by the operator. Exceeding this value triggers an alarm to prevent oil overflow causing environmental pollution or safety hazards.

7.6.11 Lower limit setpoint of Liquid Level about hydraulic station: The minimum allowable oil level in the tank of the hydraulic station set by the operator. Falling below this value triggers an alarm, indicating the need for oil replenishment to prevent the oil pump from sucking air and being damaged.

7.6.12 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast supply system - Hydraulic station - System configuration”.

7.7 Hot blast supply system - Hot blast stove - Operation monitoring

7.7.1 1#Hot stove - Flame detector fault: A fault alarm signal for the flame detector of No. 1 hot blast stove. Triggered when the flame detector itself fails or cannot detect a flame, requiring timely troubleshooting to prevent combustion safety accidents.

7.7.2 2#Hot stove - Flame detector fault: A fault alarm signal for the flame detector of No. 2 hot blast stove. Triggered when the flame detector itself fails or cannot detect a flame, requiring timely troubleshooting to prevent combustion safety accidents.

7.7.3 3#Hot stove - Flame detector fault: A fault alarm signal for the flame detector of No. 3 hot blast stove. Triggered when the flame detector itself fails or cannot detect a flame, requiring timely troubleshooting to prevent combustion safety accidents.

7.7.4 1#Hot stove - Semiautomatic operation mode activated: An indication signal that No. 1 hot blast stove is in semi-automatic operation mode, where some operations are manually intervened and some are automatically controlled by the program, used to monitor the operation mode state.

7.7.5 2#Hot stove - Semiautomatic operation mode activated: An indication signal that No. 2 hot blast stove is in semi-automatic operation mode, used to monitor the operation mode state.

7.7.6 3#Hot stove - Semiautomatic operation mode activated: An indication signal that No. 3 hot blast stove is in semi-automatic operation mode, used to monitor the operation mode state.

7.7.7 1# Hydraulic station oil pump start: The operation status feedback signal of the oil pump of the hydraulic station for No. 1 hot blast stove, indicating whether the oil pump is started, used to monitor the oil supply state of the hydraulic system.

7.7.8 2# Hydraulic station oil pump start: The operation status feedback signal of the oil pump of the hydraulic station for No. 2 hot blast stove, indicating whether the oil pump is started, used to monitor the oil supply state of the hydraulic system.
7.7.9 Pressure of main coal gas pipe before heat exchanger: The static pressure of the main gas pipe before the gas enters the heat exchanger, reflecting the pressure state of the gas supply system, and is an important input parameter for the heat exchanger and combustion system.
7.7.10 Pressure of main coal gas pipe after heat exchanger: The static pressure of the main gas pipe after the gas passes through the heat exchanger. Compared with the pressure before the heat exchanger, it can determine the heat exchanger resistance and gas pressure loss, used to evaluate the operation state of the heat exchanger.
7.7.11 Pressure of main air gas pipe before heat exchanger: The static pressure of the main combustion air pipe before the air enters the heat exchanger, reflecting the pressure of the air supply system, and is a basic parameter of the combustion air system.
7.7.12 Pressure of main air gas pipe after heat exchanger: The static pressure of the main combustion air pipe after the air passes through the heat exchanger. Compared with the pressure before the heat exchanger, it can determine the air-side resistance of the heat exchanger, used to monitor whether the heat exchanger is blocked.
7.7.13 Pressure of main flue gas pipe before heat exchanger: The static pressure of the main flue gas pipe before the flue gas enters the heat exchanger, reflecting the pressure of the flue gas system, used to judge the inlet conditions of the heat exchanger and the flow state of the flue gas.
7.7.14 Pressure of main flue gas pipe after heat exchanger: The static pressure of the main flue gas pipe after the flue gas passes through the heat exchanger. Compared with the pressure before the heat exchanger, it can determine the flue gas side resistance and blockage of the heat exchanger, used to guide heat exchanger maintenance.
7.7.15 Carbon monoxide content of flue gas at heat exchanger waste gas outlet: The volume concentration of carbon monoxide in the flue gas at the waste gas outlet of the heat exchanger, reflecting the degree of complete combustion, and is a key indicator for environmental protection emissions and combustion efficiency.
7.7.16 Temperature of main coal gas pipe before preheating: The temperature of the gas before entering the preheater, which is a basic input parameter of the preheating system, used to evaluate the preheating effect and heat recovery efficiency.
7.7.17 Temperature of main coal gas pipe after preheating: The temperature of the gas after preheating, reflecting the preheating effect, and is an important indicator for improving combustion efficiency and reducing gas consumption.
7.7.18 1#Hot stove - Inlet pressure of main cold air pipe: The static pressure of air at the inlet of the cold air main pipe of No. 1 hot blast stove. It is a pressure parameter for the cold air system supplying air to this hot blast stove, used to monitor the cold air supply state.
7.7.19 1#Hot stove - Outlet pressure of main cold air pipe: The static pressure of air at the outlet of the cold air main pipe of No. 1 hot blast stove, reflecting the pressure loss of the cold air after passing through the hot blast stove, used to judge the internal resistance of the hot blast stove.
7.7.20 1#Hot stove - Outlet pressure of main waste gas pipe: The static pressure of the flue gas at the outlet of the waste gas main pipe of No. 1 hot blast stove, reflecting the resistance of the waste gas system and the chimney draft, used to monitor the operation state of the exhaust system.
7.7.21 1#Hot stove - Inlet temperature of main flue gas pipe: The temperature of the flue gas at the inlet of

the flue gas main pipe of No. 1 hot blast stove, reflecting the exhaust gas temperature of the hot blast stove, and is an important parameter for thermal efficiency control. Too high an exhaust gas temperature indicates large heat loss.

7.7.22 1#Hot stove - Outlet temperature of main flue gas pipe: The temperature of the flue gas at the outlet of the flue gas main pipe of No. 1 hot blast stove. Compared with the inlet temperature, it can determine the flue gas cooling or waste heat recovery effect, used to evaluate the efficiency of the heat exchanger.

7.7.23 2#Hot stove - Inlet pressure of main cold air pipe: The static pressure of air at the inlet of the cold air main pipe of No. 2 hot blast stove. It is a pressure parameter for the cold air system supplying air to this hot blast stove, used to monitor the cold air supply state.

7.7.24 2#Hot stove - Outlet pressure of main cold air pipe: The static pressure of air at the outlet of the cold air main pipe of No. 2 hot blast stove, reflecting the pressure loss of the cold air after passing through the hot blast stove, used to judge the internal resistance of the hot blast stove.

7.7.25 2#Hot stove - Outlet pressure of main waste gas pipe: The static pressure of the flue gas at the outlet of the waste gas main pipe of No. 2 hot blast stove, reflecting the resistance of the waste gas system and the chimney draft, used to monitor the operation state of the exhaust system.

7.7.26 2#Hot stove - Inlet temperature of main flue gas pipe: The temperature of the flue gas at the inlet of the flue gas main pipe of No. 2 hot blast stove, reflecting the exhaust gas temperature of the hot blast stove, and is an important parameter for thermal efficiency control. Too high an exhaust gas temperature indicates large heat loss.

7.7.27 2#Hot stove - Outlet temperature of main flue gas pipe: The temperature of the flue gas at the outlet of the flue gas main pipe of No. 2 hot blast stove. Compared with the inlet temperature, it can determine the flue gas cooling or waste heat recovery effect, used to evaluate the efficiency of the heat exchanger.

7.7.28 3#Hot stove - Inlet pressure of main cold air pipe: The static pressure of air at the inlet of the cold air main pipe of No. 3 hot blast stove. It is a pressure parameter for the cold air system supplying air to this hot blast stove, used to monitor the cold air supply state.

7.7.29 3#Hot stove - Outlet pressure of main cold air pipe: The static pressure of air at the outlet of the cold air main pipe of No. 3 hot blast stove, reflecting the pressure loss of the cold air after passing through the hot blast stove, used to judge the internal resistance of the hot blast stove.

7.7.30 3#Hot stove - Outlet pressure of main waste gas pipe: The static pressure of the flue gas at the outlet of the waste gas main pipe of No. 3 hot blast stove, reflecting the resistance of the waste gas system and the chimney draft, used to monitor the operation state of the exhaust system.

7.7.31 3#Hot stove - Inlet temperature of main flue gas pipe: The temperature of the flue gas at the inlet of the flue gas main pipe of No. 3 hot blast stove, reflecting the exhaust gas temperature of the hot blast stove, and is an important parameter for thermal efficiency control. Too high an exhaust gas temperature indicates large heat loss.

7.7.32 3#Hot stove - Outlet temperature of main flue gas pipe: The temperature of the flue gas at the outlet of the flue gas main pipe of No. 3 hot blast stove. Compared with the inlet temperature, it can determine the flue gas cooling or waste heat recovery effect, used to evaluate the efficiency of the heat exchanger.

7.7.33 Temperature of main combustion air pipe before heat exchanger: The temperature of the combustion air before entering the heat exchanger, which is a basic input of the preheating system, used to evaluate the preheating effect and heat recovery efficiency.

7.7.34 Pressure of main combustion air pipe before heat exchanger: The static pressure of the combustion air before entering the heat exchanger, reflecting the air supply pressure, and is a basic parameter of the combustion air system.

7.7.35 Temperature of main combustion air pipe after heat exchanger: The temperature of the combustion air after passing through the heat exchanger, reflecting the preheating effect. The higher the temperature, the more favorable for improving the combustion efficiency of the hot blast stove.
7.7.36 Pressure of main combustion air pipe after heat exchanger: The static pressure of the combustion air after passing through the heat exchanger. Compared with the pressure before the heat exchanger, it can determine the air-side resistance of the heat exchanger, used to monitor whether the heat exchanger is blocked.
7.7.37 1#Hot stove - Dome temperature: The measured temperature at the dome of No. 1 hot blast stove. It is a key parameter for the heat storage capacity and combustion control of the hot blast stove. Too high a dome temperature may damage the refractory material, while too low a temperature indicates insufficient heat storage.
7.7.38 2#Hot stove - Dome temperature: The measured temperature at the dome of No. 2 hot blast stove. It is a key parameter for the heat storage capacity and combustion control of the hot blast stove.
7.7.39 3#Hot stove - Dome temperature: The measured temperature at the dome of No. 3 hot blast stove. It is a key parameter for the heat storage capacity and combustion control of the hot blast stove.
7.7.40 1#Hot stove - Carbon monoxide content of flue: The volume concentration of carbon monoxide in the flue gas of the flue of No. 1 hot blast stove, reflecting combustion efficiency, used to optimize the air-fuel ratio. Too high a CO content indicates incomplete combustion.
7.7.41 1#Hot stove - Carbon monoxide content of heat exchanger outlet: The volume concentration of carbon monoxide in the flue gas at the outlet of the heat exchanger of No. 1 hot blast stove. Compared with the CO content in the flue, it can determine the oxidation effect of the heat exchanger on unburned components.
7.7.42 1#Hot stove - Oxygen content of flue: The volume concentration of oxygen in the flue gas of the flue of No. 1 hot blast stove, used to control the excess air coefficient for combustion, preventing too high an oxygen content in the flue gas leading to heat loss or too low a content leading to incomplete combustion.
7.7.43 2#Hot stove - Carbon monoxide content of flue: The volume concentration of carbon monoxide in the flue gas of the flue of No. 2 hot blast stove, reflecting combustion efficiency, used to optimize the air-fuel ratio. Too high a CO content indicates incomplete combustion.
7.7.44 2#Hot stove - Carbon monoxide content of heat exchanger outlet: The volume concentration of carbon monoxide in the flue gas at the outlet of the heat exchanger of No. 2 hot blast stove. Compared with the CO content in the flue, it can determine the oxidation effect of the heat exchanger on unburned components.
7.7.45 2#Hot stove - Oxygen content of flue: The volume concentration of oxygen in the flue gas of the flue of No. 2 hot blast stove, used to control the excess air coefficient for combustion, preventing too high an oxygen content in the flue gas leading to heat loss or too low a content leading to incomplete combustion.
7.7.46 3#Hot stove - Carbon monoxide content of flue: The volume concentration of carbon monoxide in the flue gas of the flue of No. 3 hot blast stove, reflecting combustion efficiency, used to optimize the air-fuel ratio. Too high a CO content indicates incomplete combustion.
7.7.47 3#Hot stove - Carbon monoxide content of heat exchanger outlet: The volume concentration of carbon monoxide in the flue gas at the outlet of the heat exchanger of No. 3 hot blast stove. Compared with the CO content in the flue, it can determine the oxidation effect of the heat exchanger on unburned components.
7.7.48 3#Hot stove - Oxygen content of flue: The volume concentration of oxygen in the flue gas of the flue of No. 3 hot blast stove, used to control the excess air coefficient for combustion, preventing too high an oxygen content in the flue gas leading to heat loss or too low a content leading to incomplete combustion.
7.7.49 1#Hot stove - Air supply valve of main cold air pipe: The opening or status feedback signal of the air supply valve of the cold air main pipe of No. 1 hot blast stove, used to control the amount of cold air entering the hot blast stove, and is an actuator for air supply regulation of the hot blast stove.

7.7.50 2#Hot stove - Air supply valve of main cold air pipe: The opening or status feedback signal of the air supply valve of the cold air main pipe of No. 2 hot blast stove, used to control the amount of cold air entering the hot blast stove, and is an actuator for air supply regulation of the hot blast stove.
7.7.51 3#Hot stove - Air supply valve of main cold air pipe: The opening or status feedback signal of the air supply valve of the cold air main pipe of No. 3 hot blast stove, used to control the amount of cold air entering the hot blast stove, and is an actuator for air supply regulation of the hot blast stove.
7.7.52 1#Hot stove - Outlet valve of main waste gas pipe: The opening or status feedback signal of the outlet valve of the waste gas main pipe of No. 1 hot blast stove, used to control the amount of flue gas emission, adjusting the internal pressure and combustion condition of the hot blast stove.
7.7.53 2#Hot stove - Outlet valve of main waste gas pipe: The opening or status feedback signal of the outlet valve of the waste gas main pipe of No. 2 hot blast stove, used to control the amount of flue gas emission, adjusting the internal pressure and combustion condition of the hot blast stove.
7.7.54 3#Hot stove - Outlet valve of main waste gas pipe: The opening or status feedback signal of the outlet valve of the waste gas main pipe of No. 3 hot blast stove, used to control the amount of flue gas emission, adjusting the internal pressure and combustion condition of the hot blast stove.
7.7.55 1#Hot stove - Outlet valve of main flue pipe: The opening or status feedback signal of the outlet valve of the flue gas main pipe of No. 1 hot blast stove, used to adjust the chimney draft, affecting the exhaust effect of the hot blast stove and the pressure distribution inside the furnace.
7.7.56 2#Hot stove - Outlet valve of main flue pipe: The opening or status feedback signal of the outlet valve of the flue gas main pipe of No. 2 hot blast stove, used to adjust the chimney draft, affecting the exhaust effect of the hot blast stove and the pressure distribution inside the furnace.
7.7.57 3#Hot stove - Outlet valve of main flue pipe: The opening or status feedback signal of the outlet valve of the flue gas main pipe of No. 3 hot blast stove, used to adjust the chimney draft, affecting the exhaust effect of the hot blast stove and the pressure distribution inside the furnace.
7.7.58 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast supply system - Hot blast stove - Operation monitoring”.

7.8 Hot blast supply system - Hot blast stove - System configuration

7.8.1 1#Hot stove - Flame detector fault: A fault alarm signal for the flame detector of No. 1 hot blast stove. Triggered when the flame detector itself fails or cannot detect a flame, requiring timely troubleshooting to prevent combustion safety accidents.
7.8.2 2#Hot stove - Flame detector fault: A fault alarm signal for the flame detector of No. 2 hot blast stove. Triggered when the flame detector itself fails or cannot detect a flame, requiring timely troubleshooting to prevent combustion safety accidents.
7.8.3 3#Hot stove - Flame detector fault: A fault alarm signal for the flame detector of No. 3 hot blast stove. Triggered when the flame detector itself fails or cannot detect a flame, requiring timely troubleshooting to prevent combustion safety accidents.
7.8.4 1#Hot stove - Operation feedback mode: The feedback signal of the current operation mode of No. 1 hot blast stove, such as manual, automatic, semi-automatic, etc., used to monitor the operation state and control system logic judgment.
7.8.5 2#Hot stove - Operation feedback mode: The feedback signal of the current operation mode of No. 2 hot blast stove, used to monitor the operation state and control system logic judgment.
7.8.6 3#Hot stove - Operation feedback mode: The feedback signal of the current operation mode of No. 3

hot blast stove, used to monitor the operation state and control system logic judgment.
7.8.7 Setpoint value of hot stove changing time: The interval or duration for hot blast stove switching set by the operator, used to automatically control the combustion and blast cycles of the hot blast stove, ensuring continuous and stable air supply to the blast furnace.
7.8.8 Interlock mode of air/(coal gas) combustion valve: The selection of the interlock control mode between the air valve and the gas valve, such as open before close, interlock, etc., used to ensure combustion safety, preventing safety accidents caused by incorrect valve actions.
7.8.9 1#Hot stove - Operating mode selection (Combustion/Braise-oven/Blast): The currently selected operating mode of No. 1 hot blast stove, including three states: combustion, braise-oven, and blast. Used for program control, different modes have different valve group actions and control logic.
7.8.10 2#Hot stove - Operating mode selection (Combustion/Braise-oven/Blast): The currently selected operating mode of No. 2 hot blast stove, used for program control.
7.8.11 3#Hot stove - Operating mode selection (Combustion/Braise-oven/Blast): The currently selected operating mode of No. 3 hot blast stove, used for program control.
7.8.12 1#Hot stove - Upper limit setpoint of blast temperature: The maximum allowable blast temperature for No. 1 hot blast stove. Exceeding this value triggers automatic regulation or an alarm to prevent the blast furnace inlet blast temperature from being too high, affecting the furnace condition or damaging equipment.
7.8.13 2#Hot stove - Upper limit setpoint of blast temperature: The maximum allowable blast temperature for No. 2 hot blast stove. Exceeding this value triggers automatic regulation or an alarm.
7.8.14 3#Hot stove - Upper limit setpoint of blast temperature: The maximum allowable blast temperature for No. 3 hot blast stove. Exceeding this value triggers automatic regulation or an alarm.
7.8.15 1#Hot stove - Upper limit setpoint of blast pressure: The maximum allowable blast pressure for No. 1 hot blast stove. Exceeding this value triggers an alarm or regulation to prevent high pressure from damaging equipment or affecting the stability of the blast furnace.
7.8.16 2#Hot stove - Upper limit setpoint of blast pressure: The maximum allowable blast pressure for No. 2 hot blast stove. Exceeding this value triggers an alarm or regulation.
7.8.17 3#Hot stove - Upper limit setpoint of blast pressure: The maximum allowable blast pressure for No. 3 hot blast stove. Exceeding this value triggers an alarm or regulation.
7.8.18 1#Hot stove - Setpoint of blast temperature: The target blast temperature for No. 1 hot blast stove set by the operator, used to automatically control the mixing system, maintaining the blast temperature stable within the blast furnace process requirements.
7.8.19 2#Hot stove - Setpoint of blast temperature: The target blast temperature for No. 2 hot blast stove set by the operator, used to automatically control the mixing system.
7.8.20 3#Hot stove - Setpoint of blast temperature: The target blast temperature for No. 3 hot blast stove set by the operator, used to automatically control the mixing system.
7.8.21 1#Hot stove - Setpoint of blast time: The duration of a single blast cycle for No. 1 hot blast stove set by the operator, used for automatic stove changing control, ensuring the blast furnace receives a continuous and stable hot blast supply.
7.8.22 2#Hot stove - Setpoint of blast time: The duration of a single blast cycle for No. 2 hot blast stove set by the operator, used for automatic stove changing control.
7.8.23 3#Hot stove - Setpoint of blast time: The duration of a single blast cycle for No. 3 hot blast stove set by the operator, used for automatic stove changing control.
7.8.24 1#Hot stove - Upper limit value of dome temperature: The maximum allowable dome temperature for No. 1 hot blast stove. Exceeding this value triggers an alarm or automatic combustion regulation to

prevent dome overtemperature from damaging the refractory material.

7.8.25 2#Hot stove - Upper limit value of dome temperature: The maximum allowable dome temperature for No. 2 hot blast stove. Exceeding this value triggers an alarm or automatic combustion regulation.

7.8.26 3#Hot stove - Upper limit value of dome temperature: The maximum allowable dome temperature for No. 3 hot blast stove. Exceeding this value triggers an alarm or automatic combustion regulation.

7.8.27 1#Hot stove - Lower limit value of dome temperature: The minimum allowable dome temperature for No. 1 hot blast stove. Falling below this value triggers an alarm or combustion adjustment to prevent insufficient heat storage from affecting the blast temperature.

7.8.28 2#Hot stove - Lower limit value of dome temperature: The minimum allowable dome temperature for No. 2 hot blast stove. Falling below this value triggers an alarm or combustion adjustment.

7.8.29 3#Hot stove - Lower limit value of dome temperature: The minimum allowable dome temperature for No. 3 hot blast stove. Falling below this value triggers an alarm or combustion adjustment.

7.8.30 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast supply system - Hot blast stove - System configuration”.

7.9 Hot blast supply system - Hot blast stove - Stove switching operation monitoring

7.9.1 Hot blast pressure: The static pressure of hot blast in the main hot blast pipe, directly reflecting the blast pressure of the blast furnace, and is crucial for the smelting intensity, permeability, and furnace condition stability of the blast furnace.

7.9.2 Hot blast temperature A - Hot blast stove side: The measured temperature of the hot blast at the outlet of the hot blast stove, reflecting the heat supply capacity of the hot blast stove, and is an important feedback parameter for combustion control of the hot blast stove.

7.9.3 Hot blast temperature B - Blast furnace side: The measured temperature of the hot blast before entering the blast furnace, which is the actual control value of the blast furnace hot blast temperature, directly affecting the smelting intensity and furnace condition of the blast furnace.

7.9.4 Pressure of main cold air pipe: The static pressure of air in the cold air main pipe, which is a core parameter of the cold air system supply pressure, directly determining whether cold air can be smoothly delivered to the hot blast stove and blast furnace.

7.9.5 Flow rate of main cold air pipe: The volume of air flowing through the cold air main pipe per unit time, reflecting the total air supply of the cold air system, which must meet the blast furnace smelting demand.

7.9.6 Humidity of main cold air pipe: The humidity (water content) of the air in the cold air main pipe, used to monitor the impact of cold air humidity on blast furnace smelting and hot blast stove combustion. Too high a humidity may affect combustion efficiency.

7.9.7 Temperature of main cold air pipe: The temperature of the air in the cold air main pipe, which is a basic temperature parameter of the cold air system, affecting the combustion efficiency of the hot blast stove and the hot blast temperature of the blast furnace.

7.9.8 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast supply system - Hot blast stove - Stove switching operation monitoring”.

7.10 Hot blast supply system - Combustion chamber - Operation monitoring

7.10.1 1# Hot stove - Combustion mode (Manual/Automatic): The selection of the combustion control mode for No. 1 hot blast stove. In manual mode, the operator directly controls; in automatic mode, the program automatically controls the combustion process.

7.10.2 2# Hot stove - Combustion mode (Manual/Automatic): The selection of the combustion control mode for No. 2 hot blast stove.

7.10.3 3# Hot stove - Combustion mode (Manual/Automatic): The selection of the combustion control mode for No. 3 hot blast stove.

7.10.4 1# Hot stove - Air-to-fuel ratio: The ratio of air flow to gas flow during combustion of No. 1 hot blast stove. It is a core parameter for combustion control, directly affecting combustion efficiency and exhaust gas composition, and must be optimized according to operating conditions.

7.10.5 1# Hot stove - Combustion time: The duration of a single combustion cycle for No. 1 hot blast stove, used to automatically control the combustion process, ensuring that the hot blast stove obtains sufficient heat storage.

7.10.6 1# Hot stove - Dome temperature: The measured temperature at the dome of No. 1 hot blast stove, which is a key parameter for the heat storage capacity and combustion control of the hot blast stove.

7.10.7 1# Hot stove - Temperature A: The temperature at a specific measuring point of No. 1 hot blast stove, used to assist in monitoring the internal temperature distribution of the hot blast stove. The specific location may vary depending on the design.

7.10.8 1# Hot stove - Temperature B: The temperature at another specific measuring point of No. 1 hot blast stove. Combined with temperature A, it helps analyze the internal temperature distribution and thermal balance state of the hot blast stove.

7.10.9 1# Hot stove - Temperature of flue: The temperature of the flue gas in the flue of No. 1 hot blast stove, reflecting the exhaust heat loss, and is an indicator for monitoring thermal efficiency. Too high an exhaust gas temperature indicates large heat loss.

7.10.10 1# Hot stove - Oxygen content of flue: The volume concentration of oxygen in the flue gas of No. 1 hot blast stove, used to control the excess air coefficient for combustion, preventing too high or too low an oxygen content in the flue gas.

7.10.11 1# Hot stove - Temperature A of stove shell: The temperature at a certain point on the shell of No. 1 hot blast stove, used to monitor the thermal state of the shell and the insulation effect, preventing the shell from overheating or having local hot spots.

7.10.12 1# Hot stove - Temperature B of stove shell: The temperature at another point on the shell of No. 1 hot blast stove. Combined with temperature A, it helps analyze the temperature distribution of the shell and determine whether the insulation layer is intact.

7.10.13 1# Hot stove - Temperature C of stove shell: The temperature at a third point on the shell of No. 1 hot blast stove, further enriching the shell temperature monitoring data for a comprehensive assessment of the shell thermal state.

7.10.14 1# Hot stove - Temperature of combustion chamber: The measured temperature inside the combustion chamber of No. 1 hot blast stove, reflecting the combustion intensity and flame state, used to determine whether the combustion is stable.

7.10.15 1# Hot stove - Chequer brick temperature: The temperature of the checker brick in the regenerator of No. 1 hot blast stove, reflecting the heat storage capacity of the regenerator, and is a key indicator of the heat exchange efficiency of the hot blast stove.

7.10.16 1# Hot stove - Silica brick temperature: The temperature of the silica refractory material inside No. 1 hot blast stove, used to monitor the material state in the high-temperature area, preventing overtemperature from causing material phase change or damage.

7.10.17 1# Hot stove - Temperature of stove grate: The temperature of the stove grate (the structure supporting the checker brick) of No. 1 hot blast stove, reflecting the heat storage state at the bottom and the flue gas distribution, used to judge the working condition at the bottom of the regenerator.

7.10.18 2# Hot stove - Air-to-fuel ratio: The ratio of air flow to gas flow during combustion of No. 2 hot blast stove. It is a core parameter for combustion control, directly affecting combustion efficiency and exhaust gas composition, and must be optimized according to operating conditions.

7.10.19 2# Hot stove - Combustion time: The duration of a single combustion cycle for No. 2 hot blast stove, used to automatically control the combustion process, ensuring that the hot blast stove obtains sufficient heat storage.

7.10.20 2# Hot stove - Dome temperature: The measured temperature at the dome of No. 2 hot blast stove, which is a key parameter for the heat storage capacity and combustion control of the hot blast stove.

7.10.21 2# Hot stove - Temperature A: The temperature at a specific measuring point of No. 2 hot blast stove, used to assist in monitoring the internal temperature distribution of the hot blast stove.

7.10.22 2# Hot stove - Temperature B: The temperature at another specific measuring point of No. 2 hot blast stove. Combined with temperature A, it helps analyze the internal temperature distribution and thermal balance state of the hot blast stove.

7.10.23 2# Hot stove - Temperature of flue: The temperature of the flue gas in the flue of No. 2 hot blast stove, reflecting the exhaust heat loss, and is an indicator for monitoring thermal efficiency. Too high an exhaust gas temperature indicates large heat loss.

7.10.24 2# Hot stove - Oxygen content of flue: The volume concentration of oxygen in the flue gas of No. 2 hot blast stove, used to control the excess air coefficient for combustion, preventing too high or too low an oxygen content in the flue gas.

7.10.25 2# Hot stove - Temperature A of stove shell: The temperature at a certain point on the shell of No. 2 hot blast stove, used to monitor the thermal state of the shell and the insulation effect, preventing the shell from overheating or having local hot spots.

7.10.26 2# Hot stove - Temperature B of stove shell: The temperature at another point on the shell of No. 2 hot blast stove. Combined with temperature A, it helps analyze the temperature distribution of the shell and determine whether the insulation layer is intact.

7.10.27 2# Hot stove - Temperature C of stove shell: The temperature at a third point on the shell of No. 2 hot blast stove, further enriching the shell temperature monitoring data for a comprehensive assessment of the shell thermal state.

7.10.28 2# Hot stove - Temperature of combustion chamber: The measured temperature inside the combustion chamber of No. 2 hot blast stove, reflecting the combustion intensity and flame state, used to determine whether the combustion is stable.

7.10.29 2# Hot stove - Chequer brick temperature: The temperature of the checker brick in the regenerator of No. 2 hot blast stove, reflecting the heat storage capacity of the regenerator, and is a key indicator of the heat exchange efficiency of the hot blast stove.

7.10.30 2# Hot stove - Silica brick temperature: The temperature of the silica refractory material inside No. 2 hot blast stove, used to monitor the material state in the high-temperature area, preventing overtemperature from causing material phase change or damage.

7.10.31 2# Hot stove - Temperature of stove grate: The temperature of the stove grate of No. 2 hot blast stove, reflecting the heat storage state at the bottom and the flue gas distribution, used to judge the working condition at the bottom of the regenerator.

7.10.32 3# Hot stove - Air-to-fuel ratio: The ratio of air flow to gas flow during combustion of No. 3 hot blast stove. It is a core parameter for combustion control, directly affecting combustion efficiency and exhaust

gas composition, and must be optimized according to operating conditions.
7.10.33 3# Hot stove - Combustion time: The duration of a single combustion cycle for No. 3 hot blast stove, used to automatically control the combustion process, ensuring that the hot blast stove obtains sufficient heat storage.
7.10.34 3# Hot stove - Dome temperature: The measured temperature at the dome of No. 3 hot blast stove, which is a key parameter for the heat storage capacity and combustion control of the hot blast stove.
7.10.35 3# Hot stove - Temperature A: The temperature at a specific measuring point of No. 3 hot blast stove, used to assist in monitoring the internal temperature distribution of the hot blast stove.
7.10.36 3# Hot stove - Temperature B: The temperature at another specific measuring point of No. 3 hot blast stove. Combined with temperature A, it helps analyze the internal temperature distribution and thermal balance state of the hot blast stove.
7.10.37 3# Hot stove - Temperature of flue: The temperature of the flue gas in the flue of No. 3 hot blast stove, reflecting the exhaust heat loss, and is an indicator for monitoring thermal efficiency. Too high an exhaust gas temperature indicates large heat loss.
7.10.38 3# Hot stove - Oxygen content of flue: The volume concentration of oxygen in the flue gas of No. 3 hot blast stove, used to control the excess air coefficient for combustion, preventing too high or too low an oxygen content in the flue gas.
7.10.39 3# Hot stove - Temperature A of stove shell: The temperature at a certain point on the shell of No. 3 hot blast stove, used to monitor the thermal state of the shell and the insulation effect, preventing the shell from overheating or having local hot spots.
7.10.40 3# Hot stove - Temperature B of stove shell: The temperature at another point on the shell of No. 3 hot blast stove. Combined with temperature A, it helps analyze the temperature distribution of the shell and determine whether the insulation layer is intact.
7.10.41 3# Hot stove - Temperature C of stove shell: The temperature at a third point on the shell of No. 3 hot blast stove, further enriching the shell temperature monitoring data for a comprehensive assessment of the shell thermal state.
7.10.42 3# Hot stove - Temperature of combustion chamber: The measured temperature inside the combustion chamber of No. 3 hot blast stove, reflecting the combustion intensity and flame state, used to determine whether the combustion is stable.
7.10.43 3# Hot stove - Chequer brick temperature: The temperature of the checker brick in the regenerator of No. 3 hot blast stove, reflecting the heat storage capacity of the regenerator, and is a key indicator of the heat exchange efficiency of the hot blast stove.
7.10.44 3# Hot stove - Silica brick temperature: The temperature of the silica refractory material inside No. 3 hot blast stove, used to monitor the material state in the high-temperature area, preventing overtemperature from causing material phase change or damage.
7.10.45 3# Hot stove - Temperature of stove grate: The temperature of the stove grate of No. 3 hot blast stove, reflecting the heat storage state at the bottom and the flue gas distribution, used to judge the working condition at the bottom of the regenerator.
7.10.46 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Hot blast supply system - Combustion chamber - Operation monitoring”.

7.11 Hot blast supply system - Combustion chamber - System configuration

7.11.1 Main pipe - Braise-oven - Lower limit setpoint of pressure about coal gas: The minimum allowable pressure threshold of the gas pipeline when the hot blast stove is in the braise-oven state. Falling below this value triggers an alarm or interlock to prevent backfire or flame-off due to too low a pressure.

7.11.2 Main pipe - Lower limit setpoint of pressure about coal gas: The minimum allowable pressure threshold of the gas pipeline during normal operation. Falling below this value triggers an alarm to ensure stable combustion, preventing flame-out due to insufficient gas pressure.

7.11.3 Main pipe - Braise-oven - Lower limit setpoint of pressure about air: The minimum allowable pressure threshold of the air pipeline when the hot blast stove is in the braise-oven state. Falling below this value triggers an alarm or interlock to ensure safety in the braise-oven state.

7.11.4 Main pipe - Lower limit setpoint of pressure about air: The minimum allowable pressure threshold of the air pipeline during normal operation. Falling below this value triggers an alarm to ensure stable combustion air supply.

7.11.5 Main pipe - Lower limit setpoint of cold air flow: The minimum allowable flow threshold of the cold air main pipe. Falling below this value triggers an alarm or interlock to prevent insufficient blast furnace air supply from affecting smelting.

7.11.6 Main pipe - Lower limit setpoint of cold air pressure: The minimum allowable pressure threshold of the cold air main pipe. Falling below this value triggers an alarm to ensure the blast furnace blast pressure, preventing the furnace condition from being affected by too low a pressure.

7.11.7 Main pipe - Upper limit setpoint of hot blast pressure: The maximum allowable pressure threshold of the hot blast main pipe. Exceeding this value triggers an alarm or interlock to prevent high pressure from damaging equipment or affecting the stability of the blast furnace.

7.11.8 Main pipe - Lower limit setpoint of hot blast pressure: The minimum allowable pressure threshold of the hot blast main pipe. Falling below this value triggers an alarm, indicating insufficient blast furnace wind pressure, requiring timely troubleshooting.

7.11.9 Main pipe - Humidified blast off - Lower limit setpoint of hot blast pressure: The minimum allowable pressure threshold of the hot blast main pipe when the humidification (wet blast) operation is closed, used for interlock control, preventing abnormal pressure fluctuations due to the closure of humidification.

7.11.10 1#Hot stove - Upper limit setpoint of combustion - Dome temperature: The maximum allowable dome temperature during the combustion phase of No. 1 hot blast stove. Exceeding this value triggers automatic regulation or an alarm to prevent dome overtemperature during the combustion phase.

7.11.11 1#Hot stove - Upper limit setpoint of combustion - Temperature of waste gas: The maximum allowable waste gas temperature during the combustion phase of No. 1 hot blast stove. Exceeding this value triggers an alarm or combustion adjustment to prevent waste gas overheating from affecting equipment or thermal efficiency.

7.11.12 1#Hot stove - Lower limit setpoint of combustion - Dome temperature: The minimum allowable dome temperature during the blast phase of No. 1 hot blast stove. Falling below this value triggers an alarm or early stove change to ensure the blast temperature meets the blast furnace requirements.

7.11.13 1#Hot stove - Lower limit setpoint of combustion - Temperature of waste gas: The minimum allowable waste gas temperature during the blast phase of No. 1 hot blast stove. Falling below this value may indicate insufficient heat storage, requiring adjustment of the combustion or stove change cycle.

7.11.14 1#Hot stove - Upper limit setpoint of dome temperature: The overall maximum allowable dome temperature for No. 1 hot blast stove. Exceeding this value triggers an alarm or automatic regulation to prevent dome overtemperature damage.

7.11.15 1#Hot stove - Upper limit setpoint of temperature of waste gas: The maximum allowable waste gas temperature for No. 1 hot blast stove. Exceeding this value triggers an alarm or combustion adjustment to prevent waste gas overheating.

7.11.16 1#Hot stove - Temperature difference setpoint under ignition failure: The temperature difference threshold for determining ignition failure of No. 1 hot blast stove. When the temperature rise after ignition does not reach this difference, it is determined as ignition failure, triggering safety protection.

7.11.17 1#Hot stove - Time difference setpoint under ignition failure: The time threshold for determining ignition failure of No. 1 hot blast stove. If the temperature does not reach the expected value within the set time, it is determined as ignition failure, triggering safety protection.

7.11.18 1#Hot stove - Opening time of impact valve: The opening duration of the impact valve (pressurization valve) of No. 1 hot blast stove, used to control the amount of pressurization, ensuring that the hot blast stove is evenly pressurized.

7.11.19 1#Hot stove - Closing time of impact valve: The closing duration or interval time of the impact valve of No. 1 hot blast stove, used to control the pressurization cycle, working together with the opening time to achieve pressure equalization control.

7.11.20 1#Hot stove - Hydraulic setpoint under normal operation: The pressure or state setpoint for the normal operation of the hydraulic system of No. 1 hot blast stove, used for interlock protection. When the hydraulic system deviates from the setpoint, an alarm or protection action is triggered.

7.11.21 1#Hot stove - Equalizing pressure setpoint under normal operation: The pressure setpoint for the normal operation of the equalizing pressure system of No. 1 hot blast stove, used to control the pressurization process, ensuring that the pressure equalization meets the requirements.

7.11.22 1#Hot stove - Nitrogen purging time under “combustion → stop”: The duration of nitrogen purging when switching No. 1 hot blast stove from the combustion state to the stop state, used to displace combustible gas, ensuring the safety of the switching process.

7.11.23 1#Hot stove - Nitrogen purging time under “stop → combustion”: The duration of nitrogen purging before switching No. 1 hot blast stove from the stop state to the combustion state, used to displace air, preventing explosion during ignition.

7.11.24 1#Hot stove - Nitrogen purging time under ignition failure: The duration of nitrogen purging after an ignition failure of No. 1 hot blast stove, used to remove unburned gas, preparing for re-ignition and ensuring safety.

7.11.25 1#Hot stove - Ignition angle - Setpoint of air control valve opening: The preset opening of the air control valve during ignition of No. 1 hot blast stove, used to establish the air flow required for ignition, ensuring successful ignition.

7.11.26 1#Hot stove - Ignition angle - Setpoint of coal gas control valve opening: The preset opening of the gas control valve during ignition of No. 1 hot blast stove, used to establish the gas flow required for ignition, matching the air flow.

7.11.27 1#Hot stove - Ignition angle - Delay time for opening gas control valve: The delay time for opening the gas valve after opening the air valve during ignition of No. 1 hot blast stove, ensuring that air enters first to form a safe environment, then gas is introduced for ignition.

7.11.28 1#Hot stove - Lower limit value of air flow under closing coal gas control valve: The minimum allowable air flow when closing the gas control valve of No. 1 hot blast stove, preventing flame-out or backfire due to too low an air flow, ensuring safety.

7.11.29 1#Hot stove - Lower Limit Value of air control valve opening under closing coal gas control valve: The minimum allowable opening of the air control valve when closing the gas control valve of No. 1 hot blast stove, working together with the lower limit of air flow to ensure sufficient air flow when closing the gas.

7.11.30 2#Hot stove - Upper limit setpoint of combustion - Dome temperature: The maximum allowable dome temperature during the combustion phase of No. 2 hot blast stove. Exceeding this value triggers automatic regulation or an alarm to prevent dome overtemperature during the combustion phase.
7.11.31 2#Hot stove - Upper limit setpoint of combustion - Temperature of waste gas: The maximum allowable waste gas temperature during the combustion phase of No. 2 hot blast stove. Exceeding this value triggers an alarm or combustion adjustment to prevent waste gas overheating from affecting equipment or thermal efficiency.
7.11.32 2#Hot stove - Lower limit setpoint of combustion - Dome temperature: The minimum allowable dome temperature during the blast phase of No. 2 hot blast stove. Falling below this value triggers an alarm or early stove change to ensure the blast temperature meets the blast furnace requirements.
7.11.33 2#Hot stove - Lower limit setpoint of combustion - Temperature of waste gas: The minimum allowable waste gas temperature during the blast phase of No. 2 hot blast stove. Falling below this value may indicate insufficient heat storage, requiring adjustment of the combustion or stove change cycle.
7.11.34 2#Hot stove - Upper limit setpoint of dome temperature: The overall maximum allowable dome temperature for No. 2 hot blast stove. Exceeding this value triggers an alarm or automatic regulation to prevent dome overtemperature damage.
7.11.35 2#Hot stove - Upper limit setpoint of temperature of waste gas: The maximum allowable waste gas temperature for No. 2 hot blast stove. Exceeding this value triggers an alarm or combustion adjustment to prevent waste gas overheating.
7.11.36 2#Hot stove - Temperature difference setpoint under ignition failure: The temperature difference threshold for determining ignition failure of No. 2 hot blast stove. When the temperature rise after ignition does not reach this difference, it is determined as ignition failure, triggering safety protection.
7.11.37 2#Hot stove - Time difference setpoint under ignition failure: The time threshold for determining ignition failure of No. 2 hot blast stove. If the temperature does not reach the expected value within the set time, it is determined as ignition failure, triggering safety protection.
7.11.38 2#Hot stove - Opening time of impact valve: The opening duration of the impact valve of No. 2 hot blast stove, used to control the amount of pressurization, ensuring that the hot blast stove is evenly pressurized.
7.11.39 2#Hot stove - Closing time of impact valve: The closing duration or interval time of the impact valve of No. 2 hot blast stove, used to control the pressurization cycle, working together with the opening time to achieve pressure equalization control.
7.11.40 2#Hot stove - Hydraulic setpoint under normal operation: The pressure or state setpoint for the normal operation of the hydraulic system of No. 2 hot blast stove, used for interlock protection. When the hydraulic system deviates from the setpoint, an alarm or protection action is triggered.
7.11.41 2#Hot stove - Equalizing pressure setpoint under normal operation: The pressure setpoint for the normal operation of the equalizing pressure system of No. 2 hot blast stove, used to control the pressurization process, ensuring that the pressure equalization meets the requirements.
7.11.42 2#Hot stove - Nitrogen purging time under “combustion → stop”: The duration of nitrogen purging when switching No. 2 hot blast stove from the combustion state to the stop state, used to displace combustible gas, ensuring the safety of the switching process.
7.11.43 2#Hot stove - Nitrogen purging time under “stop → combustion”: The duration of nitrogen purging before switching No. 2 hot blast stove from the stop state to the combustion state, used to displace air, preventing explosion during ignition.
7.11.44 2#Hot stove - Nitrogen purging time under ignition failure: The duration of nitrogen purging after an ignition failure of No. 2 hot blast stove, used to remove unburned gas, preparing for re-ignition and

ensuring safety.

7.11.45 2#Hot stove - Ignition angle - Setpoint of air control valve opening: The preset opening of the air control valve during ignition of No. 2 hot blast stove, used to establish the air flow required for ignition, ensuring successful ignition.

7.11.46 2#Hot stove - Ignition angle - Setpoint of coal gas control valve opening: The preset opening of the gas control valve during ignition of No. 2 hot blast stove, used to establish the gas flow required for ignition, matching the air flow.

7.11.47 2#Hot stove - Ignition angle - Delay time for opening gas control valve: The delay time for opening the gas valve after opening the air valve during ignition of No. 2 hot blast stove, ensuring that air enters first to form a safe environment, then gas is introduced for ignition.

7.11.48 2#Hot stove - Lower limit value of air flow under closing coal gas control valve: The minimum allowable air flow when closing the gas control valve of No. 2 hot blast stove, preventing flame-out or backfire due to too low an air flow, ensuring safety.

7.11.49 2#Hot stove - Lower Limit Value of air control valve opening under closing coal gas control valve: The minimum allowable opening of the air control valve when closing the gas control valve of No. 2 hot blast stove, working together with the lower limit of air flow to ensure sufficient air flow when closing the gas.

7.11.50 3#Hot stove - Upper limit setpoint of combustion - Dome temperature: The maximum allowable dome temperature during the combustion phase of No. 3 hot blast stove. Exceeding this value triggers automatic regulation or an alarm to prevent dome overtemperature during the combustion phase.

7.11.51 3#Hot stove - Upper limit setpoint of combustion - Temperature of waste gas: The maximum allowable waste gas temperature during the combustion phase of No. 3 hot blast stove. Exceeding this value triggers an alarm or combustion adjustment to prevent waste gas overheating from affecting equipment or thermal efficiency.

7.11.52 3#Hot stove - Lower limit setpoint of combustion - Dome temperature: The minimum allowable dome temperature during the blast phase of No. 3 hot blast stove. Falling below this value triggers an alarm or early stove change to ensure the blast temperature meets the blast furnace requirements.

7.11.53 3#Hot stove - Lower limit setpoint of combustion - Temperature of waste gas: The minimum allowable waste gas temperature during the blast phase of No. 3 hot blast stove. Falling below this value may indicate insufficient heat storage, requiring adjustment of the combustion or stove change cycle.

7.11.54 3#Hot stove - Upper limit setpoint of dome temperature: The overall maximum allowable dome temperature for No. 3 hot blast stove. Exceeding this value triggers an alarm or automatic regulation to prevent dome overtemperature damage.

7.11.55 3#Hot stove - Upper limit setpoint of temperature of waste gas: The maximum allowable waste gas temperature for No. 3 hot blast stove. Exceeding this value triggers an alarm or combustion adjustment to prevent waste gas overheating.

7.11.56 3#Hot stove - Temperature difference setpoint under ignition failure: The temperature difference threshold for determining ignition failure of No. 3 hot blast stove. When the temperature rise after ignition does not reach this difference, it is determined as ignition failure, triggering safety protection.

7.11.57 3#Hot stove - Time difference setpoint under ignition failure: The time threshold for determining ignition failure of No. 3 hot blast stove. If the temperature does not reach the expected value within the set time, it is determined as ignition failure, triggering safety protection.

7.11.58 3#Hot stove - Opening time of impact valve: The opening duration of the impact valve of No. 3 hot blast stove, used to control the amount of pressurization, ensuring that the hot blast stove is evenly pressurized.

7.11.59 3#Hot stove - Closing time of impact valve: The closing duration or interval time of the impact valve of No. 3 hot blast stove, used to control the pressurization cycle, working together with the opening time to achieve pressure equalization control.
7.11.60 3#Hot stove - Hydraulic setpoint under normal operation: The pressure or state setpoint for the normal operation of the hydraulic system of No. 3 hot blast stove, used for interlock protection. When the hydraulic system deviates from the setpoint, an alarm or protection action is triggered.
7.11.61 3#Hot stove - Equalizing pressure setpoint under normal operation: The pressure setpoint for the normal operation of the equalizing pressure system of No. 3 hot blast stove, used to control the pressurization process, ensuring that the pressure equalization meets the requirements.
7.11.62 3#Hot stove - Nitrogen purging time under "combustion → stop": The duration of nitrogen purging when switching No. 3 hot blast stove from the combustion state to the stop state, used to displace combustible gas, ensuring the safety of the switching process.
7.11.63 3#Hot stove - Nitrogen purging time under "stop → combustion": The duration of nitrogen purging before switching No. 3 hot blast stove from the stop state to the combustion state, used to displace air, preventing explosion during ignition.
7.11.64 3#Hot stove - Nitrogen purging time under ignition failure: The duration of nitrogen purging after an ignition failure of No. 3 hot blast stove, used to remove unburned gas, preparing for re-ignition and ensuring safety.
7.11.65 3#Hot stove - Ignition angle - Setpoint of air control valve opening: The preset opening of the air control valve during ignition of No. 3 hot blast stove, used to establish the air flow required for ignition, ensuring successful ignition.
7.11.66 3#Hot stove - Ignition angle - Setpoint of coal gas control valve opening: The preset opening of the gas control valve during ignition of No. 3 hot blast stove, used to establish the gas flow required for ignition, matching the air flow.
7.11.67 3#Hot stove - Ignition angle - Delay time for opening gas control valve: The delay time for opening the gas valve after opening the air valve during ignition of No. 3 hot blast stove, ensuring that air enters first to form a safe environment, then gas is introduced for ignition.
7.11.68 3#Hot stove - Lower limit value of air flow under closing coal gas control valve: The minimum allowable air flow when closing the gas control valve of No. 3 hot blast stove, preventing flame-out or backfire due to too low an air flow, ensuring safety.
7.11.69 3#Hot stove - Lower Limit Value of air control valve opening under closing coal gas control valve: The minimum allowable opening of the air control valve when closing the gas control valve of No. 3 hot blast stove, working together with the lower limit of air flow to ensure sufficient air flow when closing the gas.
7.11.70 Material position of desulfurant level: The material level height in the desulfurizer storage tank, used to monitor the desulfurizer inventory, replenish in time, and ensure the continuous operation of the desulfurization system.
7.11.71 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Hot blast supply system - Combustion chamber - System configuration".

7.12 Hot blast supply system - Nitrogen main pipe - Operation monitoring

7.12.1 Temperature of main nitrogen pipe: The temperature of the gas in the main nitrogen pipe, reflecting the temperature change of nitrogen during transportation, used to monitor pipeline insulation and medium

state.
7.12.2 Pressure of main nitrogen pipe: The static pressure of the gas in the main nitrogen pipe, which is the pressure reference of the entire nitrogen system, determining the pressure supply capacity of each branch nitrogen pipeline.
7.12.3 Flow rate of main nitrogen pipe: The volume of nitrogen flowing through the main nitrogen pipe per unit time, reflecting the total gas supply of the nitrogen system, used to monitor nitrogen consumption and distribution.
7.12.4 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Hot blast supply system - Nitrogen main pipe - Operation monitoring".

8 Equipment maintaining

8.1 Blast supply system - Tuyere equipment - Operation monitoring

8.1.1 Blow-off condition: A signal indicating whether the blast furnace is in the blowing-off state (yes/no). When blowing off, the blast furnace stops air supply, used for operation indication and interlock control.
8.1.2 Operating mode: The current operating mode of the blast furnace or related systems, such as normal production, blow-off, maintenance, etc., used to monitor the system operation state.
8.1.3 Accumulated usage days: The cumulative operating days of equipment (such as filter bags, valves) since the last replacement or commissioning, used for life management and maintenance planning.
8.1.4 Last replacement date: The specific date of the last replacement of equipment, used to record maintenance history and guide the next replacement cycle.
8.1.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category "Blast supply system - Tuyere equipment - Operation monitoring".

8.2 Environmental protection system - Monitoring instruments at various emission points - Operation monitoring

8.2.1 Monitored value of carbon monoxide: The real-time monitored concentration of carbon monoxide, used for safety monitoring and environmental emission control.
8.2.2 Converted value of carbon monoxide: The value obtained by converting the measured carbon monoxide value under standard conditions (such as reference oxygen content), used for comparison with environmental emission standards.
8.2.3 Monitored value of oxygen content: The real-time monitored value of oxygen concentration in the flue gas, used for combustion control and emission conversion.
8.2.4 Actual value of ammonia gas: The real-time monitored concentration of ammonia (NH_3), used for denitrification system monitoring and environmental protection emissions.
8.2.5 Converted value of ammonia gas: The value obtained by converting the measured ammonia value under standard conditions, used for comparison with environmental emission standards.
8.2.6 Actual value of sulfur dioxide: The real-time monitored concentration of sulfur dioxide (SO_2), used for environmental emission control.
8.2.7 Converted value of sulfur dioxide: The value obtained by converting the measured sulfur dioxide value under standard conditions, used for comparison with environmental emission standards.
8.2.8 Actual value of flue dust: The real-time monitored concentration of particulate matter (flue dust) in the

flue gas, used for dust removal effect evaluation and environmental emissions.

8.2.9 Converted value of flue dust: The value obtained by converting the measured flue dust value under standard conditions, used for comparison with environmental emission standards.

8.2.10 Hourly value of carbon monoxide: The hourly average measured value of carbon monoxide, used for trend analysis and environmental data statistics.

8.2.11 Hourly converted value of carbon monoxide: The hourly average measured value of carbon monoxide converted under standard conditions, used for comparison with the hourly average environmental standard.

8.2.12 Emission standard of carbon monoxide: The national or local emission limit for carbon monoxide, used to determine whether emissions meet the standard.

8.2.13 Hourly value of oxygen content: The hourly average measured value of oxygen concentration, used for combustion optimization and emission conversion.

8.2.14 Emission standard of oxygen content: The specified reference value of oxygen content in flue gas, used for converting emission concentrations.

8.2.15 Hourly value of ammonia gas: The hourly average measured value of ammonia, used for denitrification system monitoring.

8.2.16 Hourly converted value of ammonia gas: The hourly average measured value of ammonia converted under standard conditions, used for comparison with the hourly average environmental standard.

8.2.17 Emission standard of ammonia gas: The national or local emission limit for ammonia, used to determine whether the ammonia slip of the denitrification system meets the standard.

8.2.18 Hourly value of sulfur dioxide: The hourly average measured value of sulfur dioxide, used for environmental data statistics and trend analysis.

8.2.19 Hourly converted value of sulfur dioxide: The hourly average measured value of sulfur dioxide converted under standard conditions, used for comparison with the hourly average environmental standard.

8.2.20 Emission standard of sulfur dioxide: The national or local emission limit for sulfur dioxide, used to determine whether emissions meet the standard.

8.2.21 Hourly value of flue dust: The hourly average measured value of flue dust, used for environmental data statistics and dust removal effect evaluation.

8.2.22 Hourly converted value of flue dust: The hourly average measured value of flue dust converted under standard conditions, used for comparison with the hourly average environmental standard.

8.2.23 Emission standard of flue dust: The national or local emission limit for flue dust, used to determine whether emissions meet the standard.

8.2.24 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Environmental protection system - Monitoring instruments at various emission points - Operation monitoring”.

8.3 Furnace baking system - Taphole - Operation monitoring

8.3.1 Bakeout temperature of 1# taphole: The measured temperature of the No. 1 taphole area during the bakeout process, used to monitor the drying and heating process of the refractory material in the taphole area, preventing material cracking due to too rapid a temperature rise.

8.3.2 Bakeout temperature of 2# taphole: The measured temperature of the No. 2 taphole area during the bakeout process, used to monitor the drying and heating process of the refractory material in the taphole area.

8.3.3 Bakeout temperature of 3# taphole: The measured temperature of the No. 3 taphole area during the bakeout process, used to monitor the drying and heating process of the refractory material in the taphole area.

8.3.4 Bakeout temperature of 4# taphole: The measured temperature of the No. 4 taphole area during the

bakeout process, used to monitor the drying and heating process of the refractory material in the taphole area.
8.3.5 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace baking system - Taphole - Operation monitoring”.

8.4 Furnace baking system - Hot blast main pipe - Operation monitoring

8.4.1 Temperature of main hot blast pipe: The measured temperature of the hot blast in the main hot blast pipe, an important reference for the blast furnace blast temperature, used to monitor the temperature change during hot blast transportation.
8.4.2 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace baking system - Hot blast main pipe - Operation monitoring”.

8.5 PLC Inspection system - Instruments & Meters - Operation monitoring

8.5.1 Fault and alarm about UPS: An alarm signal when the uninterruptible power supply system fails, indicating an abnormality in the UPS equipment, which may affect the power supply safety of key equipment.
8.5.2 Low voltage alarm of battery: An alarm signal when the UPS battery voltage is too low, indicating insufficient battery power or a charging system failure, requiring timely treatment to prevent the inability to provide backup power in case of a power outage.
8.5.3 Power failure alarm: An alarm signal when a mains power outage or power supply interruption is detected, used to prompt the operator of a power supply abnormality, requiring attention to the UPS switching status.
8.5.4 Fault and alarm about rectifier: An alarm signal when the UPS rectifier fails. A rectifier failure may cause the battery to not charge properly or the UPS to not work properly.
8.5.5 Alarm about PLC changeover switch: An alarm signal when the status of the PLC control system changeover switch is abnormal, indicating that the changeover switch position is not as expected, which may affect the control logic.
8.5.6 Q1 close of PLC changeover switch: A feedback signal that the Q1 changeover switch of the PLC control system is in the closed state, used to monitor the actual position of the changeover switch.
8.5.7 Q2 close of PLC changeover switch: A feedback signal that the Q2 changeover switch of the PLC control system is in the closed state, used to monitor the actual position of the changeover switch.
8.5.8 Q1 trip of PLC changeover switch: A feedback signal that the Q1 changeover switch of the PLC control system is in the tripped state, indicating that the switch has tripped, possibly due to overload or failure.
8.5.9 Q2 trip of PLC changeover switch: A feedback signal that the Q2 changeover switch of the PLC control system is in the tripped state, indicating that the switch has tripped.
8.5.10 Alarm about instrument changeover switch: An alarm signal when the status of the instrument control system changeover switch is abnormal, indicating that the changeover switch position is not as expected, which may affect instrument signal acquisition.
8.5.11 Q1 close of instrument changeover switch: A feedback signal that the Q1 changeover switch of the instrument control system is in the closed state, used to monitor the actual position of the changeover switch.
8.5.12 Q2 close of instrument changeover switch: A feedback signal that the Q2 changeover switch of the instrument control system is in the closed state, used to monitor the actual position of the changeover switch.
8.5.13 Q1 trip of instrument changeover switch: A feedback signal that the Q1 changeover switch of the

instrument control system is in the tripped state, indicating that the switch has tripped.
8.5.14 Q2 trip of instrument changeover switch: A feedback signal that the Q2 changeover switch of the instrument control system is in the tripped state, indicating that the switch has tripped.
8.5.15 Level-1 alarm about blast furnace main control room: The level-1 alarm signal in the blast furnace main control room, with the highest priority, usually involving major safety hazards or serious equipment failures, requiring immediate action.
8.5.16 Level-2 alarm about blast furnace main control room: The level-2 alarm signal in the blast furnace main control room, with the next highest priority, indicating equipment abnormality or process parameter deviation, requiring timely attention and action.
8.5.17 Temperature of Control Cabinet: The measured temperature inside the control cabinet, used to monitor the operating environment temperature of the equipment inside the cabinet, preventing electronic component failure due to excessive temperature.
8.5.18 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “PLC Inspection system - Instruments & Meters - Operation monitoring”.

8.6 Furnace baking system - Tuyere equipment - Operation monitoring

8.6.1 Bakeout temperature of 1# tuyere: The measured temperature of the No. 1 tuyere area during the bakeout process, used to monitor the drying and heating process of the refractory material in the tuyere area, preventing tuyere damage due to too rapid a temperature rise.
8.6.2 Bakeout temperature of 2# tuyere: The measured temperature of the No. 2 tuyere area during the bakeout process.
8.6.3 Bakeout temperature of 3# tuyere: The measured temperature of the No. 3 tuyere area during the bakeout process.
8.6.4 Bakeout temperature of 4# tuyere: The measured temperature of the No. 4 tuyere area during the bakeout process.
8.6.5 Bakeout temperature of 5# tuyere: The measured temperature of the No. 5 tuyere area during the bakeout process.
8.6.6 Bakeout temperature of 6# tuyere: The measured temperature of the No. 6 tuyere area during the bakeout process.
8.6.7 Bakeout temperature of 7# tuyere: The measured temperature of the No. 7 tuyere area during the bakeout process.
8.6.8 Bakeout temperature of 8# tuyere: The measured temperature of the No. 8 tuyere area during the bakeout process.
8.6.9 Bakeout temperature of 9# tuyere: The measured temperature of the No. 9 tuyere area during the bakeout process.
8.6.10 Bakeout temperature of 10# tuyere: The measured temperature of the No. 10 tuyere area during the bakeout process.
8.6.11 Bakeout temperature of 11# tuyere: The measured temperature of the No. 11 tuyere area during the bakeout process.
8.6.12 Bakeout temperature of 12# tuyere: The measured temperature of the No. 12 tuyere area during the bakeout process.
8.6.13 Bakeout temperature of 13# tuyere: The measured temperature of the No. 13 tuyere area during the bakeout process.

8.6.14 Bakeout temperature of 14# tuyere: The measured temperature of the No. 14 tuyere area during the bakeout process.
8.6.15 Bakeout temperature of 15# tuyere: The measured temperature of the No. 15 tuyere area during the bakeout process.
8.6.16 Bakeout temperature of 16# tuyere: The measured temperature of the No. 16 tuyere area during the bakeout process.
8.6.17 Bakeout temperature of 17# tuyere: The measured temperature of the No. 17 tuyere area during the bakeout process.
8.6.18 Bakeout temperature of 18# tuyere: The measured temperature of the No. 18 tuyere area during the bakeout process.
8.6.19 Bakeout temperature of 19# tuyere: The measured temperature of the No. 19 tuyere area during the bakeout process.
8.6.20 Bakeout temperature of 20# tuyere: The measured temperature of the No. 20 tuyere area during the bakeout process.
8.6.21 Bakeout temperature of 21# tuyere: The measured temperature of the No. 21 tuyere area during the bakeout process.
8.6.22 Bakeout temperature of 22# tuyere: The measured temperature of the No. 22 tuyere area during the bakeout process.
8.6.23 Bakeout temperature of 23# tuyere: The measured temperature of the No. 23 tuyere area during the bakeout process.
8.6.24 Bakeout temperature of 24# tuyere: The measured temperature of the No. 24 tuyere area during the bakeout process.
8.6.25 Bakeout temperature of 25# tuyere: The measured temperature of the No. 25 tuyere area during the bakeout process.
8.6.26 Bakeout temperature of 26# tuyere: The measured temperature of the No. 26 tuyere area during the bakeout process.
8.6.27 Bakeout temperature of 27# tuyere: The measured temperature of the No. 27 tuyere area during the bakeout process.
8.6.28 Bakeout temperature of 28# tuyere: The measured temperature of the No. 28 tuyere area during the bakeout process.
8.6.29 Bakeout temperature of 29# tuyere: The measured temperature of the No. 29 tuyere area during the bakeout process.
8.6.30 Bakeout temperature of 30# tuyere: The measured temperature of the No. 30 tuyere area during the bakeout process.
8.6.31 Bakeout temperature of 31# tuyere: The measured temperature of the No. 31 tuyere area during the bakeout process.
8.6.32 Bakeout temperature of 32# tuyere: The measured temperature of the No. 32 tuyere area during the bakeout process
8.6.33 Additional storage templates: Textual data such as operational experience, judgment rules, and production logs provided by workers of monitoring data category “Furnace baking system - Tuyere equipment - Operation monitoring”.